

5

Linear Transformations

SECTION 5.1 ➔ Introduction to Linear Transformations

By the end of this section you will be able to

- determine what is meant by a transformation
- prove some properties of linear transformations (mappings)

In this chapter we look at functions defined on vector spaces called transformations. Most of the theorems in linear algebra can be formulated in terms of linear transformations which we will discuss below. Transformations also give matrices a geometric flavour. For example, rotations, reflections and projections of vectors can be described by writing a transformation as a matrix.

Transformations are particularly useful in computer graphics and games. Examples of transformations which can be used to create an animation are shown in Fig. 5.1.

Sometimes linear algebra is described as the study of linear transformations on vector spaces.

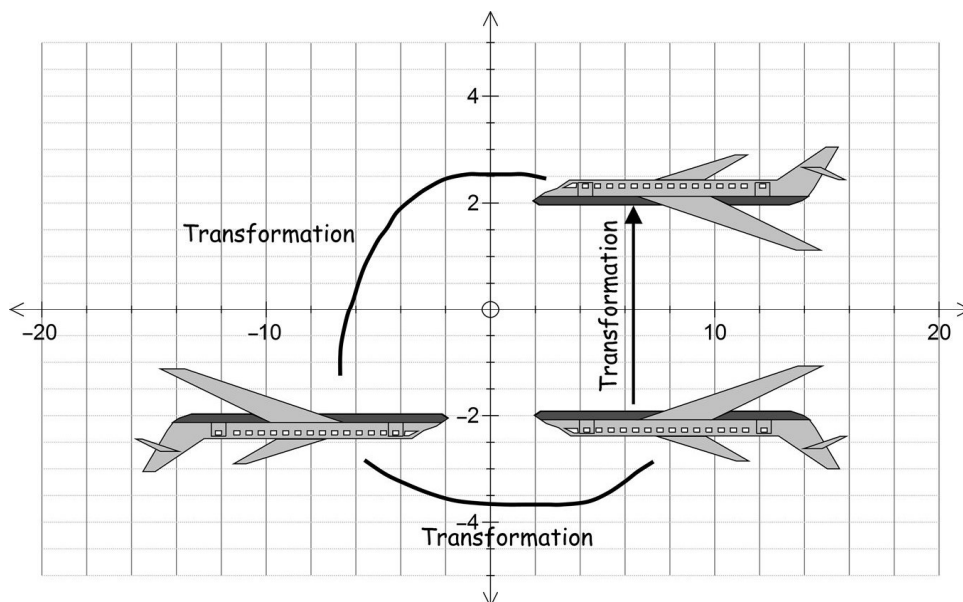


Figure 5.1

5.1.1 Functions



What does the term function mean?

In everyday language it means the purpose of something or someone. In mathematics, a function is a process that takes an input value, and generates a *unique* output value.

More formally:

Definition (5.1). A function operating from set A to set B is a rule which assigns *every* element of A to one and *only one* element in the set B .

A function f from A to B is denoted by $f : A \rightarrow B$. The set where the function starts from (A) is called the **domain** and where it arrives (B) is called the **codomain**. The function f takes a particular value, such as x , and converts it to a value y . This is denoted by

$$f(x) = y$$

The actual values that the function f takes in the set B is called the **range** or **image** (Fig. 5.2).

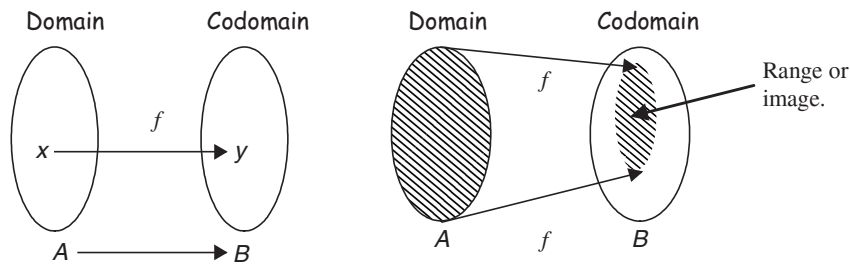


Figure 5.2

In this chapter we discuss a function between two vector spaces such as V and W . For example, $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is a function between two vector spaces (Euclidean spaces) where $\mathbb{R}^2$ is the domain and $\mathbb{R}^3$ the codomain.

In general, if for a function f we have $f : V \rightarrow W$ (a function from V to W), where V and W are vector spaces, then the function f is called a **transformation** or **map** from V to W . Transformation is just another word for function and works on vectors as well as numbers. Also we tend to use the letter T for transformation, that is $T : V \rightarrow W$.

In some mathematical literature you will see the term *map* used to mean transformation. They are *both* the same thing. We will use transformation.

Example 5.1

Consider $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ (transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$) defined by $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} -y \\ x \end{pmatrix}$.

Determine $T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right)$ and plot both $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right)$ on $\mathbb{R}^2$.

Solution

How do we find $T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right)$?

By substituting $x = 1$ and $y = 2$ into $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} -y \\ x \end{pmatrix}$:

$$T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

Plotting these on $\mathbb{R}^2$ we have (Fig. 5.3).

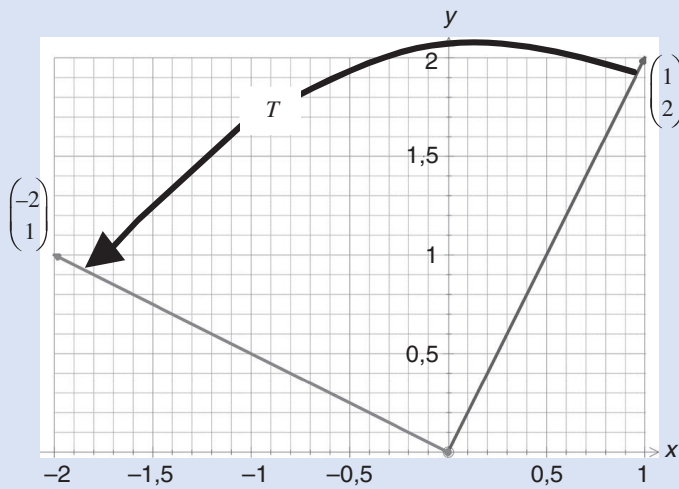


Figure 5.3

The transformation T rotates the vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ by 90° in an anti-clockwise direction with centre at the origin.

Example 5.2

Consider the transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{pmatrix} x - y \\ x - z \end{pmatrix}$. Determine

(i) $T\left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}\right)$

(ii) $T\left(\begin{bmatrix} -1 \\ -2 \\ -3 \end{bmatrix}\right)$

Solution

This type of transformation is common in computer graphics because it represents something in three dimensions ($\mathbb{R}^3$) onto a two-dimensional screen ($\mathbb{R}^2$).

(continued...)

(i) For $T\left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}\right)$ we substitute $x = 1, y = 2$ and $z = 3$ into $T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{pmatrix} x - y \\ x - z \end{pmatrix}$:

$$T\left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}\right) = \begin{pmatrix} 1 - 2 \\ 1 - 3 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \end{pmatrix}$$

(ii) Similarly we have

$$T\left(\begin{bmatrix} -1 \\ -2 \\ -3 \end{bmatrix}\right) = \begin{pmatrix} -1 - (-2) \\ -1 - (-3) \end{pmatrix} = \begin{pmatrix} -1 + 2 \\ -1 + 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Note that the transformation T maps a vector from $\mathbb{R}^3$ (3-space) to a vector in $\mathbb{R}^2$ (2-space). (Transforms a vector in 3d onto a plane.)

5.1.2 Linear transformations (linear mappings)



What do you think the term 'linear transformation' means?

A linear transformation between vector spaces is a special transformation (mapping) which preserves the fundamental linear algebra operations – scalar multiplication and vector addition.

The formal definition is:

Definition (5.2). A transformation $T : V \rightarrow W$ is called a **linear transformation** $\Leftrightarrow$ for all vectors $\mathbf{u}$ and $\mathbf{v}$ in the vector space V and for any scalar k we have

(a) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ [T preserves vector addition]

(b) $T(k\mathbf{u}) = k T(\mathbf{u})$ [T preserves scalar multiplication]

For example, the transformations given in the above examples 5.1 and 5.2 are linear. For T to be a linear transformation or a linear map it needs to satisfy *both* conditions, (a) and (b).



What do these conditions mean?

They both mean that for a linear transformation it does *not* matter whether you carry out the vector operation first and then transform, or vice versa.

A transformation is a function with an input and an output (Fig. 5.4).



Figure 5.4

A linear transformation is one where the output changes proportionately to the input. A real-life example is cooking recipes – double your ingredients and you double your output.

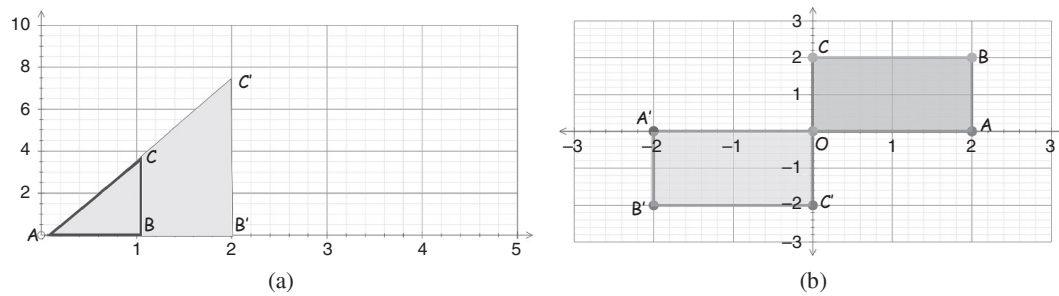


Figure 5.5

We can treat matrices as linear transformations which act on points or vectors (Fig. 5.5).

Fig. 5.5(a) shows an example of a transformation which enlarges an object. (Triangle ABC is enlarged to $A'B'C'$ – doubles each side.) The matrix transformation in this case is $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$.

Fig. 5.5(b) shows an example of a transformation which rotates the points of the rectangle anti-clockwise by 180° about the origin. The matrix transformation is $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$.

Example 5.3

Show that $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(\mathbf{x}) = \mathbf{A}\mathbf{x}$ where $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is a linear transformation.

Solution

What does this transformation do?

Let $\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$ then $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ -y \end{pmatrix}$.

The transformation T places a negative in front of the y . The result is shown in Fig. 5.6.

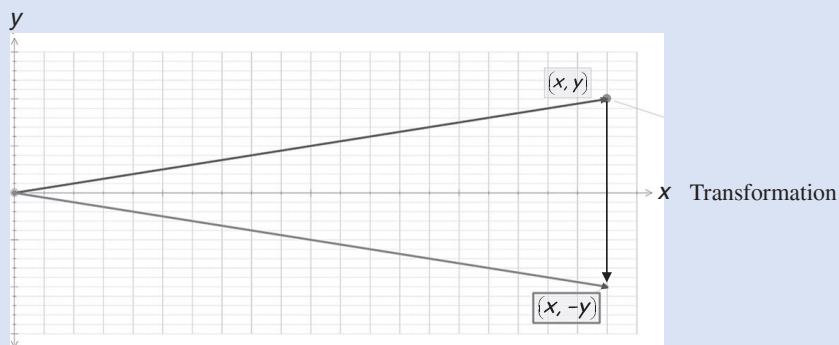


Figure 5.6

(continued...)

The given transformation *reflects* the vector in the horizontal axis. We need to show that this is a linear transformation.

Let $\mathbf{u} = \begin{bmatrix} x \\ y \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} s \\ t \end{bmatrix}$ be our vectors in $\mathbb{R}^2$.

How do we show that the given T is linear?

Show that T preserves both vector addition and scalar multiplication. This means that we need to check that both conditions (a) and (b) of the above Definition (5.2) hold.

Checking condition (a):

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= T\left(\begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} s \\ t \end{bmatrix}\right) \\ &= T\left(\begin{bmatrix} x+s \\ y+t \end{bmatrix}\right) \quad \equiv \quad \begin{pmatrix} x+s \\ -(y+t) \end{pmatrix} = \begin{pmatrix} x+s \\ -y-t \end{pmatrix} \end{aligned}$$

applying the given transformation

Now we need to show that $T(\mathbf{u}) + T(\mathbf{v})$ is equal to this vector on the right, $\begin{pmatrix} x+s \\ -y-t \end{pmatrix}$:

$$T(\mathbf{u}) + T(\mathbf{v}) = \begin{pmatrix} x \\ -y \end{pmatrix} + \begin{pmatrix} s \\ -t \end{pmatrix} = \begin{pmatrix} x+s \\ -y-t \end{pmatrix}$$

Thus we have shown condition (a), that is $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$.

In this example, linear transformation means that we can add the vectors $\mathbf{u}$ and $\mathbf{v}$ and then reflect in the horizontal axis, or we can reflect first and then add. In either case, we end up with the same vector because $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$.

Checking condition (b):

We need to demonstrate that T preserves scalar multiplication, that is $T(k\mathbf{u}) = kT(\mathbf{u})$, where k is any scalar:

$$T(k\mathbf{u}) = T\left(k\begin{bmatrix} x \\ y \end{bmatrix}\right) = T\left(\begin{bmatrix} kx \\ ky \end{bmatrix}\right) = \begin{pmatrix} kx \\ -ky \end{pmatrix} \quad \left[\text{applying } T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} x \\ -y \end{pmatrix}\right]$$

What do we do next?

Show that $kT(\mathbf{u})$ is equal to the above vector on the right, $\begin{pmatrix} kx \\ -ky \end{pmatrix}$:

$$kT(\mathbf{u}) = kT\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = k\begin{pmatrix} x \\ -y \end{pmatrix} = \begin{pmatrix} kx \\ -ky \end{pmatrix} = T(k\mathbf{u}) \quad [\text{from above}]$$

Thus we have our result. Again, we can change the order, that is first apply scalar multiplication and then transform, or vice versa. We end up with the same vector, $T(k\mathbf{u}) = kT(\mathbf{u})$.

This means the given transformation T satisfies *both* conditions (a) and (b) of definition:

(5.2) (a) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and (b) $T(k\mathbf{u}) = kT(\mathbf{u})$.

Therefore T is a linear transformation.

Let V be a vector space. Then a *linear mapping or transformation* $T : V \rightarrow V$ (the domain and codomain are the *same* vector space) is called a **linear operator**. Linear operators are important in subjects such as functional analysis and quantum mechanics because they preserve vector addition and scalar multiplication.

? Is the transformation T given in the above Example 5.3 a linear operator?

Yes, because the transform goes from $\mathbb{R}^2$ to $\mathbb{R}^2$ (same vector spaces), that is $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$.

Example 5.4

Show that the transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} x+y \\ x-y \\ y \end{pmatrix}$ is linear. This is not a linear operator because T goes from $\mathbb{R}^2$ to $\mathbb{R}^3$ (different vector spaces). The transformation T in this case transforms a vector in the plane onto 3d space (Fig. 5.7).

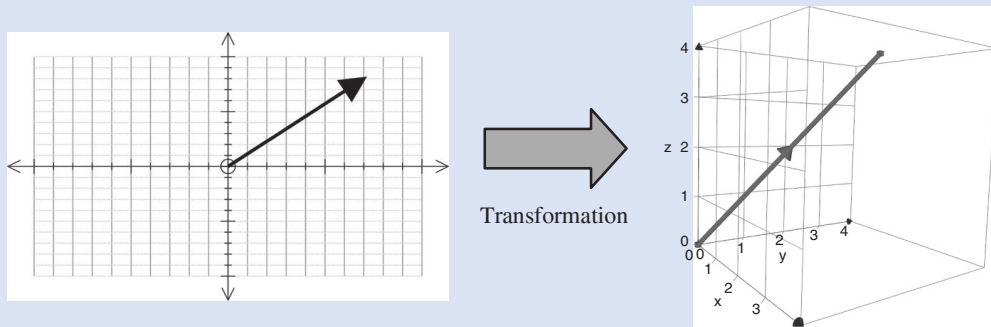


Figure 5.7

Solution

Let $\mathbf{u} = \begin{pmatrix} x \\ y \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} s \\ t \end{pmatrix}$. As for the above Example 5.3 we have:

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= T\left(\begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} s \\ t \end{bmatrix}\right) \\ &= T\left(\begin{bmatrix} x+s \\ y+t \end{bmatrix}\right) \quad \equiv \quad \begin{pmatrix} (x+s) + (y+t) \\ (x+s) - (y+t) \\ y+t \end{pmatrix} = \begin{pmatrix} x+s+y+t \\ x-y+s-t \\ y+t \end{pmatrix} \end{aligned}$$

applying the given transformation

We need to show that $T(\mathbf{u}) + T(\mathbf{v})$ is equal to the vector on the right hand side.

$$\begin{aligned} T(\mathbf{u}) + T(\mathbf{v}) &= T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) + T\left(\begin{bmatrix} s \\ t \end{bmatrix}\right) \\ &\equiv \begin{pmatrix} x+y \\ x-y \\ y \end{pmatrix} + \begin{pmatrix} s+t \\ s-t \\ t \end{pmatrix} = \begin{pmatrix} x+y+s+t \\ x-y+s-t \\ y+t \end{pmatrix} \end{aligned}$$

applying the given transformation

This is identical to $T(\mathbf{u} + \mathbf{v})$, therefore $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$, which means that T preserves vector addition. Similarly we can check $T(k\mathbf{u}) = kT(\mathbf{u})$ for any scalar k . Therefore by (5.2) (a) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and (b) $T(k\mathbf{u}) = kT(\mathbf{u})$ the given transformation T is a linear transformation.

A square function such as $f(x) = x^2$ is also a transformation. In this case, we have $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$.

? *Is f a linear transformation?*

We need to check both conditions (a) and (b) of definition (5.2):

$$\begin{aligned} f(x+y) &= (x+y)^2 = x^2 + 2xy + y^2 \\ f(x) + f(y) &= x^2 + y^2 \end{aligned}$$

Since $x^2 + 2xy + y^2 \neq x^2 + y^2$ [not equal], therefore

$$f(x+y) \neq f(x) + f(y)$$

? *What does this mean?*

It means that f is *not* a linear function or linear transformation because condition (a) of (5.2) fails. [The output does *not* change in proportion; we have an extra $2xy$ with $f(x+y)$.]

Example 5.5

Let V be an inner product space. Show that $T : V \rightarrow \mathbb{R}$ defined by $T(\mathbf{u}) = \|\mathbf{u}\|$ is *not* a linear transformation. [This transformation measures the norm or length of a vector.]

Solution

How can we show that the given transformation is not linear?

If the transformation T fails one (or both) of the conditions of (5.2), that is:

$$(a) T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \quad (b) T(k\mathbf{u}) = kT(\mathbf{u})$$

then T is *not* a linear transformation. Checking condition (a):

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= \|\mathbf{u} + \mathbf{v}\| \\ T(\mathbf{u}) &= \|\mathbf{u}\| \text{ and } T(\mathbf{v}) = \|\mathbf{v}\| \end{aligned}$$

However, by Minkowski's inequality of chapter 4:

$$(4.7) \|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\| \quad [\text{less than or equal to}],$$

we have

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= \|\mathbf{u} + \mathbf{v}\| \\ &\leq \|\mathbf{u}\| + \|\mathbf{v}\| = T(\mathbf{u}) + T(\mathbf{v}) \end{aligned}$$

Hence $T(\mathbf{u} + \mathbf{v}) \leq T(\mathbf{u}) + T(\mathbf{v})$, therefore condition (a) fails because we don't have equality and so the given transformation is *not* linear. In this case, T is *not* a linear transformation because the length of the vector $\mathbf{u} + \mathbf{v}$ is *not* equal to the length of $\mathbf{u}$ plus the length of $\mathbf{v}$.

Example 5.6

Let V be a vector space. Show that the identity transformation $T : V \rightarrow V$ defined by $T(\mathbf{u}) = \mathbf{u}$ is a linear operator.

Solution

This is the identity transformation because T maps a given vector $\mathbf{u}$ back to $\mathbf{u}$ (Fig. 5.8).

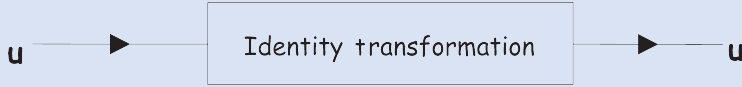


Figure 5.8

How do we show that this transformation is linear?

Clearly, the output is proportional to the input but intuition does *not* constitute a rigorous test. We use our definition of linear transformation. Check the two conditions of (5.2):

(a) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and (b) $T(k\mathbf{u}) = kT(\mathbf{u})$

Let $\mathbf{u}$ and $\mathbf{v}$ be arbitrary vectors in the vector space V . We have

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= \mathbf{u} + \mathbf{v} && \text{[as transformation has the same input and output]} \\ &= T(\mathbf{u}) + T(\mathbf{v}) \end{aligned}$$

Hence condition (a) is satisfied. Let k be any scalar, then

$$T(k\mathbf{u}) = k\mathbf{u} = kT(\mathbf{u})$$

Thus condition (b) is satisfied. Therefore by the above Definition (5.2), we conclude that the given identity transformation, T , is a linear operator.

5.1.3 Properties of linear transformations

In this subsection, we highlight some of the properties of linear transformation.

Proposition (5.3). Let V and W be vector spaces and $\mathbf{u}$ and $\mathbf{v}$ be vectors in V . Let $T : V \rightarrow W$ be a linear transformation, then we have the following:

- (a) $T(\mathbf{O}) = \mathbf{O}$ where $\mathbf{O}$ is the zero vector.
- (b) $T(-\mathbf{u}) = -T(\mathbf{u})$
- (c) $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$

Proof of (a).

We can write the zero vector $\mathbf{O}$ as $0\mathbf{u}$:

$$T(\mathbf{O}) = T(0\mathbf{u}) = \underbrace{0T(\mathbf{u})}_{\text{because } T \text{ is linear}} = \mathbf{O}$$

Thus $T(\mathbf{O}) = \mathbf{O}$. ■

Proof of (b).

We write $-\mathbf{u}$ as $(-1)\mathbf{u}$. We have

$$T(-\mathbf{u}) = T((-1)\mathbf{u}) \quad \stackrel{\text{because } T \text{ is linear}}{=} \quad (-1)T(\mathbf{u}) = -T(\mathbf{u})$$

■

Proof of (c).

We write $\mathbf{u} - \mathbf{v}$ as $\mathbf{u} + (-\mathbf{v})$:

$$\begin{aligned} T(\mathbf{u} - \mathbf{v}) &= T(\mathbf{u} + (-\mathbf{v})) \\ &= T(\mathbf{u}) + T(-\mathbf{v}) \quad [\text{because } T \text{ is linear}] \\ &= T(\mathbf{u}) - T(\mathbf{v}) \quad [\text{by part (b) } T(-\mathbf{v}) = -T(\mathbf{v})] \end{aligned}$$

■

Another important property of linear transformation is the following:

Proposition (5.4). Let V and W be vector spaces. Let $T : V \rightarrow W$ be a linear transformation and $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_n\}$ be vectors in the vector space V such that

$$\mathbf{v} = k_1\mathbf{u}_1 + k_2\mathbf{u}_2 + \dots + k_n\mathbf{u}_n$$

where the k 's are scalars. This means that the vector $\mathbf{v}$ is a linear combination of the $\mathbf{u}_{\text{subscript}}$ vectors. Then

$$T(\mathbf{v}) = k_1T(\mathbf{u}_1) + k_2T(\mathbf{u}_2) + \dots + k_nT(\mathbf{u}_n)$$



How do we prove this proposition?

By using mathematical induction. The three steps of proof by mathematical induction are:

1. Prove the given result for some base case $n = 1$ or $n = m_0$.
2. Assume that the result is true for $n = m$.
3. Prove the result for $n = m + 1$.

Proof.

1. We can take out the scalar $T(k_1\mathbf{u}_1) = k_1T(\mathbf{u}_1)$ because T is linear. The result holds for $n = 1$.
2. We assume that the result is true for $n = m$, that is

$$T(k_1\mathbf{u}_1 + k_2\mathbf{u}_2 + \dots + k_m\mathbf{u}_m) = k_1T(\mathbf{u}_1) + \dots + k_mT(\mathbf{u}_m) \quad (*)$$

3. Required to prove the result for $n = m + 1$:

$$\begin{aligned}
 & T(k_1 \mathbf{u}_1 + \cdots + k_m \mathbf{u}_m + k_{m+1} \mathbf{u}_{m+1}) \\
 &= T([k_1 \mathbf{u}_1 + \cdots + k_m \mathbf{u}_m] + k_{m+1} \mathbf{u}_{m+1}) \\
 &= T([k_1 \mathbf{u}_1 + \cdots + k_m \mathbf{u}_m]) + T(k_{m+1} \mathbf{u}_{m+1}) \quad [\text{because } T \text{ is linear}] \\
 &= \underbrace{k_1 T(\mathbf{u}_1) + \cdots + k_m T(\mathbf{u}_m)}_{\text{By (*)}} + k_{m+1} T(\mathbf{u}_{m+1})
 \end{aligned}$$

Hence the result is true for $n = m + 1$.

By mathematical induction we have our required result. ■

We can use both these propositions (5.3) and (5.4) to determine whether a given transformation is linear or not. If any one of these properties (5.3) (a), (b), (c) or (5.4) is *not* satisfied for a given transformation then we conclude that T is *not* a linear transformation.

In the exercises, you are asked to show that if $T(\mathbf{O}) \neq \mathbf{O}$ [not equal] then T is *not linear*.

Example 5.7

Let P_2 be the vector space of polynomials of degree 2 or less. The polynomials are of the form $c_2x^2 + c_1x + c_0$ where the c 's are the coefficients.

Show that the transformation $T : P_2 \rightarrow P_2$ defined by

$$T(c_2x^2 + c_1x + c_0) = c_2x^2 + c_1x + (c_0 + 1)$$

is *not* a linear transformation.

Solution

We can try testing (5.3) (a) which says that $T(\mathbf{O}) = \mathbf{O}$.

What is the zero vector in this case?

The zero vector is the zero polynomial which means that *all* the coefficients, c 's, are equal to zero.

$$\begin{aligned}
 T(\mathbf{O}) &= T(0x^2 + 0x + 0) \\
 &= 0x^2 + 0x + (0 + 1) = 1 \quad [\text{applying transformation}]
 \end{aligned}$$

Since 1 is *not* the zero vector, therefore T is *not* a linear transformation because we have $T(\mathbf{O}) \neq \mathbf{O}$ [not equal].

Example 5.8

Consider the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T(\mathbf{u}) = \mathbf{A}\mathbf{u} \text{ where } \mathbf{A} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

(continued...)

- (i) The five corners of a house are given by the column vectors in the matrix:

$$\mathbf{H} = \begin{pmatrix} 0 & 0 & 2 & 4 & 4 \\ 0 & 3 & 5 & 3 & 0 \end{pmatrix}$$

Transform each of the column vectors of $\mathbf{H}$ under T .

- (ii) Describe the effect of the transformation T on the column vectors of $\mathbf{H}$.

Solution

- (i) Remember, the matrix multiplication $\mathbf{AH}$ means matrix $\mathbf{A}$ acts on each column vector of $\mathbf{H}$:

$$\mathbf{AH} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 2 & 4 & 4 \\ 0 & 3 & 5 & 3 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 3 & 7 & 7 & 4 \\ 0 & 3 & 3 & -1 & -4 \end{pmatrix}$$

- (ii) Plotting our results (Fig. 5.9):

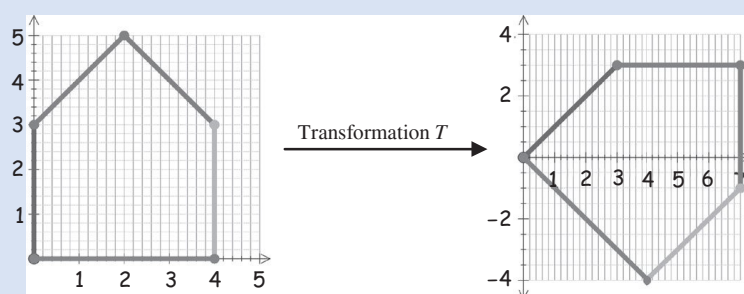


Figure 5.9

The transformation T rotates the given vectors (corners) by 45° in a clockwise direction.

Summary

A transformation or mapping is a function which assigns *every* element in the domain (start) to a *unique* element in the codomain (arrival).

A transformation between two vector spaces V and W denoted $T : V \rightarrow W$ is linear $\Leftrightarrow$

- (a) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all vectors $\mathbf{u}$ and $\mathbf{v}$
- (b) $T(k\mathbf{u}) = kT(\mathbf{u})$ for any scalar k



EXERCISES 5.1

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

1. Consider the transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T(\mathbf{u}) = \mathbf{A}\mathbf{u}$ where $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Determine the vector $T(\mathbf{u})$ where $\mathbf{u}$ is given by:

$$(a) \mathbf{u} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \quad (b) \mathbf{u} = \begin{pmatrix} -1 \\ 5 \end{pmatrix} \quad (c) \mathbf{u} = \begin{pmatrix} \sqrt{2} \\ 1 \end{pmatrix} \quad (d) \mathbf{u} = \begin{pmatrix} -2 \\ -3 \end{pmatrix}$$

For each of these vectors $\mathbf{u}$ and $T(\mathbf{u})$ plot them on the same plane $\mathbb{R}^2$.

2. Consider the transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ given by $T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} xy \\ xz \end{pmatrix}$. Determine the vector $T(\mathbf{u})$ where $\mathbf{u}$ is given by:

$$(a) \mathbf{u} = \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix} \quad (b) \mathbf{u} = \begin{pmatrix} -1 \\ -4 \\ 4 \end{pmatrix} \quad (c) \mathbf{u} = \begin{pmatrix} \sqrt{2} \\ \sqrt{8} \\ \sqrt{18} \end{pmatrix} \quad (d) \mathbf{u} = \begin{pmatrix} -2 \\ -3 \\ -4 \end{pmatrix}$$

3. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be given by $T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} x+y \\ y+z \\ z+x \end{pmatrix}$. Find $T(\mathbf{u})$ where

$$(a) \mathbf{u} = \begin{pmatrix} 2 \\ 4 \\ 7 \end{pmatrix} \quad (b) \mathbf{u} = \begin{pmatrix} -3 \\ 8 \\ -6 \end{pmatrix} \quad (c) \mathbf{u} = \begin{pmatrix} \pi \\ 2\pi \\ 5\pi \end{pmatrix} \quad (d) \mathbf{u} = \begin{pmatrix} 1/2 \\ 2/3 \\ 3/4 \end{pmatrix}$$

4. Explain why $T : U \rightarrow V$ given by $T(\mathbf{u}) = \pm\sqrt{\mathbf{u}}$ is not a transformation.

5. Determine whether the following transformations (mappings) are linear:

$$(a) T : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \text{ given by } (i) T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{pmatrix} y \\ x \end{pmatrix} \quad (ii) T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{pmatrix} x^2 \\ y^2 \end{pmatrix}$$

$$(b) T : \mathbb{R}^3 \rightarrow \mathbb{R}^2 \text{ given by } T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} xy \\ xz \end{pmatrix}$$

$$(c) T : \mathbb{R}^3 \rightarrow \mathbb{R}^3 \text{ given by } (i) T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} x+y \\ y+z \\ z+x \end{pmatrix} \quad (ii) T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} \sqrt{x} \\ \sqrt{y} \\ \sqrt{z} \end{pmatrix}$$

6. Let P_2 be the vector space of polynomials of degree 2 or less. Decide whether the following transformations are linear:

$$(a) T : P_2 \rightarrow P_2 \text{ given by } T(c_2x^2 + c_1x + c_0) = c_0x^2 + c_1x + c_2$$

$$(b) T : P_2 \rightarrow P_2 \text{ given by } T(c_2x^2 + c_1x + c_0) = c_2^2x^2 + c_1^2x + c_0^2$$

7. Let M_{nn} be the vector space of size n by n matrices. Decide whether the following transformations (mappings) are linear:

$$(a) T : M_{22} \rightarrow M_{22} \text{ given by } T(\mathbf{A}) = \mathbf{A}^T \text{ where } \mathbf{A}^T \text{ is the transpose of the matrix } \mathbf{A}.$$

$$(b) T : M_{22} \rightarrow \mathbb{R} \text{ given by } T(\mathbf{A}) = \text{tr}(\mathbf{A}) \text{ where } \text{tr}(\mathbf{A}) \text{ is the trace of the matrix } \mathbf{A}. \\ \text{[Remember, the trace of a matrix is the addition of all the leading diagonal elements of the matrix.]}$$

- (c) $T : M_{nn} \rightarrow \mathbb{R}$ given by $T(\mathbf{A}) = a_{11}a_{22}a_{33} \cdots a_{nn}$ where a_{jj} are the entries along the leading diagonal of a n by n matrix.
8. Show that the transpose of any square matrix is a linear transformation (mapping).
9. Let $T : C[0, 1] \rightarrow \mathbb{R}$ be a transformation (mapping) given by $T(f) = \int_0^1 f(x)dx$ where f is a continuous function defined in the interval $[0, 1]$. Show that T is a linear transformation.
10. Let $T : V \rightarrow W$ be a transform such that $T(\mathbf{O}) \neq \mathbf{O}$. Prove that T is *not* a linear transform.
11. Let $T : V \rightarrow W$ be the zero transform, that is $T(\mathbf{v}) = \mathbf{O}$ for all vectors $\mathbf{v} \in V$. Show that T is a linear transformation (mapping).
12. Let $T : V \rightarrow W$ be a linear transformation of an n -dimensional vector space into a vector space W . Let $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ be a basis for V . Prove that if $\mathbf{u}$ is any vector in V then we can write $T(\mathbf{u})$ as a linear combination of

$$\{T(\mathbf{v}_1), T(\mathbf{v}_2), T(\mathbf{v}_3), \dots, T(\mathbf{v}_n)\}$$

SECTION 5.2 Kernel and Range of a Linear Transformation

By the end of this section you will be able to

- understand what is meant by the kernel and range of a linear transform
- derive some properties of the kernel and range

This is a challenging section which will use some of the results from earlier chapters. In particular we will need to apply some propositions from chapter 3 to prove some of the properties of the kernel and range. Try to do the proofs on your own first, and if you get stuck then have a look at the text. This is a good way of understanding a proof.

The next two sections have similarities with subsection 3.6.1, where we defined: null space, nullity and rank of a matrix. In this section we describe the kernel (null space), nullity and range of a linear transformation.



Why is the kernel and range important?

Generally the kernel and range tell you which information has been carried over in the transform and what has been lost.



How do we know this?

By one of the most important results in linear algebra which is:

Let $T : V \rightarrow W$ be a linear transform where V is of finite dimension ($\dim$). Then

$$\dim(\text{kernel}(T)) + \dim(\text{range}(T)) = \dim(V)$$

This result says that all the information is contained in these two sets – kernel and range. This is equivalent to the dimension theorem (3.34) that we mentioned in chapter 3.



How is the kernel and range connected to a linear system of equations?

A matrix, say A , can be used to represent a linear transformation, so finding the kernel means finding all the vectors $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{0}$. The linear system $A\mathbf{x} = \mathbf{b}$ has a solution if and only if vector $\mathbf{b}$ is in the range (image) of the linear transformation.

5.2.1 Kernel of a linear transformation (mapping)

The notation $A\mathbf{x} = \mathbf{b}$ means that the matrix A transforms (maps) the vector $\mathbf{x}$ to the vector $\mathbf{b}$. Matrix A is a function with input $\mathbf{x}$ and output $\mathbf{b}$ (Fig. 5.10).



Figure 5.10

Generally a function f in mathematics acts on numbers, but the matrix A acts on vectors.

Example 5.9

Find $\mathbf{v}$ such that $T(\mathbf{v}) = \mathbf{0}$ for the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T(\mathbf{v}) = A\mathbf{v} \text{ where } A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

Solution

Let $\mathbf{v} = (x \ y)^T$ and A be the given matrix, then

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \left[\text{Using } T(\mathbf{v}) = A\mathbf{v} = \mathbf{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right]$$

Opening out the matrix gives the simultaneous equations:

$$\begin{cases} x + y = 0 \\ -x + y = 0 \end{cases} \Rightarrow x = 0 \quad \text{and} \quad y = 0$$

Hence $\mathbf{v} = (x \ y)^T = (0 \ 0)^T = \mathbf{0}$.

The set of vectors in the starting vector space which are transformed to the zero vector is called the **kernel** (pronounced 'kur-nl') of the transformation and is denoted by $\ker(T)$.

For the above Example 5.9 we have $\ker(T) = \{\mathbf{O}\}$, that is the kernel of T is *only* the zero vector. There are *no* other vectors in $\mathbb{R}^2$ which get transformed to the zero vector under T . (This $\ker(T) = \{\mathbf{O}\}$ is the **null space** of the matrix $\mathbf{A}$.) Note that $T(\mathbf{v}) = \mathbf{O}$ is equivalent to solving $\mathbf{A}\mathbf{v} = \mathbf{O}$, because matrix $\mathbf{A}$ represents the transform T .

We will show later that if the kernel is just the zero vector, $\mathbf{O}$, then we can move things back, that is the linear transform has an inverse. It means that all the information was carried over. In general, there may exist other vectors besides the zero vector which are also transformed to the zero vector as Example 5.10 below demonstrates.

For a general linear transform, $T : V \rightarrow W$, the set of vectors in the domain V which arrive at the zero vector $\mathbf{O}$ in W is called the kernel of T and is illustrated in Fig. 5.11.

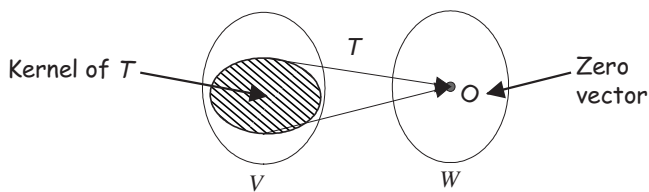


Figure 5.11

In everyday English language, kernel means softer part inside a shell. In linear algebra, kernel represents those parts which get transformed to the zero vector.

Definition (5.5). Let $T : V \rightarrow W$ be a linear transform (map). The set of *all* vectors $\mathbf{v}$ in V that are transformed to the zero vector in W is called the **kernel** of T , denoted by $\ker(T)$. It is the set of vectors $\mathbf{v}$ in V such that $T(\mathbf{v}) = \mathbf{O}$.

Example 5.10

Consider the zero linear transformation $T : V \rightarrow W$ such that $T(\mathbf{v}) = \mathbf{O}$ for all vectors $\mathbf{v}$ in V . Find $\ker(T)$.

Solution

All the vectors in V arrive at the zero vector under T , that is $T(\mathbf{v}) = \mathbf{O}$, therefore $\ker(T) = V$. This means *all* of the input set V is the kernel of T (Fig. 5.12).

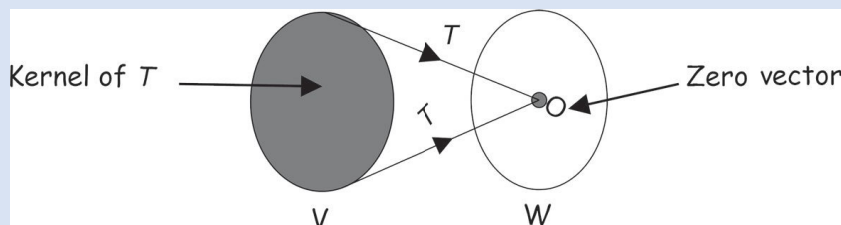


Figure 5.12

Example 5.11

- (i) Consider the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by $T \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} \right) = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$. Find $\ker(T)$.
- (ii) Show that $\mathbf{u} - \mathbf{v}$ is in $\ker(T)$ where $\mathbf{u} = (1 \ 2 \ 3)^T$ and $\mathbf{v} = (1 \ 2 \ 5)^T$.

Solution

(i) We need to find the vectors in $\mathbb{R}^3$ which arrive at the zero vector under T .
Which vectors are transformed (mapped) to the zero vector?

Inspecting the given transformation $T \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} \right) = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$, we arrive at the zero vector if $x = 0, y = 0$ and $z = r$, where r is any real number. Therefore

$$T \left(\begin{pmatrix} 0 \\ 0 \\ r \end{pmatrix} \right) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \mathbf{0}$$

Thus the kernel of T in this case is $\left\{ \begin{pmatrix} 0 \\ 0 \\ r \end{pmatrix} \mid r \text{ is a real number} \right\}$. The kernel of the given transformation T is the z axis in $\mathbb{R}^3$ as illustrated in Fig. 5.13.

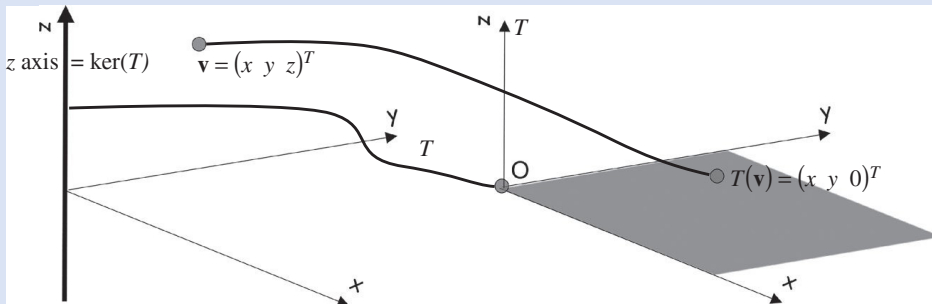


Figure 5.13

The given T transforms any point in $\mathbb{R}^3$ to the shaded plane shown in Fig. 5.13.

(ii) We have

$$\mathbf{u} - \mathbf{v} = (1 \ 2 \ 3)^T - (1 \ 2 \ 5)^T = (0 \ 0 \ -2)^T$$

Transforming this vector $\mathbf{u} - \mathbf{v}$:

$$T(\mathbf{u} - \mathbf{v}) = T((0 \ 0 \ -2)^T) = (0 \ 0 \ 0)^T = \mathbf{0}$$

Since $T(\mathbf{u} - \mathbf{v}) = \mathbf{0}$ the vector $\mathbf{u} - \mathbf{v}$ is in $\ker(T)$. Using T is linear on $T(\mathbf{u} - \mathbf{v}) = \mathbf{0}$ gives

$$T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v}) = \mathbf{0} \text{ implies } T(\mathbf{u}) = T(\mathbf{v})$$

(continued...)

$\mathbf{u} - \mathbf{v}$ is in $\ker(T)$ means that vectors $\mathbf{u}$ and $\mathbf{v}$ arrive at the same point under T (Fig. 5.14).

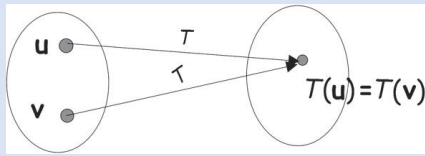


Figure 5.14

Hence the kernel detects when two points are transformed to the same point.

The linear transformation in Example 5.11 projects vectors in $\mathbb{R}^3$ onto a plane in $\mathbb{R}^3$. This type of linear transformation crops up in computer graphics. In computer graphics the concept of two vectors being transformed to the same point means that one object is ‘blocking’ the view of the other on the screen.

If the vector $\mathbf{u} - \mathbf{v}$ is in $\ker(T)$ for a linear transformation T , which projects vectors onto a screen, then $T(\mathbf{u})$ is blocking $T(\mathbf{v})$ or vice versa on the screen.

5.2.2 Properties of the kernel

Proposition (5.6). Let $T : V \rightarrow W$ be a linear transformation (mapping) between the vector spaces V and W . Then the kernel of T is a subspace of the vector space V .

? *What does this mean?*

The set of vectors $\mathbf{v}$ in V such that $T(\mathbf{v}) = \mathbf{0}$ is a subspace of V which can be illustrated as shown in Fig. 5.15.

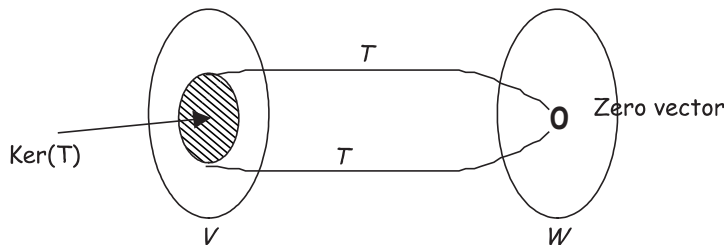


Figure 5.15

? *How do we prove this result?*

By using Proposition (3.7) of chapter 3 which states:

A non-empty subset S with vectors $\mathbf{u}$ and $\mathbf{v}$ is a subspace $\Leftrightarrow k\mathbf{u} + c\mathbf{v}$ is also in S .

In this case, the set S in question is $\ker(T)$.

Proof.

? *How do we know that $\ker(T)$ is non-empty?*

Because by Proposition (5.3) (a) of the last section we have $T(\mathbf{0}) = \mathbf{0}$ for a linear transform T . Hence the zero vector $\mathbf{0}$ is in $\ker(T)$ so it *cannot* be empty.



What else do we need to show?

(3.7) says that if $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\ker(T)$ then any linear combination $k\mathbf{u} + c\mathbf{v}$ (k and c are scalars) is also in $\ker(T)$.

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in $\ker(T)$, consider the vector $k\mathbf{u} + c\mathbf{v}$:

$$\begin{aligned} T(k\mathbf{u} + c\mathbf{v}) &= kT(\mathbf{u}) + cT(\mathbf{v}) && \left[\text{because } T \text{ is linear} \right] \\ &= k\mathbf{0} + c\mathbf{0} && \left[\begin{array}{l} T(\mathbf{u}) = T(\mathbf{v}) = \mathbf{0} \text{ because} \\ \mathbf{u} \text{ and } \mathbf{v} \text{ are in } \ker(T) \end{array} \right] \\ &= \mathbf{0} \end{aligned}$$

We have $T(k\mathbf{u} + c\mathbf{v}) = \mathbf{0}$, therefore $k\mathbf{u} + c\mathbf{v}$ is in $\ker(T)$.

Any linear combination of vectors $\mathbf{u}$ and $\mathbf{v}$ are in $\ker(T)$, therefore $\ker(T)$ is a subspace of V . ■

Since $\ker(T)$ is a subspace of the vector space V we have the following definition:

Definition (5.7). Let $T : V \rightarrow W$ be a linear transform. Then $\ker(T)$ is also called the **null space** of T and the dimension of $\ker(T)$ is called the **nullity** of T which is denoted by $\text{nullity}(T)$.

This definition is identical to the definition of null space of a matrix given in chapter 3.

Let $T(\mathbf{u}) = \mathbf{A}\mathbf{u}$ be a linear transformation then $\ker(T)$, or the null space of T , is the general solution $\mathbf{u}$ such that $\mathbf{A}\mathbf{u} = \mathbf{0}$. [Definition (3.31).]

This means that $\ker(T)$ is the set of input vectors $\mathbf{u}$ that arrive at the zero vector by applying matrix $\mathbf{A}$. Remember, this is the same as the null space of matrix $\mathbf{A}$. The null space definition given above is more generic because it applies to any linear transformation as you will see in later examples.

Example 5.12

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be given by $T(\mathbf{x}) = \mathbf{A}\mathbf{x}$ where $\mathbf{x}$ is in $\mathbb{R}^3$ and

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

Find (i) $\ker(T)$ (ii) null space of T (iii) $\text{nullity}(T)$

Solution

(i) $\ker(T)$ is the solution space of

$$T(\mathbf{x}) = \mathbf{A}\mathbf{x} = \mathbf{0}$$

Hence it is the null space of matrix $\mathbf{A}$. We found null spaces of matrices in section 3.6.

(continued...)

How?

By placing the matrix $\mathbf{A}$ into (reduced) row echelon form $\mathbf{R}$ and then solving $\mathbf{R}\mathbf{x} = \mathbf{0}$:

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \longrightarrow \mathbf{R} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}$$

Letting $\mathbf{x} = (x \ y \ z)^T$ then $\mathbf{R}\mathbf{x} = \mathbf{0}$ gives:

$$\begin{aligned} x - z &= 0 \\ y + 2z &= 0 \end{aligned}$$

From the first equation, we have $x = z$ and the bottom equation gives $y = -2z$. Let $z = r$, where r is any real number, then $x = r$, $y = -2r$ and $z = r$. This means that $\ker(T)$ is the solution:

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} r \\ -2r \\ r \end{pmatrix} = r \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

We have $\ker(T) = \{r\mathbf{u} \mid \mathbf{u} = (1 \ -2 \ 1)^T \text{ and } r \text{ is any real number}\}$.

(ii) The null space of T and matrix $\mathbf{A}$ is $\ker(T) = \{r\mathbf{u} \mid \mathbf{u} = (1 \ -2 \ 1)^T\}$.

(iii) What does $\text{nullity}(T)$ mean?

It is the dimension of the null space or $\ker(T)$. The dimension is the number of vectors in the basis of $\ker(T)$. There is only one vector in the basis of $\ker(T)$ which is $\mathbf{u} = (1 \ -2 \ 1)^T$, therefore $\text{nullity}(T) = 1$.

The challenge is finding $\ker(T)$, the kernel of the given linear transformation (mapping). Once we have $\ker(T)$ then the dimension of this subspace, given by $\text{nullity}(T)$ is normally straightforward, we only need to find the number of basis vectors for $\ker(T)$.

However, finding a basis can be more challenging.

Example 5.13

Let P_2 and P_3 be the vector space of polynomials of degrees less than or equal to 2 and 3 respectively. Consider the linear transformation $T : P_3 \rightarrow P_2$ given by

$$T(\mathbf{p}) = p'(x)$$

where $p(x)$ is a cubic polynomial and $p'(x)$ represents the first derivative of $p(x)$.

Determine $\text{nullity}(T)$.

Solution

How do we find $\text{nullity}(T)$?

First we find $\ker(T)$.

What is $\ker(T)$ equal to?

$\ker(T)$ is the set of polynomials that arrive at the zero vector under the given linear transform T . This transform T differentiates a given polynomial $p(x)$. We have

$$T(\mathbf{p}) = p'(x) = 0$$

Which polynomials gives zero after differentiation?

The constant polynomials, that is

$$T(c) = c' = 0 \text{ where } c \text{ is a constant}$$

Thus $\ker(T) = \{c \mid c \text{ is any real number}\}$.

What is the dimension of this?

A basis for $\ker(T)$ is $\{1\}$, therefore dimension of $\ker(T)$ is 1 because there is only *one* vector in the basis. We have

$$\text{nullity}(T) = 1$$

5.2.3 The range (image) of a linear transformation

We briefly defined what is meant by the range (image) of a function in section 5.1.



What is the range of a linear transformation $T : V \rightarrow W$?

Figure 5.16 illustrates the meaning of the range or image of a linear transform.

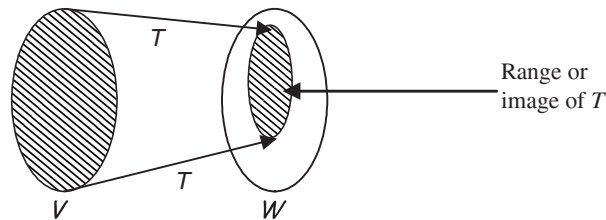


Figure 5.16

The range of a linear transformation T is the set of vectors we arrive at after applying T .

Definition (5.8). Let $T : V \rightarrow W$ be a linear transform. The **range** (image) of the linear transform is the set of *all* the output vectors $\mathbf{w}$ in W for which there are input vectors $\mathbf{v}$ in V such that

$$T(\mathbf{v}) = \mathbf{w}$$

The range is the set of output vectors in W that are images of the input vectors in V under the transform T . We can write the $\text{range}(T)$ of $T : V \rightarrow W$ in set theory notation as follows:

$$(5.9) \quad \text{range}(T) = \{T(\mathbf{v}) \mid \mathbf{v} \text{ in } V\}$$

Note that the $\text{range}(T)$ may *not* include *all* of the arrival vector space W as you can see in the above Fig. 5.16.

Example 5.14

Consider the zero linear transformation $T : V \rightarrow W$ such that $T(\mathbf{v}) = \mathbf{O}$ for all vectors $\mathbf{v}$ in V . Find $\text{range}(T)$ or in other words the image of T .

Solution

All vectors arrive at a single destination – the zero vector under T , therefore:

$$\text{range}(T) = \mathbf{O} \quad [\text{zero vector}]$$

Thus the range or image is a single element $\{\mathbf{O}\}$. This is illustrated in Fig. 5.17.

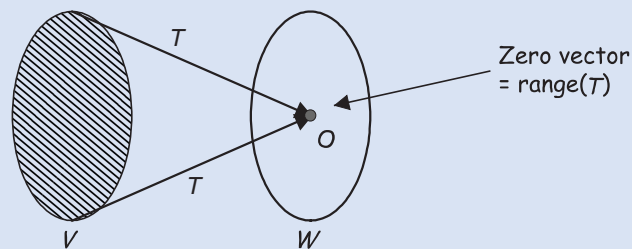


Figure 5.17

Note that the $\ker(T)$ is the whole of the start vector space V , $\ker(T) = V$, but range of T is just the zero vector, $\mathbf{O}$.

It is important to note the difference between kernel and range. The kernel of a transformation lies in the domain (start) and range lies in the codomain (arrival).

A common term for *range* is *image*. We will use both.

Example 5.15

Consider the linear transformation $T : P_2 \rightarrow \mathbb{R}$ such that $T(\mathbf{p}) = \int_0^1 p(x)dx$ where P_2 is the vector space of polynomials of degree 2 or less. Determine $\text{range}(T)$. [You can check that T is indeed a linear transformation.]

Solution

Let $p(x) = ax^2 + bx + c$ because $p(x)$ is a polynomial of degree 2 or less, then

$$\begin{aligned} T(p(x)) &= \int_0^1 (ax^2 + bx + c)dx \\ &\equiv \underset{\text{integrating}}{\left[\frac{ax^3}{3} + \frac{bx^2}{2} + cx \right]_0^1} = \frac{a}{3} + \frac{b}{2} + c \quad \left[\begin{array}{l} \text{substituting limits} \\ x = 1 \text{ and } x = 0 \end{array} \right] \end{aligned}$$

We have a, b and c are any real numbers, therefore $T(\mathbf{p}) = \frac{a}{3} + \frac{b}{2} + c$ can be any real number.
 What is $\text{range}(T)$ equal to?
 The set of all the real numbers, that is $\text{range}(T) = \mathbb{R}$. We have all of P_2 being transformed to the set of real numbers $\mathbb{R}$ (Fig. 5.18).

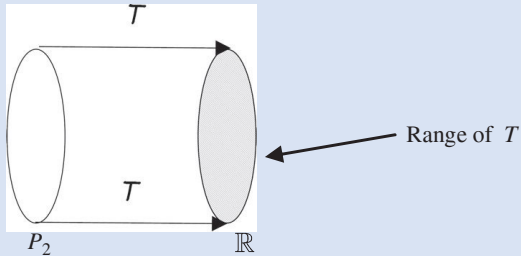


Figure 5.18

5.2.4 Properties of the range of a linear transformation

Note that the range of a linear transform $T : V \rightarrow W$ is composed of vectors in the arrival vector space W . (The vectors we arrive at in W after applying T to vectors in V .) Next we show that the range of T is a subspace of the arrival vector space W .

Proposition (5.10). Let V and W be vector spaces and $T : V \rightarrow W$ be a linear transformation. The range or image of the transformation T is a subspace of the arrival vector space W .



What does this mean?

The shaded part in W in Fig. 5.16 is a subspace of W .



How do we prove this proposition?

Again, by using Proposition (3.7) of chapter 3:

A non-empty subset S with vectors $\mathbf{u}$ and $\mathbf{v}$ is a subspace $\Leftrightarrow k\mathbf{u} + c\mathbf{v}$ is also in S .

Proof.

For a linear transform we have $T(\mathbf{O}) = \mathbf{O}$, therefore $\mathbf{O}$ is in the set of $\text{range}(T)$, which means the range is non-empty.



How do we prove that $\text{range}(T)$ is a subspace of the arrival space W ?

By showing any linear combination is also in the range. Let $\mathbf{u}$ and $\mathbf{w}$ be vectors in $\text{range}(T)$ then we need to show that

$$k\mathbf{u} + c\mathbf{w} \text{ is also in } \text{range}(T) \quad [k \text{ and } c \text{ are scalars}]$$

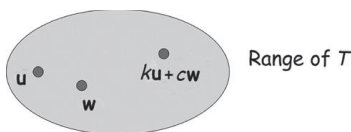


Figure 5.19

Required to prove that the set $\text{range}(T)$ is *closed*, which means that $k\mathbf{u} + c\mathbf{w}$ cannot escape from $\text{range}(T)$.

Since $\mathbf{u}$ and $\mathbf{w}$ are in $\text{range}(T)$, there must exist input vectors $\mathbf{x}$ in V and $\mathbf{y}$ in V such that

$$T(\mathbf{x}) = \mathbf{u} \text{ and } T(\mathbf{y}) = \mathbf{w}$$

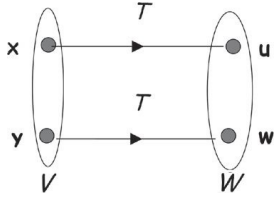


Figure 5.20

Consider the vector $k\mathbf{u} + c\mathbf{w}$:

$$\begin{aligned} k\mathbf{u} + c\mathbf{w} &= kT(\mathbf{x}) + cT(\mathbf{y}) \\ &= T(k\mathbf{x}) + T(c\mathbf{y}) && \text{[because } T \text{ is linear]} \\ &= T(k\mathbf{x} + c\mathbf{y}) && \text{[because } T \text{ is linear]} \end{aligned}$$

V is a vector space, therefore the linear combination $k\mathbf{x} + c\mathbf{y}$ is in V . We have

$$T(k\mathbf{x} + c\mathbf{y}) = k\mathbf{u} + c\mathbf{w}$$

therefore $k\mathbf{u} + c\mathbf{w}$ is in $\text{range}(T)$. Hence by (3.7), we conclude that $\text{range}(T)$ is a subspace of the vector space W . ■



Summary

Let $T : V \rightarrow W$ be a linear transformation. We have the following:

(5.5) The set of *all* the vectors $\mathbf{v}$ in V such that $T(\mathbf{v}) = \mathbf{0}$ is called the **kernel** of T .

(5.9) The **range** (or **image**) of T in W is defined by $\text{range}(T) = \{T(\mathbf{v}) \mid \mathbf{v} \text{ in } V\}$.



EXERCISES 5.2

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

1. Find the kernel of the following linear transformations:

(a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T(\mathbf{v}) = \mathbf{A}\mathbf{v}$ where $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

(b) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T(\mathbf{v}) = \mathbf{A}\mathbf{v}$ where $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$.

(c) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

(d) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by $T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 0 \\ x \end{pmatrix}$.

(e) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ given by $T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} y - z \\ x - z \end{pmatrix}$.

2. Let $T : M_{nn} \rightarrow M_{nn}$ be the linear transformation given by the transpose, that is

$$T(\mathbf{A}) = \mathbf{A}^T$$

Determine $\ker(T)$ and $\text{range}(T)$. [M_{nn} is an n by n square matrix].

3. Let $T : P_2 \rightarrow P_1$ be the linear transformation of differentiation

$$T(\mathbf{p}) = p'(x)$$

Determine $\ker(T)$ and $\text{range}(T)$.

4. Let $T : V \rightarrow W$ be a linear transformation. Show that if $\mathbf{u} \in \ker(T)$ and $\mathbf{v} \in \ker(T)$ then for any scalars k and c the vector $(k\mathbf{u} + c\mathbf{v}) \in \ker(T)$.
5. Let $T : V \rightarrow W$ be a linear transformation. Prove that if $\mathbf{v} \in \ker(T)$ then $-\mathbf{v} \in \ker(T)$.
6. Consider the zero linear transformation $T : V \rightarrow W$ defined by $T(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v}$ in the domain V . Prove that $\ker(T) = V$.
7. Consider the identity linear transformation $T : V \rightarrow W$ defined by $T(\mathbf{v}) = \mathbf{v}$. Prove the following results:
 - (a) $\ker(T) = \{\mathbf{0}\}$
 - (b) $\text{range}(T) = V$
8. Let $T : V \rightarrow W$ be a linear transformation and vectors $\mathbf{u}, \mathbf{v}$ be in V .
Prove that if $T(\mathbf{u}) = \mathbf{x}$ and $T(\mathbf{v}) = \mathbf{x}$ then $(\mathbf{u} - \mathbf{v}) \in \ker(T)$.
9. Let $T : V \rightarrow W$ be a linear transformation. Prove that if $S_1 = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ spans the domain V then $S_2 = \{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\}$ spans $\text{range}(T)$.
10. Let $T : V \rightarrow W$ be a linear transformation. Prove that if $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for the domain V then $S = \{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\}$ is a basis for $\text{range}(T)$.
11. Let $T : V \rightarrow W$ be a linear transformation. Let S be a subspace of V and $T(S) = \{T(s) \mid s \in S\}$. Prove that $T(S)$ is a subspace of $\text{range}(T)$.

SECTION 5.3 Rank and Nullity

By the end of this section you will be able to

- find the rank and nullity of a linear transformation

5.3.1 Rank of a linear transformation (mapping)

In chapter 3 we discussed what is meant by the rank of a matrix and why it is important. The rank of a matrix is the maximum number of linearly independent rows in the matrix, or the dimension of the row space of the matrix. Remember, in some matrices many of the rows vanish because they are linear combinations of the other rows (linearly dependent). The rank is important in transferring data because a transformation with a lower rank takes up less memory and time to be transferred. Low rank transformations are much more computationally efficient.

The rank of a linear transformation tells us how much information has been transformed over and is measured as the dimension of the range. The rank also tells us whether information has been lost by the linear transform.

Definition (5.11). Let $T : V \rightarrow W$ be a linear transform (map) and $\text{range}(T)$ be the range. Then the dimension of $\text{range}(T)$ is called the **rank** of T denoted $\text{rank}(T)$ (Fig. 5.21).

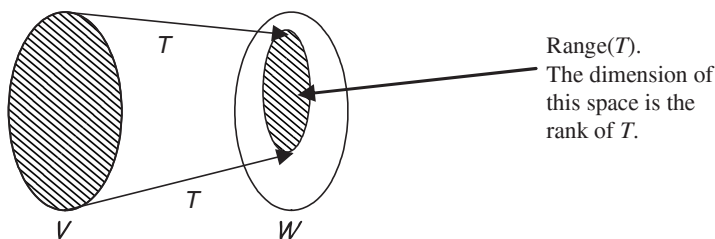


Figure 5.21

Next we state one of the most important results of linear algebra. The proof of this theorem is given towards the end of the section because it is long and requires you to recall some definitions given in previous chapters.

Dimension theorem (also called the rank-nullity theorem) (5.12). Let $T : V \rightarrow W$ be a linear transformation from an n -dimensional vector space V to a vector space W . Then

$$\text{rank}(T) + \text{nullity}(T) = n$$



What does this formula mean?

It means

$$\dim(\text{range}(T)) + \dim(\ker(T)) = n$$

where $\dim$ represents dimension. This is same as the dimension theorem (3.34) of chapter 3. This suggests that adding the dimension of the range and kernel gives the dimension of the start vector space V . As stated earlier, this result says that all the information is contained in these two sets – kernel and range.



Why is the rank of a linear transformation important?

The rank gives us how much information has been carried over by the transform. If the rank of the linear transform $T : V \rightarrow W$ is equal to the dimension of the start vector space V then *all* the information has been moved over and we can go back; that is, the linear transform has an inverse. We have:

1. If $\text{rank}(T) = \dim(V)$ then all the information has been carried over by T .
2. If $\text{rank}(T) < \dim(V)$ then some information has been lost by T .
3. If $\text{rank}(T) = 0 < \dim(V)$ then virtually all the information has been lost by T .

Example 5.16

Determine the rank and nullity of the linear transformation (mapping) $T : P_2 \rightarrow \mathbb{R}$:

$$T(\mathbf{p}) = \int_0^1 p(x) dx$$

where P_2 is the vector space of polynomials of degree 2 or less.

Solution

What do you notice about the given linear transform?

It's the same transform as in Example 5.15 of the last section. We have already established that $\text{range}(T) = \mathbb{R}$ in Example 5.15.

What is the rank of T ?

By the above definition (5.11) we have

$$\text{rank}(T) = \dim(\text{range}(T)) = \dim(\mathbb{R})$$

$\dim(\mathbb{R}) = 1$ because of the general rule $\dim(\mathbb{R}^m) = m$. Thus $\text{rank}(T) = 1$.

We also need to find the nullity of T which is the dimension of the kernel.

How?

By using the above formula (5.12):

$$\dim(\text{range}(T)) + \dim(\ker(T)) = n$$

What is the value of n , the dimension of P_2 equal to?

From chapter 3, Table 3.1 states that $\dim(P_k) = k + 1$ so $\dim(P_2) = 3$.

Substituting $n = \dim(P_2) = 3$ and $\dim(\text{range}(T)) = 1$ into

$$\dim(\text{range}(T)) + \dim(\ker(T)) = n$$

gives

$$1 + \dim(\ker(T)) = 3 \quad \text{which yields} \quad \dim(\ker(T)) = 2$$

(continued...)

Hence the nullity of T is 2 and the rank of T is 1.

Since $\text{rank}(T) = 1$ but dimension of the start vector space P_2 is 3, some of the information has been lost by the linear transform T .

We are given that $T : P_2 \rightarrow \mathbb{R}$ and we know that $\dim(P_2) = 3$. However, the dimension of the arrival vector space $\mathbb{R}$ is 1. Clearly the given transform T has lost some information by going from dimension 3 to 1.

Example 5.17

Find the rank and nullity of the linear transform, $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 2x - y \\ 6x - 3y \end{pmatrix}$$

Solution

Since T is linear we can find the rank and nullity of T .

How do we find these?

Apply the dimension theorem formula (5.12):

$$\dim(\text{range}(T)) + \dim(\ker(T)) = n$$

What is n equal to?

Remember, n is the dimension of the domain (start vector space) which is $\mathbb{R}^2$ because we are given $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$. Hence $n = \dim(\mathbb{R}^2) = 2$.

We need to find the dimension of the kernel of T .

What is $\ker(T)$ equal to?

Remember, $\ker(T)$ represents those vectors in $\mathbb{R}^2$ which arrive at the zero vector under the transformation T :

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 2x - y \\ 6x - 3y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

We have the simultaneous equations:

$$\begin{cases} 2x - y = 0 \\ 6x - 3y = 0 \end{cases} \text{ gives } y = 2x$$

Let $x = s$ where s is any real number, then $y = 2s$. Thus

$$\ker(T) = \left\{ s \begin{pmatrix} 1 \\ 2 \end{pmatrix} \mid s \text{ is a real number} \right\}$$

What is the dimension of $\ker(T)$?

A basis (vector representing axis) for $\ker(T)$ is $\{(1 \ 2)^T\}$, therefore $\dim(\ker(T)) = 1$ because we only have *one* vector in the basis. Hence nullity $(T) = 1$.

To find the rank which is the dimension of the range, we substitute $n = 2$ and $\dim(\ker(T)) = 1$ into the above formula (5.12):

$$\dim(\text{range}(T)) + \dim(\ker(T)) = n$$

to get $\dim(\text{range}(T)) + 1 = 2$. This gives $\dim(\text{range}(T)) = 1$. Therefore $\text{rank}(T) = 1$.

Hence $\text{rank}(T) = \text{nullity}(T) = 1$.

$\text{rank}(T) = 1$ means that the actual arrival space is of dimension 1. However, the dimension of the start vector space $\mathbb{R}^2$ is 2, so some information has been lost by T because T transforms vectors from dimension 2 to 1.

5.3.2 Kernel and rank of the linear transformation $T(\mathbf{x}) = \mathbf{Ax}$

Proposition (5.13). Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation given by $T(\mathbf{x}) = \mathbf{Ax}$. Then $\text{range}(T)$ is the column space of $\mathbf{A}$.

Proof.

See Exercises 5.3. ■

In the next example, we convert the given matrix into its (reduced) row echelon form. The method of conversion has been covered so many times in previous chapters that we will not detail the steps, but simply write down the final (reduced) row echelon form.

Example 5.18

Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ be given by $T(\mathbf{x}) = \mathbf{Ax}$ where $\mathbf{x}$ is in $\mathbb{R}^4$ and

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 4 & 5 \\ 2 & 6 & -8 & -6 \end{pmatrix}$$

Find a basis for **(i)** $\ker(T)$ **(ii)** $\text{range}(T)$

Solution

(i) The kernel is a subspace of $\mathbb{R}^4$ because $\mathbb{R}^4$ is the start vector space of T .

What is $\ker(T)$ equal to?

It is the null space $\mathbf{x}$ of matrix $\mathbf{A}$.

How do we find a basis for $\ker(T)$?

We need to place the above matrix $\mathbf{A}$ into (reduced) row echelon form $\mathbf{R}$ and then solve $\mathbf{Rx} = \mathbf{0}$. The general solution $\mathbf{x}$ gives a basis (axes) for the kernel. We can use MATLAB or hand calculations to obtain the reduced row echelon form:

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 4 & 5 \\ 2 & 6 & -8 & -6 \end{pmatrix} \quad \longrightarrow \quad \begin{pmatrix} 1 & 3 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \mathbf{R}$$

Let $\mathbf{x} = (x \ y \ z \ w)^T$. Expanding the rows of $\mathbf{Rx} = \mathbf{0}$ gives the simultaneous equations:

$$\begin{aligned} x + 3y + w &= 0 & (1) \\ z + w &= 0 & (2) \end{aligned}$$

(continued...)

From the bottom equation (2) we have $z = -w$. Let $w = s$ where s is any real number. We have $z = -w = -s$. Substituting $w = s$ into the top equation (1) gives

$$x + 3y + s = 0 \quad \text{implies} \quad x = -3y - s$$

Let $y = t$ where t is any real number. We have $x = -3y - s = -3t - s$. Hence $x = -3t - s$, $y = t$, $z = -s$ and $w = s$:

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} -3t - s \\ t \\ -s \\ s \end{pmatrix} = s \begin{pmatrix} -1 \\ 0 \\ -1 \\ 1 \end{pmatrix} + t \begin{pmatrix} -3 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

Therefore a set B of basis (axes) vectors for the kernel of T is given by

$$B = \{(-1 \ 0 \ -1 \ 1)^T, (-3 \ 1 \ 0 \ 0)^T\}$$

Hence $\text{nullity}(T) = 2$ because we have two basis (axes) vectors for $\ker(T)$.

(ii) What is $\text{range}(T)$ equal to?

By the above Proposition (5.13):

$\text{range}(T)$ is the column space of $\mathbf{A}$.

The column space of $\mathbf{A}$ gives $\text{range}(T)$. We convert the columns of matrix $\mathbf{A}$ into rows by taking the transpose of $\mathbf{A}$.

A set B' of basis (axes) vectors for the range of T can be found by determining the reduced row echelon form of $\mathbf{A}^T$ (matrix $\mathbf{A}$ transposed). The non-zero rows of this reduced row echelon form give a basis for the range of T . We have

$$\mathbf{A}^T = \begin{pmatrix} 1 & 2 \\ 3 & 6 \\ 4 & -8 \\ 5 & -6 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{array}{l} \longleftarrow \text{non-zero row} \\ \longleftarrow \text{non-zero row} \\ = \mathbf{R}' \end{array}$$

The first two rows of the reduced row echelon form matrix $\mathbf{R}'$ is a basis for the range of T :

$$B' = \{(1 \ 0)^T, (0 \ 1)^T\}$$

Since the range of T has two basis (axes) vectors so $\text{rank}(T) = 2$.

Remember, adding the rank and nullity of T gives the dimension of the start vector space:

$$\text{rank}(T) + \text{nullity}(T) = 2 + 2 = 4 = \dim(\mathbb{R}^4)$$

5.3.3 Proof of the dimension theorem

Now we prove the dimension theorem which was:

(5.12). Let $T : V \rightarrow W$ be a linear transformation from an n dimensional vector space V to a vector space W . Then $\text{rank}(T) + \text{nullity}(T) = n$.

Proof.

Let $\dim(\ker(T)) = k$ and $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_k\}$ be a set of basis (axes) vectors for $\ker(T)$. Since $\ker(T)$ is a subspace of the n -dimensional space V , therefore $k \leq n$. We consider the two cases, $k = n$ and $k < n$:

Case 1: $k = n$.



What does $k = n$ mean?

It means that *all* the vectors in V are transformed to the zero vector $\mathbf{O}$, that is

$$T(\mathbf{v}) = \mathbf{O} \text{ for all } \mathbf{v} \text{ in } V$$



What is the range of T ?

All the vectors arrive at the zero vector $\mathbf{O}$ so $\text{range}(T) = \{\mathbf{O}\}$.



What is dimension of $\text{range}(T) = \{\mathbf{O}\}$?

By section 3.4 of chapter 3 we have $\dim(\{\mathbf{O}\}) = 0$.

Hence we have $\dim(\ker(T)) = n$ and $\dim(\text{range}(T)) = 0$:

$$\dim(\text{range}(T)) + \dim(\ker(T)) = 0 + n = n$$

This is our required result.

Case 2: Consider $k < n$. Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{v}_{k+1}, \dots, \mathbf{v}_n\}$ be a set of basis (axes) vectors for the vector space V where $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_k\}$ is a basis (axes) for $\ker(T)$ as stated above. This is possible because we are given that the dimension of the vector space V is n and $\dim(\ker(T)) = k < n$.

Let $\mathbf{v}$ be an arbitrary vector in the start vector space V , then we can write this vector $\mathbf{v}$ as a linear combination of the basis (axes) vectors, that is

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k + \dots + c_n\mathbf{v}_n$$

where the c_j 's are scalars.

Thus the vector $T(\mathbf{v})$ is in $\text{range}(T)$ and we have

$$\begin{aligned} T(\mathbf{v}) &= T(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k + c_{k+1}\mathbf{v}_{k+1} + \dots + c_n\mathbf{v}_n) \\ &= T(\underbrace{c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k}_{\text{is in } \ker(T)}) + T(c_{k+1}\mathbf{v}_{k+1} + \dots + c_n\mathbf{v}_n) \quad [\text{because } T \text{ is linear}] \\ &= \mathbf{O} + T(c_{k+1}\mathbf{v}_{k+1} + \dots + c_n\mathbf{v}_n) \\ &= c_{k+1}T(\mathbf{v}_{k+1}) + \dots + c_nT(\mathbf{v}_n) \quad [\text{because } T \text{ is linear}] \end{aligned}$$

We have

$$T(\mathbf{v}) = c_{k+1}T(\mathbf{v}_{k+1}) + \dots + c_nT(\mathbf{v}_n) \quad (*)$$

where $\mathbf{v}$ is an arbitrary vector in the start vector space V , and $T(\mathbf{v})$ is in $\text{range}(T)$. Thus the vectors on the right hand side of (*), that is $S = \{T(\mathbf{v}_{k+1}), T(\mathbf{v}_{k+2}), \dots, T(\mathbf{v}_n)\}$ span (or generate) $\text{range}(T)$. We need to show that these vectors in the set S form a basis for $\text{range}(T)$.

? We have already established that these vectors span $\text{range}(T)$, but what do we need to show for these vectors in S to form a basis for $\text{range}(T)$?

Show that they are linearly independent.

? How?

Consider the linear combination

$$d_{k+1}T(\mathbf{v}_{k+1}) + d_{k+2}T(\mathbf{v}_{k+2}) + \cdots + d_nT(\mathbf{v}_n) = \mathbf{0}$$

where the d_j 's are scalars.

? How do we show linear independence?

We need to prove that *all* the scalars d_j 's are zero.

Since T is linear we have

$$d_{k+1}T(\mathbf{v}_{k+1}) + d_{k+2}T(\mathbf{v}_{k+2}) + \cdots + d_nT(\mathbf{v}_n) = \mathbf{0}$$

$$T(d_{k+1}\mathbf{v}_{k+1} + d_{k+2}\mathbf{v}_{k+2} + \cdots + d_n\mathbf{v}_n) = \mathbf{0}$$

This means that the vector $d_{k+1}\mathbf{v}_{k+1} + d_{k+2}\mathbf{v}_{k+2} + \cdots + d_n\mathbf{v}_n$ is in $\ker(T)$. Since $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_k\}$ is a basis for $\ker(T)$, we have

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 + \cdots + c_k\mathbf{v}_k = d_{k+1}\mathbf{v}_{k+1} + d_{k+2}\mathbf{v}_{k+2} + \cdots + d_n\mathbf{v}_n$$

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 + \cdots + c_k\mathbf{v}_k - d_{k+1}\mathbf{v}_{k+1} - d_{k+2}\mathbf{v}_{k+2} - \cdots - d_n\mathbf{v}_n = \mathbf{0}$$

Remember, at the start of the proof we had $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \dots, \mathbf{v}_n\}$ as a set of basis (axes) vectors for the vector space V , therefore they are linearly independent so we have

$$c_1 = c_2 = \cdots = c_k = d_{k+1} = d_{k+2} = \cdots = d_n = 0$$

All the d_j 's are zero, therefore $S = \{T(\mathbf{v}_{k+1}), T(\mathbf{v}_{k+2}), \dots, T(\mathbf{v}_n)\}$ is a linearly independent set of vectors. Thus $S = \{T(\mathbf{v}_{k+1}), T(\mathbf{v}_{k+2}), \dots, T(\mathbf{v}_n)\}$ forms a set of basis (axes) vectors for $\text{range}(T)$.

? How many vectors are there in this basis?

There are $n - k$ vectors in this basis. Hence the dimension of this space is given by $\dim(\text{range}(T)) = n - k$. We have our result

$$\dim(\text{range}(T)) + \dim(\ker(T)) = (n - k) + k = n$$

This completes our proof. ■

We can apply the dimension theorem to check whether a given transformation is linear or not.

Example 5.19

Consider the transform $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ given by $T \left[\begin{pmatrix} x \\ y \\ z \end{pmatrix} \right] = \begin{pmatrix} xz \\ xy \end{pmatrix}$. Test this transform for linearity.

Solution

The value of n in the dimension theorem (5.12) is dimension of the start vector space:

$$n = \dim(\mathbb{R}^3) = 3$$

What is the kernel equal to in this case?

It is the set of vectors in the start vector space which are transformed to the zero vector:

$$T \left[\begin{pmatrix} x \\ y \\ z \end{pmatrix} \right] = \begin{pmatrix} xz \\ xy \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow x = 0, y = r \text{ and } z = s \text{ where } r, s \text{ are any real numbers}$$

The entries of vector $\mathbf{x}$ such that $T(\mathbf{x}) = \mathbf{0}$ gives $x = 0, y = r$ and $z = s$:

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ r \\ s \end{pmatrix} = r \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

A set of basis (axes) vectors for the kernel of T is $\{(0 \ 1 \ 0)^T, (0 \ 0 \ 1)^T\}$. This means $\text{nullity}(T) = 2$.

The range of T is given by $\begin{pmatrix} xz \\ xy \end{pmatrix}$ which is $\mathbb{R}^2$. Hence $\dim(\mathbb{R}^2) = \text{rank}(T) = 2$.

We have

$$\text{rank}(T) + \text{nullity}(T) = 2 + 2 = 4 \neq 3 = n$$

Since the dimension theorem fails, the given transformation cannot be linear.

 Summary

(5.11) The dimension of $\text{range}(T)$ is called the rank of T .

(5.12) Dimension or rank-nullity theorem is

$$\text{rank}(T) + \text{nullity}(T) = n$$

where n is the dimension of the domain V .

 EXERCISES 5.3

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

1. Determine

(i) $\ker(T)$

(ii) $\text{nullity}(T)$

(iii) $\text{range}(T)$

(iv) $\text{rank}(T)$

for the following linear transformations without using the dimension theorem:

- (a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T(\mathbf{v}) = \mathbf{A}\mathbf{v}$ where $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$.
- (b) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by $T(\mathbf{v}) = \mathbf{A}\mathbf{v}$ where $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.
- (c) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by $T(\mathbf{v}) = \mathbf{A}\mathbf{v}$ where $\mathbf{A} = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 6 & 10 \\ 4 & 12 & 20 \end{pmatrix}$.
- (d) $T : P_3 \rightarrow P_3$ given by $T(\mathbf{p}) = x\mathbf{p}'$.
- (e) $T : P_3 \rightarrow P_2$ given by $T(\mathbf{p}) = \mathbf{p}'$.
- (f) $T : P_3 \rightarrow \mathbb{R}$ given by $T(\mathbf{p}) = \int_0^1 p(x) dx$.
- (g) $T : M_{22} \rightarrow P_1$ given by $T\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = (a+c)x + (b+d)$

2. For each of the examples in parts (a) to (g) in question 1 verify the dimension theorem (5.12) which says $\text{nullity}(T) + \text{rank}(T) = n$ where n is the dimension of the domain of the linear transformation T .
3. Let $T : \mathbb{R}^5 \rightarrow \mathbb{R}^3$ be given by $T(\mathbf{u}) = \mathbf{A}\mathbf{u}$ where $\mathbf{u}$ is in $\mathbb{R}^5$ and

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 4 & 2 & 1 \\ 2 & 6 & -7 & -2 & 5 \\ 4 & 12 & 6 & 4 & 6 \end{pmatrix}$$

Find a basis for (i) $\ker(T)$ (ii) $\text{range}(T)$

4. Prove Proposition (5.13).

SECTION 5.4 Inverse Linear Transformations

By the end of this section you will be able to

- prove properties of a one-to-one and onto transformations
- test which transforms have an inverse
- understand what is meant by isomorphism

In this section we state the conditions that allow an inverse transformation to exist. Inverse transformations are important because we often want to be able to undo a transformation. We need a way back from our destination to our starting point and this can only be achieved by the inverse transformation.

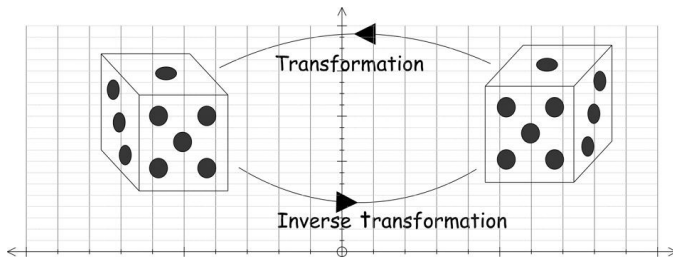


Figure 5.22

Figure 5.22 illustrates what is meant by an inverse transform.

Transformations are used in cryptography, where a message is encoded by a linear transform T . In order to decode this message we use the inverse transform of T .

Before we examine inverse transformations we need to discuss what is meant by **one-to-one** and **onto** because they are closely related to the inverse. Having established these definitions we prove some properties of these concepts.

This is a challenging section because you are required to prove a number of results. However, if you learn the definitions thoroughly and can apply these with confidence then the proofs should be straightforward.

5.4.1 One-to-one (injective) transformations



What do you think a one-to-one linear transformation is?

It's a linear transformation T in which every vector in the domain (start) arrives at a *different* vector in the range under T (Fig. 5.23).

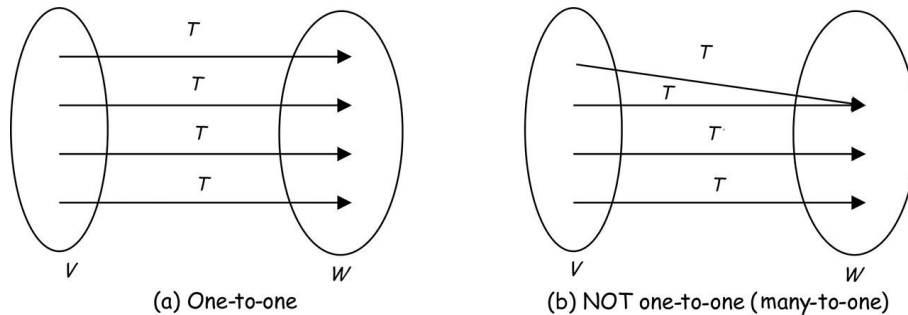


Figure 5.23

Fig. 5.23(a) shows a *one-to-one* transformation while Fig. 5.23(b) is *not* one-to-one (it is actually a many-to-one) transformation. Next, we give a formal definition of one-to-one.

Definition (5.14). Let $T : V \rightarrow W$ be a linear transform, $\mathbf{u}$ and $\mathbf{v}$ be in the domain V . The transform T is **one-to-one** $\Leftrightarrow$

$$\mathbf{u} \neq \mathbf{v} \quad \text{implies} \quad T(\mathbf{u}) \neq T(\mathbf{v})$$



What does this definition mean?

Different vectors such as $\mathbf{u}$ and $\mathbf{v}$ where $\mathbf{u} \neq \mathbf{v}$ have *distinct* images $T(\mathbf{u}) \neq T(\mathbf{v})$ under the linear transformation T . Two *different* start vectors arrive at *different* destinations.

In other literature on linear algebra, terms such as **1–1** or **injective** are used for one-to-one. We will stick with one-to-one.

Example 5.20

Consider the linear identity transformation $T : V \rightarrow W$ defined by $T(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v}$ in V . Show that this transformation T is one-to-one (injective).

Solution

Let $\mathbf{u}$ and $\mathbf{v}$ be different vectors in V , that is $\mathbf{u} \neq \mathbf{v}$. Applying the given identity transformation we have

$$T(\mathbf{u}) = \mathbf{u} \text{ and } T(\mathbf{v}) = \mathbf{v}$$

We have $\mathbf{u} \neq \mathbf{v}$, therefore $T(\mathbf{u}) = \mathbf{u} \neq \mathbf{v} = T(\mathbf{v})$. Hence $T(\mathbf{u}) \neq T(\mathbf{v})$, so by the above Definition (5.14):

The transform T is one-to-one $\Leftrightarrow \mathbf{u} \neq \mathbf{v}$ implies $T(\mathbf{u}) \neq T(\mathbf{v})$

The given transformation T is one-to-one because different vectors, $\mathbf{u} \neq \mathbf{v}$, arrive at different destinations, $T(\mathbf{u}) \neq T(\mathbf{v})$.

Example 5.21

Let $T : V \rightarrow W$ be the zero linear transformation defined by $T(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v}$ in V . Show that this transformation is *not* one-to-one (*not* injective).

Solution

Let $\mathbf{u}$ and $\mathbf{v}$ be different vectors in V , that is $\mathbf{u} \neq \mathbf{v}$. Then applying the given transformation:

$$T(\mathbf{u}) = \mathbf{0} \text{ and } T(\mathbf{v}) = \mathbf{0}$$

This gives $T(\mathbf{u}) = T(\mathbf{v}) = \mathbf{0}$. We have different vectors, $\mathbf{u} \neq \mathbf{v}$, arriving at the same destination, $T(\mathbf{u}) = T(\mathbf{v})$, therefore by (5.14) we conclude that the given zero linear transformation is *not* one-to-one.

Another test for one-to-one is:

T is one-to-one $\Leftrightarrow$

$$(5.15) \quad T(\mathbf{u}) = T(\mathbf{v}) \text{ implies } \mathbf{u} = \mathbf{v}$$

[If we have arrived at the same destination $T(\mathbf{u}) = T(\mathbf{v})$ then we have started with the same vectors $\mathbf{u} = \mathbf{v}$.]

Example 5.22

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation defined by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} x - y \\ x + y \end{pmatrix}$$

Show that this transformation is one-to-one (injective).

Solution

This time we use result (5.15) for the given T .

Let $\mathbf{u} = \begin{bmatrix} a \\ b \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} c \\ d \end{bmatrix}$. Applying the given transformation to vectors $\mathbf{u}$ and $\mathbf{v}$ we have

$$T(\mathbf{u}) = T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = \begin{pmatrix} a - b \\ a + b \end{pmatrix} \text{ and } T(\mathbf{v}) = T\left(\begin{bmatrix} c \\ d \end{bmatrix}\right) = \begin{pmatrix} c - d \\ c + d \end{pmatrix}$$

If $T(\mathbf{u}) = T(\mathbf{v})$ then we have $\begin{pmatrix} a - b \\ a + b \end{pmatrix} = \begin{pmatrix} c - d \\ c + d \end{pmatrix}$ and equating corresponding entries gives:

$$\left. \begin{array}{l} a - b = c - d \\ a + b = c + d \end{array} \right\} \text{ implies } a = c \text{ and } b = d$$

Thus the solution of the above simultaneous equations is $a = c$ and $b = d$ which gives

$$\mathbf{u} = \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} c \\ d \end{bmatrix} = \mathbf{v}$$

We have $T(\mathbf{u}) = T(\mathbf{v})$ implies $\mathbf{u} = \mathbf{v}$.

This means that the same arrival vectors $T(\mathbf{u}) = T(\mathbf{v})$ implies that we started with the same vectors $\mathbf{u} = \mathbf{v}$ so by the above result (5.15) the given transformation T is one-to-one.

5.4.2 Properties of one-to-one transformations

An easier check for one-to-one transformation is the following:

Proposition (5.16). Let $T : V \rightarrow W$ be a linear transformation between the vector spaces V and W . Then T is one-to-one $\Leftrightarrow \ker(T) = \{\mathbf{O}\}$.



Why do we need another test for one-to-one?

This is a much simpler test for one-to-one. The proposition means that $\ker(T) = \{\mathbf{O}\}$ is equivalent to T is one-to-one. The kernel of T measures one-to-one.



How do we prove this result?

By going both $(\Rightarrow)$ and $(\Leftarrow)$ ways. $(\Rightarrow)$ First assume T is one-to-one and derive $\ker(T) = \{\mathbf{O}\}$ and then $(\Leftarrow)$ assume $\ker(T) = \{\mathbf{O}\}$ and derive T is one-to-one.

Proof.

$(\Rightarrow)$ Assume T is one-to-one. Since T is linear, by result (5.3) part (a):

If T is linear then $T(\mathbf{O}) = \mathbf{O}$

We have $T(\mathbf{O}) = \mathbf{O}$. Suppose there is a vector $\mathbf{u}$ in V such that $T(\mathbf{u}) = \mathbf{O}$ then

$$T(\mathbf{u}) = \mathbf{O} = T(\mathbf{O}) \text{ implies } T(\mathbf{u}) = T(\mathbf{O})$$

Since we are assuming T is one-to-one, by

(5.15) $T(\mathbf{u}) = T(\mathbf{v})$ implies $\mathbf{u} = \mathbf{v}$
we have

$$T(\mathbf{u}) = T(\mathbf{O}) \text{ implies } \mathbf{u} = \mathbf{O}$$

Thus the *only* vector transformed under T to the zero vector is $\mathbf{O}$, which gives

$$\ker(T) = \{\mathbf{O}\}$$

($\Leftarrow$) Now going the other way, we assume $\ker(T) = \{\mathbf{O}\}$ and we need to prove T is one-to-one. Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in V which arrive at the same destination, $T(\mathbf{u}) = T(\mathbf{v})$. We have

$$\begin{aligned} T(\mathbf{u}) &= T(\mathbf{v}) \\ T(\mathbf{u}) - T(\mathbf{v}) &= \mathbf{O} \\ T(\mathbf{u} - \mathbf{v}) &= \mathbf{O} \quad \text{[because } T \text{ is Linear]} \end{aligned}$$

This means that $(\mathbf{u} - \mathbf{v})$ is in $\ker(T) = \{\mathbf{O}\}$. Thus

$$\mathbf{u} - \mathbf{v} = \mathbf{O} \Rightarrow \mathbf{u} = \mathbf{v}$$

By statement (5.15):

T is one-to-one $\Leftrightarrow T(\mathbf{u}) = T(\mathbf{v})$ implies $\mathbf{u} = \mathbf{v}$

We have $T(\mathbf{u}) = T(\mathbf{v})$ implies $\mathbf{u} = \mathbf{v}$, therefore T is a one-to-one transformation.

We have proved our result both ways, therefore T is one-to-one $\Leftrightarrow \ker(T) = \{\mathbf{O}\}$. ■

Example 5.23

Show that the linear transformation T given in Example 5.22 is one-to-one by using proposition (5.16).

Solution

What do we need to show?

Required to prove that the kernel of T of Example 5.22 is the zero vector.

How?

Applying the given transformation and equating to zero (because we are trying to find the kernel) yields

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x - y \\ x + y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ implies } \begin{cases} x - y = 0 \\ x + y = 0 \end{cases} \text{ gives } x = 0, y = 0$$

Thus the kernel of T is $(0 \ 0)^T = \mathbf{O}$. Therefore $\ker(T) = \{\mathbf{O}\}$ (is the zero vector) which means that T is one-to-one.

There are many examples where the kernel of a transformation T is *not* equal to the zero vector. For example, if we consider the derivative transformation S of polynomials then the kernel of S is the set of all constants which are *not* the zero vector.

Another test for one-to-one is the following:

Proposition (5.17). Let $T : V \rightarrow W$ be a linear transformation. T is one-to-one $\Leftrightarrow \text{nullity}(T) = 0$.

(Remember, the nullity of the transformation T is the dimension of the kernel of T .)

Proof – Exercises 5.4. ■

5.4.3 Onto (surjective) linear transformations

? What do you think the term *onto transformation* means?

An illustration of an onto transformation is shown in Fig. 5.24.

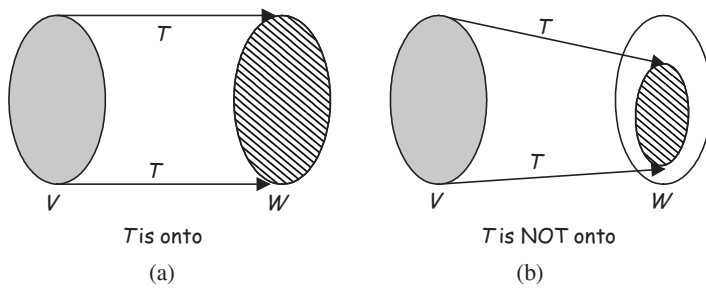


Figure 5.24

An *onto* transformation is when all the information carried over by T fills the whole arrival vector space W .

? How can we write this in mathematical terms?

Definition (5.18). Let $T : V \rightarrow W$ be a linear transform. The transform T is **onto** $\Leftrightarrow$ for every $\mathbf{w}$ in the arrival vector space W there exists at least one $\mathbf{v}$ in the start vector space V such that

$$\mathbf{w} = T(\mathbf{v})$$

In other words $T : V \rightarrow W$ is an *onto* transformation $\Leftrightarrow \text{range}(T) = W$. This means the arriving vectors of T fill *all* of W . We can write this as a proposition:

Proposition (5.19). A linear transformation $T : V \rightarrow W$ is *onto* $\Leftrightarrow \text{range}(T) = W$.

Proof – Exercises 5.4. ■

In other mathematical literature, or your lecture notes, you might find the term **surjective** to mean onto. We will use onto.

Example 5.24

Show that the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$T(\mathbf{v}) = \mathbf{A}\mathbf{v} \text{ where } \mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

is an onto (surjective) transformation.

Solution

How do we show that the given transformation is onto?

Show that the range of T is $\mathbb{R}^2$.

Let $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ then applying the given transformation to this vector yields:

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

What is the range of T ?

Since $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x \\ y \end{pmatrix}$ we have $\text{range}(T) = \mathbb{R}^2$, which means that the range of T fills the whole arrival vector space $\mathbb{R}^2$. Hence the given transformation T is onto.

Example 5.25

Show that the linear differentiation transformation $T : P_3 \rightarrow P_3$ given by

$$T(\mathbf{p}) = \mathbf{p}'$$

is not onto (not surjective).

Solution

Let $\mathbf{p} = ax^3 + bx^2 + cx + d$ because $\mathbf{p}$ is a member of P_3 which is the vector space of polynomials of degree 3 or less. Applying the given transformation to $\mathbf{p} = ax^3 + bx^2 + cx + d$ yields:

$$\begin{aligned} T(ax^3 + bx^2 + cx + d) &= (ax^3 + bx^2 + cx + d)' \\ &= 3ax^2 + 2bx + c \quad [\text{differentiating}] \end{aligned}$$

We have $T(\mathbf{p}) = 3ax^2 + 2bx + c$ and this is a quadratic (degree 2) polynomial (not a cubic, degree 3, polynomial) which means that it is a member of P_2 . (Remember, P_2 is the vector space of polynomials of degree 2 or less.) Thus the range of T is P_2 and because $P_2 \neq P_3$ (not equal) the given transformation is not onto (or not surjective) (Fig. 5.25).

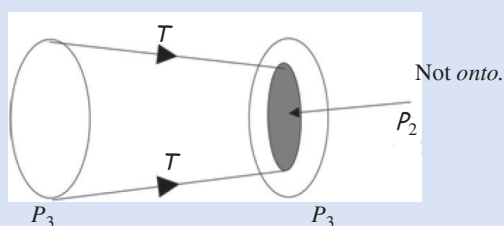


Figure 5.25

The given transform T does *not* fill the whole of the arrival vector space P_3 .

5.4.4 Properties of onto (surjective) linear transformations

Proposition (5.20). Let $T : V \rightarrow W$ be a linear transformation. Then T is onto $\Leftrightarrow$

$$\text{rank}(T) = \dim(W)$$

This proposition means that if the dimension of the range is equal to the dimension of the arrival vector space then the transformation T is onto. Also, if T is onto then the dimensions of the range and codomain are equal. (The range = codomain.)

Proof – Exercises 5.4. ■

Proposition (5.21). If $T : V \rightarrow W$ is a linear transformation and $\dim(V) = \dim(W)$ then T is a one-to-one transformation $\Leftrightarrow T$ is onto.

This means that if the start and arrival vector spaces are of equal dimension then a transformation which is *one-to-one* is also *onto* and an *onto* transformation is also *one-to-one*. We get both (one-to-one and onto) or neither provided the start and arrival sets have the same dimension (Fig. 5.26).

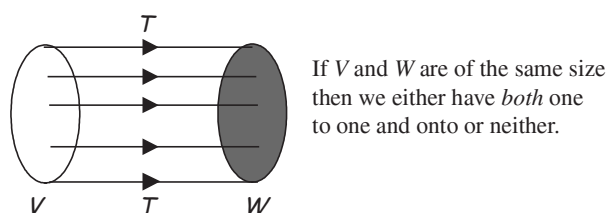


Figure 5.26



How do we prove the given result?

($\Rightarrow$) First we assume that T is one-to-one and deduce T is onto and then ($\Leftarrow$) we assume T is onto and deduce that T is one-to-one.

Proof.

Let the start and arrival vector spaces have the same dimension, $\dim(V) = \dim(W) = n$.

($\Rightarrow$) Assume T is one-to-one. By Proposition (5.17):

T is one-to-one $\Leftrightarrow \text{nullity}(T) = 0$

We have $\text{nullity}(T) = 0$. By the dimension theorem (5.12):

$\text{rank}(T) + \text{nullity}(T) = n$ (where n is the dimension of V)

We have $\text{rank}(T) = n$ because $\text{nullity}(T) = 0$. Since $\dim(W) = n$ so

$$\text{rank}(T) = \dim(W) = n$$

By the above Proposition (5.20):

$$T : V \rightarrow W \text{ is onto} \Leftrightarrow \text{rank}(T) = \dim(W)$$

We conclude that T is onto.

($\Leftarrow$) Conversely we assume that T is onto and prove it is one-to-one.



How?

We prove $\text{nullity}(T) = 0$ then by Proposition (5.17) we have one-to-one:

T is one-to-one $\Leftrightarrow \text{nullity}(T) = 0$

We are assuming T is onto, therefore $W = \text{range}(T)$. We have

$$\text{rank}(T) = \dim(W) = n \quad [\text{remember, at the start we let } \dim(W) = n]$$

We also have $\dim(V) = n$, therefore substituting this and $\text{rank}(T) = n$ into the dimension theorem

$$\text{rank}(T) + \text{nullity}(T) = \dim(V)$$

gives

$$n + \text{nullity}(T) = n \quad \Rightarrow \quad \text{nullity}(T) = 0$$

Since the $\text{nullity}(T) = 0$ so by (5.17) we conclude that T is one-to-one. ■

Proposition (5.22). If $T : V \rightarrow W$ is a linear transformation and $\dim(V) = \dim(W)$ then T is *both* one-to-one and onto $\Leftrightarrow \ker(T) = \{\mathbf{O}\}$.

Proof – Exercises 5.4. ■

Proposition (5.22) makes life easier.



Why?

It means that to prove a linear transformation $T : V \rightarrow W$ with $\dim(V) = \dim(W)$ is *both* one-to-one and onto, we only need to show $\ker(T) = \{\mathbf{O}\}$.



Why is this important?

If the kernel is the zero vector, $\mathbf{O}$, then you know that everything was carried over by the transform, and you can move things back (i.e. your linear transformation has an inverse).

Example 5.26

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation given by $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x - y \\ x + y \end{pmatrix}$.

Show that T is *both* onto and one-to-one.

Solution

What is the dimension of $\mathbb{R}^2$?

2. Both our start and arrival vector space have the same dimension:

$$\dim(\mathbb{R}^2) = \dim(\mathbb{R}^2) = 2$$

We can use the above Proposition (5.22) to show that kernel of T is the zero vector, $\ker(T) = \{\mathbf{O}\}$. We have

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x - y \\ x + y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Solving these simultaneous equations:

$$\left. \begin{matrix} x - y = 0 \\ x + y = 0 \end{matrix} \right\} \text{ gives } x = 0 \text{ and } y = 0 \text{ implies } \ker(T) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \mathbf{O}$$

Since $\ker(T) = \{\mathbf{O}\}$, by (5.22) the given transformation is both *one-to-one* and *onto*.

The above example shows a simple test for a transformation to be one-to-one and onto. Just check $\ker(T) = \{\mathbf{O}\}$ provided the dimension of start and arrival vector spaces are equal.

A transformation which is *both* one-to-one and onto is called a **bijection**, or we say the transform is **bijective**.

5.4.5 Inverse linear transformations

For inverse transformations to exist we need the given transformation to be **bijective** (one-to-one and onto). This is why the work on these topics preceded this subsection.



What does the term 'inverse linear transform' mean?

In everyday language, inverse means opposite or in reverse. An inverse linear transform undoes the linear transform. Figure 5.27 overleaf illustrates this.

It shows a linear transform $T : V \rightarrow W$ in which T takes the vector $\mathbf{u}$ to $T(\mathbf{u})$ and *inverse* T takes $T(\mathbf{u})$ back to $\mathbf{u}$.



How do we denote inverse T ?

T^{-1} denotes inverse T (recall T^{-1} does not equal $1/T$.)

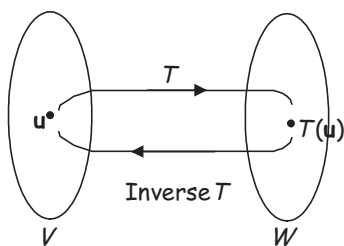


Figure 5.27

Next we give the formal definition of T^{-1} .

Definition (5.23). Let $T : V \rightarrow W$ be a *bijective* linear transform. The inverse transformation $T^{-1} : W \rightarrow V$ is defined as

$$\mathbf{v} = T^{-1}(\mathbf{w}) \Leftrightarrow T(\mathbf{v}) = \mathbf{w}$$

If T has an inverse transform, we refer to T as being **invertible**. To be invertible, T must be *both* one-to-one *and* onto (bijective) as the above proposition states. This means that if a given linear transform is bijective (one-to-one and onto) then it has an inverse transform.

Example 5.27

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation given by

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x - y \\ x + y \end{pmatrix}$$

Find the inverse transformation T^{-1} .

Solution

We can only find T^{-1} if T is *both* one-to-one and onto.

How do we show this?

We have already established that the given T is one-to-one and onto in Example 5.26, because we have the same transformation T .

How do we determine T^{-1} ?

Let our arrival points be $a = x - y$ and $b = x + y$, then express our starting points x and y in terms of a and b .

Why?

Because we have the situation shown in Fig. 5.28.

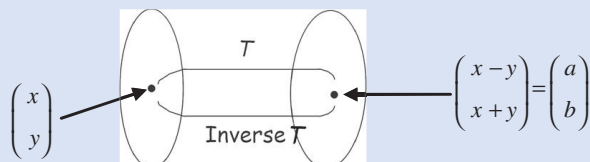


Figure 5.28

Adding these equations:

$$\begin{array}{r} x - y = a \\ + \quad x + y = b \\ \hline 2x + 0 = a + b \text{ which gives } x = \frac{a+b}{2} = \frac{1}{2}(a+b) \end{array}$$

Subtracting these equations:

$$\begin{array}{r} x - y = a \\ - \quad (x + y = b) \\ \hline 0 - 2y = a - b \text{ which gives } y = \frac{b-a}{2} = \frac{1}{2}(b-a) \end{array}$$

Thus the inverse transformation $T^{-1} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is given by

$$T^{-1} \left(\begin{pmatrix} a \\ b \end{pmatrix} \right) = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2}(a+b) \\ \frac{1}{2}(b-a) \end{pmatrix} = \frac{1}{2} \begin{pmatrix} a+b \\ b-a \end{pmatrix}$$

Hence we have $T^{-1} \left(\begin{pmatrix} a \\ b \end{pmatrix} \right) = \frac{1}{2} \begin{pmatrix} a+b \\ b-a \end{pmatrix}.$

Example 5.28

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation given by $T(\mathbf{x}) = \mathbf{Ax}$ where $\mathbf{A} = \begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix}.$

Find the inverse transformation $T^{-1}.$

Solution

Since the start and arrival vector spaces are of *same* dimension $\mathbb{R}^2$, there is a chance that the given transform is *invertible*.

If we can show that the kernel of T is the zero vector, $\mathbf{O}$, then by (5.22) we can say T is a bijection and so it is invertible.

How can we find the kernel of T ?

Let vector $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$ be in the start vector space, which arrives at the zero vector under T . We need to find $\mathbf{x}$ such that $T(\mathbf{x}) = \mathbf{Ax} = \mathbf{O}$. The augmented matrix is given by:

$$\left(\begin{array}{cc|c} 1 & 3 & 0 \\ 2 & 6 & 0 \end{array} \right) \quad \longrightarrow \quad \begin{array}{cc} x & y \\ \left(\begin{array}{cc|c} 1 & 3 & 0 \\ 0 & 0 & 0 \end{array} \right) \end{array}$$

We have $x + 3y = 0$ implies $x = -3y$. Let $y = r$, where r is any real number, then $x = -3r$:

$$\ker(T) = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -3r \\ r \end{pmatrix} = r \begin{pmatrix} -3 \\ 1 \end{pmatrix} \neq \mathbf{O}$$

Hence $\ker(T) \neq \mathbf{O}$ [not zero] so the given transform does *not* have an inverse.

Proposition (5.24). Let $T : V \rightarrow W$ be a linear transform which is *both* one-to-one and onto. Then the inverse transform $T^{-1} : W \rightarrow V$ is also linear.

Proof – Exercises 5.4. ■

5.4.6 Isomorphism

At the end of section 4.3.4 of chapter 4 we described isomorphic vector spaces, but next we give the formal definition:

Definition (5.25). If the linear transformation $T : V \rightarrow W$ is *invertible* then we say that the vector spaces V and W are **isomorphic**. Such a transformation T is called an **isomorphism**.

 *What does isomorphism say about the vector spaces?*

An isomorphism between vector spaces means these spaces are identical from a mathematical viewpoint, even though they are different spaces.

 *Well, what does this mean?*

It means that *isomorphic* vector spaces have an *identical* structure. They have similar properties with respect to the fundamental linear algebra operations of vector addition and scalar multiplication.

We can make an analogy with music. Consider a low C note on a piano or a guitar, as an example. Although the piano and guitar have different sounds, both the low C notes on the piano and on the guitar will be in tune and is a representation of the same vibration. Both instruments will produce a sound vibration with the same characteristics and properties. We can think of the piano and guitar as being isomorphic on a particular range of notes.

 *Why is isomorphism useful?*

If we didn't look for isomorphism then we would be reinventing the wheel each time by studying the same things over and over again for different vector spaces. An isomorphism between two vector spaces preserves the structure.

Example 5.29

Show that vector spaces P_n and $\mathbb{R}^{n+1}$ are isomorphic.

Solution

Remember, P_n is the vector space of polynomials of degree n or less and $\mathbb{R}^{n+1}$ is the $n + 1$ Euclidean space. We need to create a linear transformation between P_n and $\mathbb{R}^{n+1}$.

How?

We can write every polynomial $\mathbf{p}$ in P_n as

$$\mathbf{p} = p(x) = c_0 + c_1x + c_2x^2 + \cdots + c_nx^n$$

The set of standard basis (axes) vectors for P_n is $\{1, x, x^2, \dots, x^n\}$. We can describe this vector $\mathbf{p}$ by writing it in terms of the coordinates $(c_0, c_1, c_2, \dots, c_n)$. This coordinate can also correspond to our Euclidean space $\mathbb{R}^{n+1}$. Hence, we write the transformation as $T : P_n \rightarrow \mathbb{R}^{n+1}$ given by

$$T(c_0 + c_1x + c_2x^2 + \dots + c_nx^n) = \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{pmatrix}$$

Verify that this transformation is indeed linear.

In order to show P_n and $\mathbb{R}^{n+1}$ are isomorphic we need to prove T is invertible.

How?

Since the dimensions of the two vector spaces are equal, $\dim(P_n) = \dim(\mathbb{R}^{n+1}) = n + 1$, we only need to prove that kernel of T is the zero vector, that is $\ker(T) = \{\mathbf{0}\}$.

Remember, the kernel of T are those vectors in P_n which arrive at the zero vector under T . The zero vector in the arrival vector space $\mathbb{R}^{n+1}$ is given by

$$(c_0 \ c_1 \ \dots \ c_n)^T = (0 \ 0 \ \dots \ 0)^T = \mathbf{0}$$

Hence all the coefficient c 's are equal to zero:

$$T(c_0 + c_1x + c_2x^2 + \dots + c_nx^n) = \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \Rightarrow \begin{matrix} c_0 = 0 \\ c_1 = 0 \\ \vdots \\ c_n = 0 \end{matrix}$$

This gives $\mathbf{p} = 0 + 0x + 0x^2 + \dots + 0x^n = \mathbf{0}$. Hence $\ker(T) = \{\mathbf{0}\}$, so T is one-to-one and onto which means it is invertible.

Since T is invertible, the vector spaces connecting T are isomorphic. Hence P_n and $\mathbb{R}^{n+1}$ are isomorphic.

Lemma (5.26). Let V and W be finite-dimensional real vector spaces of equal dimension and $T : V \rightarrow W$ be an isomorphism. If $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis (axes) for V then

$$\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\}$$

is a basis (axes) for W .

Proof – Exercises 5.4. ■

Next we prove a powerful result.

Theorem (5.27). Let V and W be finite-dimensional real vector spaces. Then V is isomorphic to $W \Leftrightarrow$

$$\dim(V) = \dim(W)$$



Why is this result important?

Because it says that any two vector spaces of the same dimension are identical in structure. Given two vector spaces V and W then are they essentially the same. We can think of V and W as identical if they have the same structure and only the nature of their elements differs.

Proof.

If V and W are of dimension 0 then clearly the result holds. Assume the dimensions of V and W are not zero.

($\Leftarrow$). Let $T : V \rightarrow W$ be a transformation. Required to prove V and W are isomorphic which means that we have to show that T is invertible.



How do we show that T is invertible?

Prove that $\ker(T) = \{\mathbf{O}\}$.

Since both the vector spaces are of same dimension, n say, they have the same number of vectors in a basis (axes). Let $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ be a basis for V . By the above Lemma (5.27) we have $\{T(\mathbf{v}_1), T(\mathbf{v}_2), T(\mathbf{v}_3), \dots, T(\mathbf{v}_n)\}$ is a basis (axes) for W .

Let $T : V \rightarrow W$ be the transformation:

$$T(k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \dots + k_n\mathbf{v}_n) = k_1T(\mathbf{v}_1) + k_2T(\mathbf{v}_2) + \dots + k_nT(\mathbf{v}_n)$$

where k 's are scalars.

Verify that this transformation is linear.

Let $\mathbf{u}$ be a vector in vector space V which is transformed to the zero vector under T , that is $T(\mathbf{u}) = \mathbf{O}$. Required to prove $\mathbf{u} = \mathbf{O}$ because we need $\ker(T) = \{\mathbf{O}\}$.

We can write the vector $\mathbf{u}$ as a linear combination of the basis (axes) vectors of V :

$$\mathbf{u} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \dots + k_n\mathbf{v}_n \quad (*)$$

We have

$$\begin{aligned} T(\mathbf{u}) &= T(k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \dots + k_n\mathbf{v}_n) \\ &= k_1T(\mathbf{v}_1) + k_2T(\mathbf{v}_2) + \dots + k_nT(\mathbf{v}_n) \\ &= \mathbf{O} \quad [\text{because } T(\mathbf{u}) = \mathbf{O}] \end{aligned}$$

Since $\{T(\mathbf{v}_1), T(\mathbf{v}_2), T(\mathbf{v}_3), \dots, T(\mathbf{v}_n)\}$ is a basis for W , these vectors are linearly independent, which means that all the scalars are equal to zero:

$$k_1 = k_2 = k_3 = \dots = k_n = 0$$

Substituting these scalars into (*) gives

$$\mathbf{u} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \dots + k_n\mathbf{v}_n = \mathbf{O}$$

Hence $\ker(T) = \{\mathbf{O}\}$, so the transformation is invertible, which implies that start V and arrival W vector spaces are isomorphic.

($\Rightarrow$). Exercises 5.4 question 19(c).



By the above Theorem (5.27) we have:

Proposition (5.28). Every n -dimensional real vector space is isomorphic to $\mathbb{R}^n$.

This means that all the non-Euclidean vector spaces we have looked at in previous chapters, such as the set of polynomials P_n , matrices M_{mn} and continuous functions $C[a, b]$, are identical in structure to $\mathbb{R}^n$, provided they have the same dimension. This is the most useful isomorphism in linear algebra, because any n -dimensional vector space can be recast in terms of the familiar Euclidean space $\mathbb{R}^n$.

For example, the vector space M_{22} is isomorphic to $\mathbb{R}^4$. However, M_{23} is *not* isomorphic to $\mathbb{R}^5$ because M_{23} is of dimension 6 but $\mathbb{R}^5$ is of dimension 5.

Summary

Let $T : V \rightarrow W$.

The transform T is one-to-one $\Leftrightarrow \ker(T) = \{\mathbf{0}\}$.

The transform T is onto $\Leftrightarrow \text{range}(T) = W$.

(5.23) If T is bijective then the inverse transformation $T^{-1} : W \rightarrow V$ is defined as

$$\mathbf{v} = T^{-1}(\mathbf{w}) \Leftrightarrow T(\mathbf{v}) = \mathbf{w}$$

Vector spaces V and W are isomorphic if there is an invertible linear transformation between V and W .



EXERCISES 5.4

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

1. Show that the following linear transformations are one-to-one by using Propositions (5.14) or (5.15):

(a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T(\mathbf{v}) = \mathbf{I}\mathbf{v}$ where $\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

(b) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} y \\ x \end{pmatrix}$.

(c) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x+y \\ x-y \end{pmatrix}$.

2. Let M_{mn} be the vector space of m by n matrices. Consider the linear transformation $T : M_{mn} \rightarrow M_{nm}$ defined by $T(\mathbf{A}) = \mathbf{A}^T$. Show that T is one-to-one. (Remember, $\mathbf{A}^T$ is the transpose of the matrix $\mathbf{A}$).

3. Determine whether the following linear transformations are one-to-one and/or onto:

(a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x \\ x \end{pmatrix}$.

(b) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by $T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} 0 \\ y \\ z \end{pmatrix}$.

(c) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by $T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} z \\ y \\ x \end{pmatrix}$.

(d) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ given by $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 2x + 3y \\ x + y \\ 0 \end{pmatrix}$.

4. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation defined by

$$T(\mathbf{u}) = \mathbf{A}\mathbf{u} \text{ where } \mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Show that T is one-to-one and onto.

5. Show that the linear transformation $T : P_3 \rightarrow P_2$ given by $T(\mathbf{p}) = \mathbf{p}'$ where $\mathbf{p}'$ is the derivative of $\mathbf{p}$, is *not* one-to-one but is onto.
6. Show that the linear transformation $T : P_n \rightarrow P_n$ given by $T(\mathbf{p}) = \mathbf{p}''$ where $\mathbf{p}''$ is the second derivative of $\mathbf{p}$ is *not* one-to-one nor onto.

7. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the linear transformation defined by $T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} y \\ z \end{pmatrix}$.

Show that T is *not* one-to-one but is onto.

8. Show that the linear transformation $T : P_2 \rightarrow P_3$ given by

$$T(ax^2 + bx + c) = ax + (b + c)$$

is neither one-to-one nor onto.

9. Show that the linear transformation $T : P_n \rightarrow P_n$ given by

$$T(\mathbf{p}) = \mathbf{p}'$$

is neither one-to-one nor onto.

10. Prove Proposition (5.17).

11. Prove the following:

If $T : V \rightarrow W$ is a linear one-to-one transformation then for every vector $\mathbf{w}$ in $\text{range}(T)$ there exists a *unique* vector $\mathbf{v}$ in V such that $T(\mathbf{v}) = \mathbf{w}$.

12. Prove Proposition (5.19).

13. Prove Proposition (5.20).
 14. Prove Proposition (5.22).
 15. Prove Proposition (5.24).
 16. Prove Lemma (5.26).
 17. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation given by

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 2x + y \\ x - y \end{pmatrix}$$

Find the inverse linear transformation.

18. Show that $T : M_{22} \rightarrow \mathbb{R}^4$, given by

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a \ b \ c \ d)^T$$

is an isomorphism.

19. Prove the following properties of isomorphism.
- A linear transform $T : V \rightarrow W$ is an isomorphism $\Leftrightarrow \ker(T) = \{\mathbf{0}\}$.
 - If $T : V \rightarrow W$ is an isomorphism then $T^{-1} : W \rightarrow V$ is also an isomorphism.
 - If vector spaces V and W are isomorphic then $\dim(V) = \dim(W)$.

SECTION 5.5 The Matrix of a Linear Transformation

By the end of this section you will be able to

- find the standard matrix of a linear transformation
- determine the transformation matrix for non-standard bases
- apply the transformation matrix

Two of the most powerful tools in mathematics are linear algebra and calculus. In this section we establish a link between them by representing a derivative in terms of a matrix.

We start by showing that any linear transformation can be written in matrix form. This simply requires us to write our linear transform equations in the form $\mathbf{A}\mathbf{u}$, where the variables $x, y, z, \dots$ are stored in the vector $\mathbf{u}$, leaving behind their coefficients in the matrix $\mathbf{A}$. Writing a linear transformation in this way has untold benefits. With the numerical components of the transformation held in isolation, we can use some conventional arithmetic to manipulate them.

For example, if the matrix of coefficients performs a rotation and a number of rotations need to be performed in succession to create an animation, then you can just multiply by the matrix as many times as required. Each application of the matrix performs another rotation.

Using the transformation equations or using the matrix alternative are equivalent. You won't lose information if you use the matrix, and in many cases, using the matrix is simply much more efficient.

In the next section we cover composition of transformations. We find that if transformations are written as matrices, then this radically reduces the work involved in the composition of linear transformations.

5.5.1 The standard matrix for a linear transformation (mapping)

We can describe a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ by a matrix as the following example demonstrates.

Example 5.30

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear operator (start and arrival spaces are identical) given by

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 2x - y \\ x + 3y \end{pmatrix}$$

Write this transformation as $T(\mathbf{u}) = \mathbf{A}\mathbf{u}$ where $\mathbf{u} = (x \ y)^T$.

Solution

Writing $\begin{pmatrix} 2x - y \\ x + 3y \end{pmatrix}$ in the form $\mathbf{A}\mathbf{u}$ where $\mathbf{u} = \begin{pmatrix} x \\ y \end{pmatrix}$ is given by

$$\begin{pmatrix} 2x - y \\ x + 3y \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{A}\mathbf{u} \text{ where } \mathbf{A} = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$$

Thus $T(\mathbf{u}) = \mathbf{A}\mathbf{u}$. Since $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ so the matrix $\mathbf{A}$ is of size 2 by 2.

In this section we write a general linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ as $T(\mathbf{u}) = \mathbf{A}\mathbf{u}$. We say that $\mathbf{A}$ is the *matrix representation* of the linear transformation T . The transformation T acts on the vector $\mathbf{u}$. Similarly the matrix $\mathbf{A}$ acts on the vector $\mathbf{u}$ to give $\mathbf{A}\mathbf{u}$.



Is it always going to be possible to write any linear transformation in matrix representation?

Yes, provided we have finite-dimensional vector spaces.



How are we going to show this?

Consider a general linear transformation $T : V \rightarrow W$. We examine the basis (axes) vectors in V and look at the images of these under the given transformation T . To find $T(\mathbf{v})$ for any $\mathbf{v}$ in V we only need to know how T behaves towards these basis (axes) vectors because $\mathbf{v}$ can be written as a linear combination of the basis (axes) vectors.

First we consider a linear transformation with respect to the standard basis for $\mathbb{R}^n$; $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \dots, \mathbf{e}_n\}$ which represents our normal $x, y, z, \dots$ axes in $\mathbb{R}^n$. Remember, $\mathbf{e}_1$ is the unit vector in the x direction, $\mathbf{e}_2$ is the unit vector in the y direction. . .

Proposition (5.29). Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation and $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \dots, \mathbf{e}_n\}$ be the standard basis for $\mathbb{R}^n$. If

$$T(\mathbf{e}_1) = \begin{pmatrix} c_{11} \\ c_{21} \\ \vdots \\ c_{m1} \end{pmatrix}, \quad T(\mathbf{e}_2) = \begin{pmatrix} c_{12} \\ c_{22} \\ \vdots \\ c_{m2} \end{pmatrix}, \dots \text{ and } T(\mathbf{e}_n) = \begin{pmatrix} c_{1n} \\ c_{2n} \\ \vdots \\ c_{mn} \end{pmatrix}$$

then we can write the transformation $T(\mathbf{u})$ as

$$T(\mathbf{u}) = \mathbf{A}\mathbf{u}$$

$$\text{where } \mathbf{A} = \begin{pmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \dots & T(\mathbf{e}_n) \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \vdots & \vdots & \dots & \vdots \\ c_{m1} & c_{m2} & \dots & c_{mn} \end{pmatrix} \text{ and } \mathbf{u} \text{ is in } \mathbb{R}^n.$$

$\mathbf{A}$ is called the **standard matrix** and is of size m by n .

Note that the first column of matrix $\mathbf{A}$ is the column vector $T(\mathbf{e}_1)$, the second column of matrix $\mathbf{A}$ is the column vector $T(\mathbf{e}_2)$, . . . and the last column is the column vector $T(\mathbf{e}_n)$.

Proof.

Let $\mathbf{u}$ be an arbitrary vector in $\mathbb{R}^n$ and $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ be the standard basis (axes) vectors of the n -dimensional vector space $\mathbb{R}^n$. By the definition of standard basis we can write

$$\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} = u_1 \underbrace{\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{=\mathbf{e}_1} + u_2 \underbrace{\begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}}_{=\mathbf{e}_2} + \dots + u_n \underbrace{\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}}_{=\mathbf{e}_n} = u_1\mathbf{e}_1 + u_2\mathbf{e}_2 + \dots + u_n\mathbf{e}_n$$

where u_j 's are scalars and are the coordinates of $\mathbf{u}$ with respect to the standard basis (axes) $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \dots, \mathbf{e}_n\}$. Applying the linear transformation T we have

$$\begin{aligned} T(\mathbf{u}) &= T(u_1\mathbf{e}_1 + u_2\mathbf{e}_2 + \dots + u_n\mathbf{e}_n) && [\text{because } \mathbf{u} = u_1\mathbf{e}_1 + u_2\mathbf{e}_2 + \dots + u_n\mathbf{e}_n] \\ &= u_1T(\mathbf{e}_1) + u_2T(\mathbf{e}_2) + \dots + u_nT(\mathbf{e}_n) && [\text{because } T \text{ is linear}] \\ &\equiv u_1 \underbrace{\begin{pmatrix} c_{11} \\ c_{21} \\ \vdots \\ c_{m1} \end{pmatrix}}_{=T(\mathbf{e}_1)} + u_2 \underbrace{\begin{pmatrix} c_{12} \\ c_{22} \\ \vdots \\ c_{m2} \end{pmatrix}}_{=T(\mathbf{e}_2)} + \dots + u_n \underbrace{\begin{pmatrix} c_{1n} \\ c_{2n} \\ \vdots \\ c_{mn} \end{pmatrix}}_{=T(\mathbf{e}_n)} && (1) \end{aligned}$$

Given in
Proposition

Let the matrix $\mathbf{A} = (T(\mathbf{e}_1) \ T(\mathbf{e}_2) \ \cdots \ T(\mathbf{e}_n)) = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{pmatrix}.$

Applying the matrix $\mathbf{A}$ to the vector $\mathbf{u}$ gives $\mathbf{A}\mathbf{u}$, which is the linear combination of the column vectors of $\mathbf{A}$:

$$\mathbf{A}\mathbf{u} = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} = u_1 \begin{pmatrix} c_{11} \\ c_{21} \\ \vdots \\ c_{m1} \end{pmatrix} + u_2 \begin{pmatrix} c_{12} \\ c_{22} \\ \vdots \\ c_{m2} \end{pmatrix} + \cdots + u_n \begin{pmatrix} c_{1n} \\ c_{2n} \\ \vdots \\ c_{mn} \end{pmatrix} \quad (2)$$

The right hand sides of (1) and (2) are identical, so we have our result $T(\mathbf{u}) = \mathbf{A}\mathbf{u}$. ■

Example 5.31

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear operator given by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} 2x - y \\ x + 3y \end{pmatrix}$$

Determine the standard matrix $\mathbf{A}$ which represents T .

Solution

What is the standard basis for $\mathbb{R}^2$?

$\mathbf{e}_1 = (1 \ 0)^T$ is the unit vector in the x direction and $\mathbf{e}_2 = (0 \ 1)^T$ is the unit vector in the y direction.

Applying the given linear operator to these basis (axes) vectors $\{\mathbf{e}_1, \mathbf{e}_2\}$ yields:

$$T(\mathbf{e}_1) = T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{pmatrix} 2(1) - 0 \\ 1 + 3(0) \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \left[\text{using } T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} 2x - y \\ x + 3y \end{pmatrix} \right]$$

$$T(\mathbf{e}_2) = T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{pmatrix} 2(0) - 1 \\ 0 + 3(1) \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \end{pmatrix} \quad \left[\text{using } T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} 2x - y \\ x + 3y \end{pmatrix} \right]$$

Note, $T(\mathbf{e}_1)$ gives the x coefficients and $T(\mathbf{e}_2)$ gives the y coefficients, because $\mathbf{e}_1$ and $\mathbf{e}_2$ are the unit vectors in these directions respectively.

What is the standard matrix $\mathbf{A}$ equal to in this case?

By the above Proposition (5.29) we have $\mathbf{A} = (T(\mathbf{e}_1) \ T(\mathbf{e}_2)) = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}.$

Note that we have the same matrix as in Example 5.30, which represents the given transformation T .

We have $T(\mathbf{u}) = \mathbf{A}\mathbf{u}$ where $\mathbf{u} = (x \ y)^T$.

Example 5.32

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the linear transformation given by

$$T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} 3x + y + z \\ x - 3y - z \end{pmatrix}$$

Determine the standard matrix $\mathbf{A}$ which represents T .

Solution

What is the standard basis for $\mathbb{R}^3$?

$$\mathbf{e}_1 = [1 \ 0 \ 0]^T, \mathbf{e}_2 = [0 \ 1 \ 0]^T \text{ and } \mathbf{e}_3 = [0 \ 0 \ 1]^T$$

where $\mathbf{e}_1$, $\mathbf{e}_2$ and $\mathbf{e}_3$ are unit vectors in the x , y and z directions respectively.

We first examine how T acts on these basis (axes) vectors. Applying the given linear transformation to the basis (axis) vector $\mathbf{e}_1$ yields:

$$T(\mathbf{e}_1) = T \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right) = \begin{pmatrix} 3(1) + 0 + 0 \\ 1 - 3(0) - 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad \left[\text{because } T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} 3x + y + z \\ x - 3y - z \end{pmatrix} \right]$$

Similarly, applying T to $\mathbf{e}_2$ and $\mathbf{e}_3$ gives $T(\mathbf{e}_2) = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$ and $T(\mathbf{e}_3) = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ respectively.

What is the standard matrix $\mathbf{A}$ equal to?

By (5.29) which says $\mathbf{A} = (T(\mathbf{e}_1) \ T(\mathbf{e}_2) \ \cdots \ T(\mathbf{e}_n))$, we write these vectors as the first, second and last columns of matrix $\mathbf{A}$:

$$\mathbf{A} = (T(\mathbf{e}_1) \ T(\mathbf{e}_2) \ T(\mathbf{e}_3)) = \begin{pmatrix} 3 & 1 & 1 \\ 1 & -3 & -1 \end{pmatrix}$$

We can check this result by evaluating $\mathbf{A}\mathbf{u}$, where $\mathbf{u} = (x \ y \ z)^T$:

$$\mathbf{A}\mathbf{u} = \begin{pmatrix} 3 & 1 & 1 \\ 1 & -3 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3x + y + z \\ x - 3y - z \end{pmatrix}$$

This is the given transformation, therefore matrix $\mathbf{A}$ represents transformation T .

Normally the standard matrix can be found by inspection.

**How?**

From Example 5.32, we have the transformation and standard matrix as

$$T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} 3x + y + z \\ x - 3y - z \end{pmatrix} \text{ and standard matrix is } \mathbf{A} = \begin{pmatrix} 3 & 1 & 1 \\ 1 & -3 & -1 \end{pmatrix}$$

$\mathbf{A}$ is the standard matrix because we use the standard basis to write $\mathbf{A}$.

In the above example $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ so the matrix $\mathbf{A}$ is of size 2 by 3.

5.5.2 Transformation matrix for non-standard bases



Why use non-standard bases?

Examining a vector in a different basis (axes) may bring out structure related to that basis, which is hidden in the standard representation. It may be a relevant and useful structure. For example, we used to measure the motion of the planets in a basis (axes) with the earth at the centre. Then we discovered that putting the sun at the centre made life simpler – orbits were measured against a basis with the sun at the focus.

For some motions, such as projectiles, our standard basis (xy axes) may be the most suitable, but for studying other kinds of motions, such as orbits, a polar basis (r, θ) may work better.

If we use latitudes and longitudes to work out a map then we have been effectively using spherical polar coordinates (r, θ, φ) rather than our standard xyz axes.

Another example is trying to find the forces on an aeroplane as shown in Fig. 5.29. The components parallel and perpendicular to the aeroplane are a lot more useful than the horizontal and vertical components.

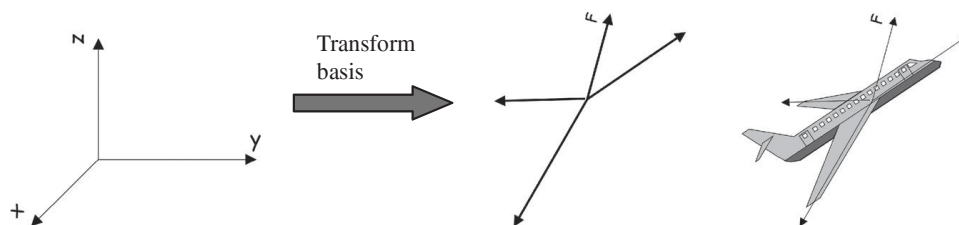


Figure 5.29

In computer games and 3D design software we often want to rotate our xyz axes (basis) to obtain new axes (basis) which are a lot more useful. (See question 7 of Exercises 5.5.)

In crystal structures, we need to use a basis which gives a cleaner set of coordinates called Miller indices. The Miller indices are coordinates used to specify direction and planes in a crystal or lattice. A vector from the origin to the lattice point is normally written in appropriate basis (axes) vectors and then the coordinates are given by the Miller indices.

Many problems in physics can be simplified due to their symmetrical properties if the right basis (axes) is chosen. Choosing a basis (axes) wisely can greatly reduce the amount of arithmetic you have to do.

For the remaining part of this section we will need to use the notation $[\mathbf{u}]_B$, which means the coordinates of the vector $\mathbf{u}$ with respect to a basis (axes) B .

Example 5.33

Write the vector $\mathbf{u} = \begin{pmatrix} 3 \\ 9 \end{pmatrix}$ of $\mathbb{R}^2$ in terms of new basis (axes) vectors $B = \left\{ \mathbf{b}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \mathbf{b}_2 = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \right\}$.

Solution

This means find $[\mathbf{u}]_B$. Let k and c be scalars, then we write the linear combination

$$k \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 9 \end{pmatrix}$$

Solving the simultaneous equations

$$\begin{cases} k + c = 3 \\ k + 4c = 9 \end{cases} \quad \text{gives } k = 1, c = 2$$

We have $\mathbf{u} = \begin{pmatrix} 3 \\ 9 \end{pmatrix} = 1\mathbf{b}_1 + 2\mathbf{b}_2 = 1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ 4 \end{pmatrix}$ or coordinates with respect to basis B is

$$\begin{bmatrix} 3 \\ 9 \end{bmatrix}_B = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

The coordinates of the vector $\begin{pmatrix} 3 \\ 9 \end{pmatrix}$ is $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ with respect to the basis $B = \left\{ \mathbf{b}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \mathbf{b}_2 = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \right\}$ (Fig. 5.30).

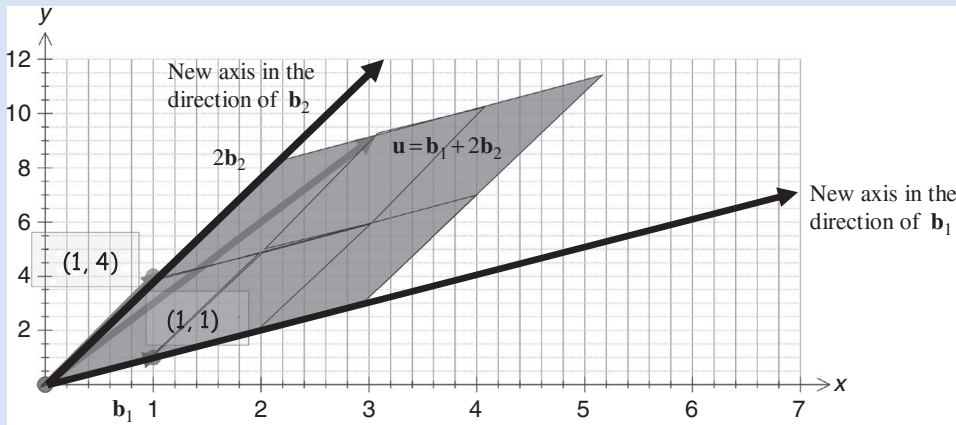


Figure 5.30

Note that $\mathbf{u} = (3 \ 9)^T$ coordinates are calculated with respect to a standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ or our normal xy axes, but the coordinates with respect to the new basis $B = \{\mathbf{b}_1, \mathbf{b}_2\}$ is $(1 \ 2)^T$.

We assume that if we have the vector $k_1 \mathbf{u}_1 + k_2 \mathbf{u}_2 + \cdots + k_n \mathbf{u}_n$ where k 's are scalars, then

$$(5.30) \quad [k_1 \mathbf{u}_1 + k_2 \mathbf{u}_2 + \cdots + k_n \mathbf{u}_n]_B = k_1 [\mathbf{u}_1]_B + k_2 [\mathbf{u}_2]_B + \cdots + k_n [\mathbf{u}_n]_B$$

Proposition (5.31). Let $T: V \rightarrow W$ be a linear transformation and V and W be finite-dimensional vector spaces with basis $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ and $C = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ respectively. Then we can write the transformation as

$$[T(\mathbf{u})]_C = \mathbf{A} [\mathbf{u}]_B$$

where $[T(\mathbf{u})]_C$ are the coordinates of $T(\mathbf{u})$ with respect to the basis (axes) C , $[\mathbf{u}]_B$ are the coordinates of $\mathbf{u}$ with respect to the basis (axes) B and $\mathbf{A}$ is a matrix given by

$$\mathbf{A} = ([T(\mathbf{v}_1)]_C \ [T(\mathbf{v}_2)]_C \ \cdots \ [T(\mathbf{v}_n)]_C)$$



What does this notation mean?

Figure 5.31 illustrates this:

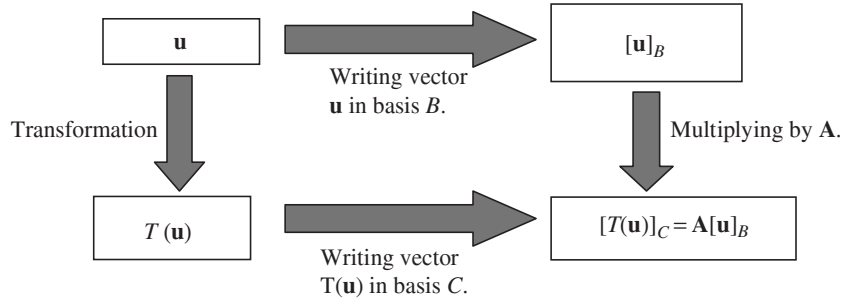


Figure 5.31

Proof.

Let $\mathbf{u}$ be an arbitrary vector in the n -dimensional vector space V with basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. By the definition of basis we can write $\mathbf{u}$ as a linear combination of basis (axes) vectors:

$$\mathbf{u} = k_1 \mathbf{v}_1 + k_2 \mathbf{v}_2 + \cdots + k_n \mathbf{v}_n$$

where k 's are scalars.

The coordinates of vector $\mathbf{u}$ with respect to basis (axes) $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is

$$[\mathbf{u}]_B = \begin{pmatrix} k_1 \\ \vdots \\ k_n \end{pmatrix}.$$

Applying the rules of linear transformation we have

$$T(\mathbf{u}) = T(k_1 \mathbf{v}_1 + k_2 \mathbf{v}_2 + \cdots + k_n \mathbf{v}_n) \quad [\text{because } \mathbf{u} = k_1 \mathbf{v}_1 + k_2 \mathbf{v}_2 + \cdots + k_n \mathbf{v}_n]$$

$$\stackrel{\text{because } T \text{ is linear}}{=} k_1 T(\mathbf{v}_1) + k_2 T(\mathbf{v}_2) + \cdots + k_n T(\mathbf{v}_n) \quad (*)$$

Note that $T(\mathbf{v}_1)$, $T(\mathbf{v}_2)$, $\dots$, and $T(\mathbf{v}_n)$ are all in the arrival vector space W . Let $C = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ be a set of basis (axes) vectors for W then we can write

$$\begin{aligned} T(\mathbf{v}_1) &= c_{11} \mathbf{w}_1 + c_{21} \mathbf{w}_2 + \cdots + c_{m1} \mathbf{w}_m \\ T(\mathbf{v}_2) &= c_{12} \mathbf{w}_1 + c_{22} \mathbf{w}_2 + \cdots + c_{m2} \mathbf{w}_m \\ &\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ T(\mathbf{v}_n) &= c_{1n} \mathbf{w}_1 + c_{2n} \mathbf{w}_2 + \cdots + c_{mn} \mathbf{w}_m \end{aligned}$$

where the c 's are scalars. We can write these as coordinates with respect to the basis (axes) C :

$$[T(\mathbf{v}_1)]_C = \begin{pmatrix} c_{11} \\ c_{21} \\ \vdots \\ c_{m1} \end{pmatrix}, [T(\mathbf{v}_2)]_C = \begin{pmatrix} c_{12} \\ c_{22} \\ \vdots \\ c_{m2} \end{pmatrix}, \dots \text{ and } [T(\mathbf{v}_n)]_C = \begin{pmatrix} c_{1n} \\ c_{2n} \\ \vdots \\ c_{mn} \end{pmatrix} \quad (\dagger)$$

Let $\mathbf{A} = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \vdots & \vdots & & \vdots \\ c_{m1} & c_{m2} & & c_{mn} \end{pmatrix}$ and from above $[\mathbf{u}]_B = \begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_n \end{pmatrix}$ then $\mathbf{A}[\mathbf{u}]_B$ is given by

$$\mathbf{A}[\mathbf{u}]_B = \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \vdots & \vdots & & \vdots \\ c_{m1} & c_{m2} & & c_{mn} \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_n \end{pmatrix} \quad (**)$$

Examining $[T(\mathbf{u})]_C$ we have

$$\begin{aligned} [T(\mathbf{u})]_C &= [k_1 T(\mathbf{v}_1) + k_2 T(\mathbf{v}_2) + \dots + k_n T(\mathbf{v}_n)]_C \quad [\text{from } (*)] \\ &\stackrel{\text{by (5.30)}}{=} k_1 [T(\mathbf{v}_1)]_C + k_2 [T(\mathbf{v}_2)]_C + \dots + k_n [T(\mathbf{v}_n)]_C \\ &= k_1 \begin{pmatrix} c_{11} \\ c_{21} \\ \vdots \\ c_{m1} \end{pmatrix} + k_2 \begin{pmatrix} c_{12} \\ c_{22} \\ \vdots \\ c_{m2} \end{pmatrix} + \dots + k_n \begin{pmatrix} c_{1n} \\ c_{2n} \\ \vdots \\ c_{mn} \end{pmatrix} \quad [\text{by } (\dagger)] \\ &= \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \vdots & \vdots & & \vdots \\ c_{m1} & c_{m2} & & c_{mn} \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_n \end{pmatrix} \stackrel{\text{by } (**)}{=} \mathbf{A}[\mathbf{u}]_B \end{aligned}$$

Hence we have our result $[T(\mathbf{u})]_C = \mathbf{A}[\mathbf{u}]_B$. ■

Example 5.34

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the linear transformation defined by

$$T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} 3x + y + z \\ x - 3y - z \end{pmatrix}$$

(continued...)

Find the transformation matrix $\mathbf{A}$ with respect to the bases for $\mathbb{R}^3$ and $\mathbb{R}^2$ given by:

$$B = \left\{ \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \text{ and } C = \left\{ \mathbf{w}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathbf{w}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\} \text{ respectively}$$

Note that B is a set of basis (axes) vectors for the start vector space $\mathbb{R}^3$ while C is a set of basis (axes) vectors for the arrival vector space $\mathbb{R}^2$.

Solution

Applying the given linear transformation T to each of the basis (axes) vectors $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$:

$$T(\mathbf{v}_1) = T\left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 3(1) + 1 + 1 \\ 1 - 3(1) - 1 \end{pmatrix} = \begin{pmatrix} 5 \\ -3 \end{pmatrix} \quad \left[\text{because } T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} 3x + y + z \\ x - 3y - z \end{pmatrix} \right]$$

Similarly (verify this) we have

$$T(\mathbf{v}_2) = \begin{pmatrix} -2 \\ -2 \end{pmatrix} \text{ and } T(\mathbf{v}_3) = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

What else do we need to find?

We need to write each of these arriving vectors, $T(\mathbf{v}_1)$, $T(\mathbf{v}_2)$ and $T(\mathbf{v}_3)$, as the coordinates of the basis (axes) $C = \left\{ \mathbf{w}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathbf{w}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$:

$$T(\mathbf{v}_1) = \begin{pmatrix} 5 \\ -3 \end{pmatrix} = a\mathbf{w}_1 + b\mathbf{w}_2 = a\begin{pmatrix} 1 \\ 2 \end{pmatrix} + b\begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad (1)$$

$$T(\mathbf{v}_2) = \begin{pmatrix} -2 \\ -2 \end{pmatrix} = c\mathbf{w}_1 + d\mathbf{w}_2 = c\begin{pmatrix} 1 \\ 2 \end{pmatrix} + d\begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad (2)$$

$$T(\mathbf{v}_3) = \begin{pmatrix} 1 \\ -1 \end{pmatrix} = e\mathbf{w}_1 + f\mathbf{w}_2 = e\begin{pmatrix} 1 \\ 2 \end{pmatrix} + f\begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad (3)$$

How can we find the matrix $\mathbf{A}$?

By the above Proposition (5.31) we have $\mathbf{A} = ([T(\mathbf{v}_1)]_C \ [T(\mathbf{v}_2)]_C \ [T(\mathbf{v}_3)]_C)$ which in this case is

$$\mathbf{A} = \begin{pmatrix} a & c & e \\ b & d & f \end{pmatrix}.$$

Why?

Because by (1), (2) and (3) we have

$$[T(\mathbf{v}_1)]_C = \begin{pmatrix} a \\ b \end{pmatrix}, [T(\mathbf{v}_2)]_C = \begin{pmatrix} c \\ d \end{pmatrix} \text{ and } [T(\mathbf{v}_3)]_C = \begin{pmatrix} e \\ f \end{pmatrix}$$

Remember, the scalars in the linear combination of $\mathbf{w}_1$ and $\mathbf{w}_2$ give the coordinates of vectors $T(\mathbf{v}_1)$, $T(\mathbf{v}_2)$ and $T(\mathbf{v}_3)$ with respect to basis (axes) C .

How can we determine the unknowns $a, b, c, \dots, f$?

We need to solve the three pairs of simultaneous equations (1), (2) and (3) in the above. Consider the first pair (1):

$$\begin{pmatrix} 5 \\ -3 \end{pmatrix} = a \begin{pmatrix} 1 \\ 2 \end{pmatrix} + b \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} a \\ 2a \end{pmatrix} + \begin{pmatrix} -b \\ b \end{pmatrix} = \begin{pmatrix} a - b \\ 2a + b \end{pmatrix}$$

What is the solution to this pair of simultaneous equations?

$$\begin{cases} a - b = 5 \\ 2a + b = -3 \end{cases} \text{ gives } a = \frac{2}{3} \text{ and } b = -\frac{13}{3}$$

Similarly, we can find the solutions of the other two simultaneous equations. They are

$$(2) \quad \begin{cases} c - d = -2 \\ 2c + d = -2 \end{cases} \text{ gives } c = -\frac{4}{3} \text{ and } d = \frac{2}{3}$$

$$(3) \quad \begin{cases} e - f = 1 \\ 2e + f = -1 \end{cases} \text{ gives } e = 0 \text{ and } f = -1$$

What is the matrix $\mathbf{A}$ equal to?

$$\mathbf{A} = \begin{pmatrix} a & c & e \\ b & d & f \end{pmatrix} = \begin{pmatrix} 2/3 & -4/3 & 0 \\ -13/3 & 2/3 & -1 \end{pmatrix}$$

This means that $[T(\mathbf{u})]_C = \mathbf{A} [\mathbf{u}]_B$ which is

$$[T(\mathbf{u})]_C = \begin{pmatrix} 2/3 & -4/3 & 0 \\ -13/3 & 2/3 & -1 \end{pmatrix} [\mathbf{u}]_B$$

To write the vector $T(\mathbf{u})$ in terms of basis (axes) C , denoted $[T(\mathbf{u})]_C$, we multiply the vector $\mathbf{u}$ written in basis (axes) B , denoted $[\mathbf{u}]_B$, by the matrix $\mathbf{A}$.

Example 5.35

For the linear transformation given in the above Example 5.34, use the matrix $\mathbf{A}$ to find $T(\mathbf{u})$ where $\mathbf{u} = (1 \ 0 \ 2)^T$ with respect to the same bases

$$B = \left\{ \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \text{ and } C = \left\{ \mathbf{w}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathbf{w}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$$

Solution

We need to find $[T(\mathbf{u})]_C = \mathbf{A} [\mathbf{u}]_B$. First we determine $[\mathbf{u}]_B$.

What does $[\mathbf{u}]_B$ represent?

Remember, the entries in the given vector $\mathbf{u} = (1 \ 0 \ 2)^T$ are the coordinates with respect to the standard basis (xyz axes). However, we need to write $\mathbf{u}$ with respect to the given basis B because $[\mathbf{u}]_B$ is the coordinates of the vector $\mathbf{u}$ with respect to the basis (axes) vectors B .

(continued...)

$$\mathbf{u} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3 = a \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + b \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + c \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (*)$$

These a , b and c are *not* the same values as for Example 5.34.

How can we find these?

Solve the simultaneous equations by expanding (*):

$$\left. \begin{array}{rcl} a - b & = & 1 \\ a & = & 0 \\ a + b + c & = & 2 \end{array} \right\} \text{ gives } a = 0, b = -1 \text{ and } c = 3$$

What are the coordinates of $\mathbf{u}$ with respect to the basis (axes) B , $[\mathbf{u}]_B$, equal to?

$$[\mathbf{u}]_B = \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix}$$

Note that $\mathbf{u} = (1 \ 0 \ 2)^T$ gives the coordinates of $\mathbf{u}$ with respect to our standard basis (xyz axes) but $[\mathbf{u}]_B = (0 \ -1 \ 3)^T$ gives the coordinates of $\mathbf{u}$ with respect to the given basis $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

From the above Proposition (5.31): $[T(\mathbf{u})]_C = \mathbf{A} [\mathbf{u}]_B$

We found matrix $\mathbf{A}$ in the previous example, so using this $\mathbf{A}$ we have

$$[T(\mathbf{u})]_C = \mathbf{A} [\mathbf{u}]_B = \begin{pmatrix} 2/3 & -4/3 & 0 \\ -13/3 & 2/3 & -1 \end{pmatrix} [\mathbf{u}]_B$$

and evaluating this right hand side gives

$$[T(\mathbf{u})]_C = \begin{pmatrix} 2/3 & -4/3 & 0 \\ -13/3 & 2/3 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 4/3 \\ -11/3 \end{pmatrix}$$

We can illustrate the above process as shown in Fig. 5.32.

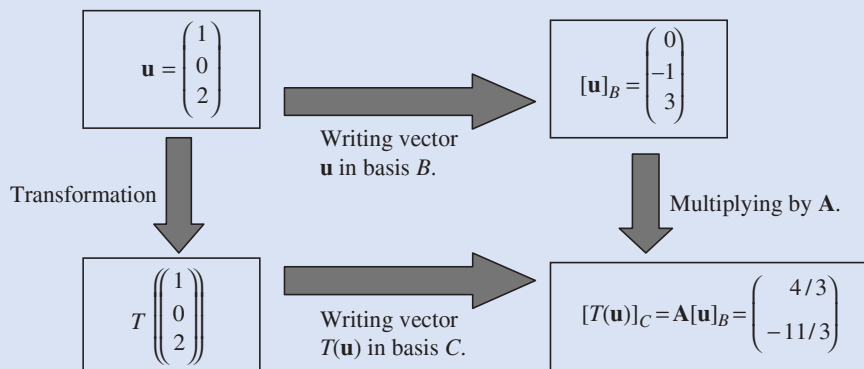


Figure 5.32

The coordinates of $T(\mathbf{u})$ where $\mathbf{u} = (1 \ 0 \ 2)^T$ with respect to the given basis C are $\begin{pmatrix} 4/3 \\ -11/3 \end{pmatrix}$. We have measured the vector $T(\mathbf{u})$ against the new axes represented by $\mathbf{w}_1$ and $\mathbf{w}_2$:

$$T(\mathbf{u}) = \frac{4}{3}\mathbf{w}_1 - \frac{11}{3}\mathbf{w}_2$$

This is illustrated in Fig. 5.33.

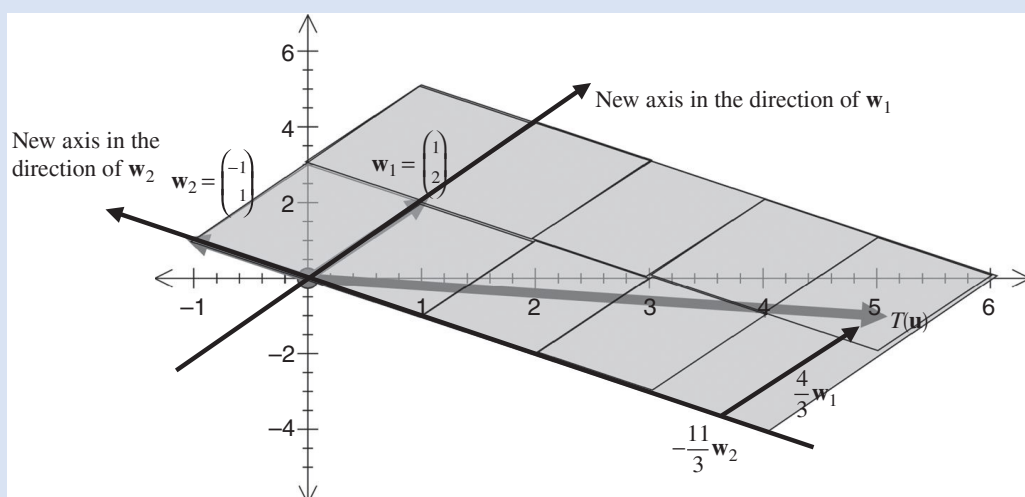


Figure 5.33

However, the coordinates of $T(\mathbf{u})$ with respect to our standard basis (xy axes) are given by:

$$\begin{aligned} T\left(\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}\right) &= \frac{4}{3}\mathbf{w}_1 - \frac{11}{3}\mathbf{w}_2 = \frac{4}{3}\begin{pmatrix} 1 \\ 2 \end{pmatrix} - \frac{11}{3}\begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad \left[\begin{array}{l} \text{because } \mathbf{w}_1 \text{ and } \mathbf{w}_2 \\ \text{are vectors in basis } C \end{array} \right] \\ &= \begin{pmatrix} 4/3 \\ 8/3 \end{pmatrix} - \begin{pmatrix} -11/3 \\ 11/3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \end{pmatrix} \end{aligned}$$

You can check this $T(\mathbf{u}) = \begin{pmatrix} 5 \\ -1 \end{pmatrix}$ by applying the given transformation directly with

$$\mathbf{u} = (1 \ 0 \ 2)^T.$$

5.5.3 A matrix for the derivative

Example 5.36

Let the linear transformation $T: P_2 \rightarrow P_1$ be defined by

$$T(\mathbf{p}) = \mathbf{p}'$$

(continued...)

where $\mathbf{p}'$ is the derivative of $\mathbf{p}$. Determine the matrix $\mathbf{A}$ which represents the given transformation with respect to the ordered bases

$$B = [1, x, x^2] \text{ and } C = [1, x]$$

for P_2 and P_1 respectively. Find $T(2x^2 + 3x + 1)$ by using this matrix $\mathbf{A}$.

Solution

First we look at how T behaves towards the basis (axes) vectors. Applying the given linear transformation to the vectors in basis (axes) $B = [1, x, x^2]$ we have

$$T(1) = 1' = 0 \quad [\text{differentiating}]$$

$$T(x) = x' = 1 \quad [\text{differentiating}]$$

$$T(x^2) = (x^2)' = 2x \quad [\text{differentiating}]$$

We need to write each of these as the coordinates of the basis (axes) $C = [1, x]$:

$$T(1) = 0 = a(1) + b(x) \quad \text{gives } a = 0 \text{ and } b = 0$$

$$T(x) = 1 = c(1) + d(x) \quad \text{gives } c = 1 \text{ and } d = 0$$

$$T(x^2) = 2x = e(1) + f(x) \quad \text{gives } e = 0 \text{ and } f = 2$$

What is our matrix $\mathbf{A}$ equal to?

$$\mathbf{A} = ([T(1)]_C \quad [T(x)]_C \quad [T(x^2)]_C) = \begin{pmatrix} a & c & e \\ b & d & f \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

For $T(2x^2 + 3x + 1)$ we have $\mathbf{p} = 2x^2 + 3x + 1$ so $[\mathbf{p}]_B = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$ (coefficients of 1, x and x^2 , and it

must be in this order, because the given basis is ordered, $B = [1, x, x^2]$. Remember, a basis is a set of vectors which represent axes, so the order of the basis vectors matters.):

$$[T(\mathbf{p})]_C = \mathbf{A}[\mathbf{p}]_B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

The entries 3 and 4 in the right hand column vector are the coefficients of the basis $C = [1, x]$, which means that we have

$$T(2x^2 + 3x + 1) = 3(1) + 4(x) = 3 + 4x$$

Checking this result by differentiating the quadratic:

$$T(2x^2 + 3x + 1) = (2x^2 + 3x + 1)' = 4x + 3.$$

Thus the matrix $\mathbf{A}$ does give the derivative of the quadratic polynomial $2x^2 + 3x + 1$.

We can find the derivative of any quadratic polynomial by using the matrix **A**. For example, to find $(3x^2 + 5x + 7)'$ we use the above matrix **A** as follows:

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} 7 \\ 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$$

Hence $(3x^2 + 5x + 7)' = 5 + 6x$.

If we consider the derivative of the general quadratic $(ax^2 + bx + c)'$ then

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} c \\ b \\ a \end{pmatrix} = \begin{pmatrix} b \\ 2a \end{pmatrix} \text{ implies } (ax^2 + bx + c)' = b + 2ax$$

There is another advantage of using matrix form: we can write a matrix for a different non-standard basis such as $B = [1, 1 - x, (1 - x)^2]$. You are asked to find such a matrix in Exercises 5.5. It is not clear what happens using calculus, whereas the matrix form makes it transparent.

Note that a problem in differentiation can be converted to a problem in arithmetic of matrices. This is always going to be the case as long as we have a linear transformation T between two vector spaces U and V because by Proposition (5.31) we have

$$[T(\mathbf{u})]_C = \mathbf{A}[\mathbf{u}]_B$$

where B and C are bases for U and V respectively.

Summary

(5.31) Let $T : V \rightarrow W$ be a linear transformation and V and W be finite-dimensional vector spaces with bases $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ and $C = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \dots, \mathbf{w}_m\}$ respectively. Then

$$[T(\mathbf{u})]_C = \mathbf{A}[\mathbf{u}]_B$$

where $\mathbf{u}$ is a vector in V and $\mathbf{A} = ([T(\mathbf{v}_1)]_C \ [T(\mathbf{v}_2)]_C \ \cdots \ [T(\mathbf{v}_n)]_C)$.

EXERCISES 5.5

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

1. Determine the standard matrix for the following linear transformations:

$$(a) \ T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x + y \\ 2x + 2y \end{pmatrix} \quad (b) \ T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x + y \\ -x - y \end{pmatrix}$$

$$(c) T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 2x + 3y \\ x - 5y \end{pmatrix}$$

$$(d) T((x \ y \ z)^T) = (x + y + z \ x - y - z \ 2x + y - z)^T$$

$$(e) T((x \ y \ z)^T) = (2x - y \ 4x - y + 3z \ 7x - y - z)^T$$

$$(f) T((x \ y \ z \ w)^T) = (x - y + z - 3w \ -x + 3y - 7z - w \ 9x + 5y + 6z + 12w \ x)^T$$

$$(g) T((x \ y \ z \ w)^T) = \mathbf{O}$$

2. Determine the standard matrix of the following linear transformations *without* reading off the coefficients of x, y, z .

$$(a) T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x - y \\ x + 2y \end{pmatrix}$$

$$(b) T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 3x - 2y \\ 5y - x \end{pmatrix}$$

$$(c) T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} x - y - z \\ x + y + z \end{pmatrix}$$

$$(d) T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$(e) T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} -3x - 5y - 6z \\ -2x + 7y + 5z \\ 0 \end{pmatrix}$$

3. The linear transform $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ satisfies the following:

$$T\left(\begin{bmatrix} 1 & 0 \end{bmatrix}^T\right) = (1 \ 2 \ 3)^T \text{ and } T\left(\begin{bmatrix} 0 & 1 \end{bmatrix}^T\right) = (4 \ 5 \ 6)^T$$

Determine the matrix $\mathbf{A}$ which represents this transformation T .

4. For the linear transformation $T : P_2 \rightarrow P_1$ given by

$$T(\mathbf{p}) = \mathbf{p}'$$

where $\mathbf{p}'$ is the derivative of $\mathbf{p}$, determine the matrix $\mathbf{A}$ which represents the given transformation with respect to the ordered bases

$$B = [1, 1 - x, (1 - x)^2] \text{ and } C = [1, x]$$

for P_2 and P_1 respectively. Find $T(2x^2 + 3x + 1)$ by using this matrix $\mathbf{A}$.

5. For the linear transformation $T : P_3 \rightarrow P_2$ given by

$$T(\mathbf{p}) = \mathbf{p}'$$

where $\mathbf{p}'$ is the derivative of $\mathbf{p}$. Determine the matrix $\mathbf{A}$ which represents the given transformation with respect to the ordered bases in each of the following cases:

- (i) $B = [1, x, x^2, x^3]$ and $C = [1, x, x^2]$ for P_3 and P_2 respectively.
- (ii) $B = [1, x, x^2, x^3]$ and $C = [x^2, x, 1]$ for P_3 and P_2 respectively.
- (iii) $B = [x^3, x^2, x, 1]$ and $C = [1, x, x^2]$ for P_3 and P_2 respectively.



What do you notice about the matrix $\mathbf{A}$ for parts (i), (ii) and (iii)?

In each case find $T(-1 + 3x - 7x^2 - 2x^3)$ by using this matrix $\mathbf{A}$.

6. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear operator defined by $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x + y \\ x - y \end{pmatrix}$.

Find the transformation matrix $\mathbf{A}$ with respect to the basis

$$B = \left\{ \mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 0 \\ -1 \end{pmatrix} \right\}$$

Find $T\left(\begin{pmatrix} -3 \\ 1 \end{pmatrix}\right)$ by using this matrix $\mathbf{A}$ and directly by using the given transformation.

7. In a computer game we want to change our standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ by a rotation of 45° anti-clockwise which is given by the matrix $\mathbf{A} = \begin{pmatrix} \cos(45^\circ) & -\sin(45^\circ) \\ \sin(45^\circ) & \cos(45^\circ) \end{pmatrix}$.

Determine the new basis and write the coordinates of the vector $\mathbf{v} = (2 \ 1)^T$ in terms of the new basis. Sketch the new basis and the vector $\mathbf{v}$.

8. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be defined by

$$T\left(\begin{bmatrix} x & y \end{bmatrix}^T\right) = (-x \quad -y \quad x + 3y)^T$$

Determine the transformation matrix $\mathbf{A}$ with respect to the bases

$$B = \left\{ \mathbf{v}_1 = (1 \ 2)^T, \mathbf{v}_2 = (1 \ 1)^T \right\} \text{ and } \\ C = \left\{ \mathbf{w}_1 = (1 \ 0 \ 1)^T, \mathbf{w}_2 = (1 \ 2 \ 0)^T, \mathbf{w}_3 = (0 \ 1 \ 1)^T \right\}$$

for $\mathbb{R}^2$ and $\mathbb{R}^3$ respectively. Find $T\left(\begin{bmatrix} 2 & 1 \end{bmatrix}^T\right)$ by using this matrix $\mathbf{A}$.

9. Let $T : M_{22} \rightarrow M_{22}$ be defined by

$$T(\mathbf{X}) = \mathbf{X}^T \quad (\text{where } \mathbf{X}^T \text{ is the transpose of the matrix } \mathbf{X})$$

Determine the transformation matrix $\mathbf{A}$ with respect to the bases

$$B = C = \left\{ \mathbf{m}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \mathbf{m}_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \mathbf{m}_3 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \mathbf{m}_4 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

for M_{22} where M_{22} is the 2 by 2 matrices. Find $T\left(\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\right)$ by using this matrix $\mathbf{A}$.

10. Let V be vector space spanned by the set $\{\sin(x), \cos(x)\}$. Let T be the differential linear operator $T : V \rightarrow V$ given by

$$T(\mathbf{f}) = \mathbf{f}' \text{ where } \mathbf{f}' \text{ is the derivative of } \mathbf{f}$$

Determine the transformation matrix $\mathbf{A}$ with respect to the ordered bases

$$B = [\sin(x), \cos(x)]$$

By using this transformation matrix $\mathbf{A}$ find $T(\mathbf{g})$ where

$$(i) \mathbf{g} = 5 \sin(x) + 2 \cos(x) \quad (ii) \mathbf{g} = m \sin(x) + n \cos(x)$$

11. Let $T : P_2 \rightarrow P_2$ be the linear operator given by

$$T(\mathbf{p}) = \mathbf{p}(x+3)$$

This means that $T(ax^2 + bx + c) = a(x+3)^2 + b(x+3) + c$. Determine the transformation matrix $\mathbf{A}$ with respect to the ordered basis $B = [1, x, x^2]$.

Find $T(q + nx + mx^2)$ by using this matrix $\mathbf{A}$ and directly by applying the given transformation.

12. Let V be a vector space spanned by the set $\{\sin(x), \cos(x), e^x\}$. Let T be the differential linear operator $T : V \rightarrow V$ given by

$$T(\mathbf{f}) = \mathbf{f}' \text{ where } \mathbf{f}' \text{ is the derivative of } \mathbf{f}$$

Determine the transformation matrix $\mathbf{A}$ with respect to the ordered basis:

$$B = [\sin(x), \cos(x), e^x]$$

By using this transformation matrix $\mathbf{A}$, find $T(\mathbf{g})$ where

$$(i) \mathbf{g} = -\sin(x) + 4 \cos(x) - 2e^x \quad (ii) \mathbf{g} = m \sin(x) + n \cos(x) + pe^x$$

13. Let V be a vector space spanned by the set $\{e^{2x}, xe^{2x}, x^2e^{2x}\}$. Let T be the differential linear operator $T : V \rightarrow V$ given by

$$T(\mathbf{f}) = \mathbf{f}' \text{ where } \mathbf{f}' \text{ is the derivative of } \mathbf{f}$$

Determine the transformation matrix $\mathbf{A}$ with respect to the ordered basis:

$$B = [e^{2x}, xe^{2x}, x^2e^{2x}]$$

By using this matrix find $T(ae^{2x} + bxe^{2x} + cx^2e^{2x})$.

14. Let $T : V \rightarrow V$ be the identity linear operator on an n -dimensional vector space V which is given by

$$T(\mathbf{v}) = \mathbf{v} \text{ for all vectors } \mathbf{v} \text{ in } V$$

Prove that the matrix $\mathbf{A}$ for the linear transformation T with respect to any basis B is the identity matrix $\mathbf{I}_n$.

SECTION 5.6 Composition and Inverse Linear Transformations

By the end of this section you will be able to

- understand what is meant by composition of linear transforms
- prove some properties of composition of transformations
- prove that the inverse matrix represents the inverse transform T^{-1}

5.6.1 Composition of linear transformations (mappings)

Remember that linear transforms are functions and you should be familiar with the concept of a function.

 *What does composition mean?*

Composition means making something by combining parts.

 *What do you think composition of linear transformation means?*

It is the linear transformation created by putting together two or more linear transformations.

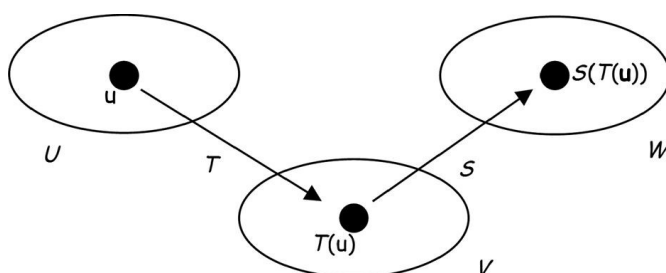


Figure 5.34

Fig. 5.34 shows a linear transformation $T : U \rightarrow V$ which takes the vector $\mathbf{u}$ to $T(\mathbf{u})$ and then the linear transformation $S : V \rightarrow W$ takes $T(\mathbf{u})$ to $S(T(\mathbf{u}))$. This composition of T and S is denoted by $S \circ T$. The formal definition is the following:

Definition (5.32). Let $T : U \rightarrow V$ and $S : V \rightarrow W$ be linear transforms and $\mathbf{u}$ and $T(\mathbf{u})$ be in the domain (start vector spaces) of U and V respectively. Then the **composition** of these two transforms, S and T , denoted by $S \circ T$ is defined by $S \circ T : U \rightarrow W$ (from U to W) and

$$(S \circ T)(\mathbf{u}) = S(T(\mathbf{u}))$$

 *What does $S(T(\mathbf{u}))$ mean?*

Means *first* apply the transform T to the vector $\mathbf{u}$ to get $T(\mathbf{u})$ and *then* apply the transform S to this vector to give $S(T(\mathbf{u}))$.

Example 5.37

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $S : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be linear operators given by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} x - y \\ x + y \end{pmatrix} \text{ and } S\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} 2x + 3y \\ x - 5y \end{pmatrix}$$

Determine $(S \circ T)\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right)$ and $(T \circ S)\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right)$.

Solution

$(S \circ T)\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right)$ means apply the transform T to the vector $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and then apply transform S to this result. By using the above Definition (5.32) we have

$$\begin{aligned} (S \circ T)\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) &= S\left(T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right)\right) \\ &= S\left(\begin{bmatrix} 2 - 3 \\ 2 + 3 \end{bmatrix}\right) && \left[\text{applying } T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} x - y \\ x + y \end{pmatrix}\right] \\ &= S\left(\begin{bmatrix} -1 \\ 5 \end{bmatrix}\right) = \begin{pmatrix} 2(-1) + 3(5) \\ -1 - 5(5) \end{pmatrix} && \left[\text{applying } S\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} 2x + 3y \\ x - 5y \end{pmatrix}\right] \\ &= \begin{pmatrix} 13 \\ -26 \end{pmatrix} \end{aligned}$$

Similarly, carrying out the transformations the other way, first S then T , we have [verify this]:

$$(T \circ S)\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = T\left(S\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right)\right) = \begin{pmatrix} 26 \\ 0 \end{pmatrix}$$

These vectors are illustrated in Fig. 5.35.

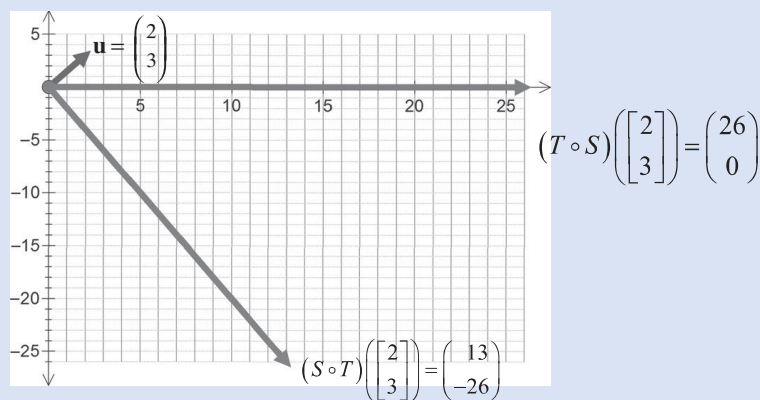


Figure 5.35

Note that $(S \circ T)\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{pmatrix} 13 \\ -26 \end{pmatrix}$ does not equal $(T \circ S)\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{pmatrix} 26 \\ 0 \end{pmatrix}$.

In general, $(S \circ T) \neq (T \circ S)$ [not equal]. This means that *changing* the order of transformation gives a *different* result.

Next, we prove that if S and T are linear transformations then the composite of these $S \circ T$ is also a linear transformation.

Proposition (5.33). Let $T : U \rightarrow V$ and $S : V \rightarrow W$ be linear transforms. Then the *composition transformation* $S \circ T : U \rightarrow W$ is also linear.

Remember, linear transformations preserve the fundamental linear algebra operations of vector addition and scalar multiplication. This proposition means that combining two linear transformations preserves scalar multiplication and vector addition. This is a useful property to have because we don't have to worry about the order of doing things.



How do we prove this result?

To prove that $S \circ T$ is a *linear* transformation we need to show two things:

1. $(S \circ T)(\mathbf{u} + \mathbf{v}) = (S \circ T)(\mathbf{u}) + (S \circ T)(\mathbf{v})$ [vector addition]
2. $(S \circ T)(k\mathbf{u}) = k(S \circ T)(\mathbf{u})$ [scalar multiplication]

where $\mathbf{u}$ and $\mathbf{v}$ are in the start vector space U of transform T and k is a scalar.

Proof.

- 1) Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in U . Then by the above definition of composition (5.32) we have

$$\begin{aligned}
 (S \circ T)(\mathbf{u} + \mathbf{v}) &= S[T(\mathbf{u} + \mathbf{v})] \\
 &= S[T(\mathbf{u}) + T(\mathbf{v})] && \text{[because } T \text{ is linear]} \\
 &= S[T(\mathbf{u})] + S[T(\mathbf{v})] && \text{[because } S \text{ is linear]} \\
 &= (S \circ T)(\mathbf{u}) + (S \circ T)(\mathbf{v}) && \text{[because } S(T(\mathbf{u})) = (S \circ T)(\mathbf{u})]
 \end{aligned}$$

- 2) Similarly we have

$$\begin{aligned}
 (S \circ T)(k\mathbf{u}) &= S(T(k\mathbf{u})) \\
 &= S(kT(\mathbf{u})) && \text{[because } T \text{ is linear]} \\
 &= kS(T(\mathbf{u})) && \text{[because } S \text{ is linear]} \\
 &= k(S \circ T)(\mathbf{u}) && \text{[because } S(T(\mathbf{u})) = (S \circ T)(\mathbf{u})]
 \end{aligned}$$

Since both the above conditions, 1 and 2, are satisfied, $S \circ T$ is a linear transformation. ■

5.6.2 Matrix of composition linear transformations

In the last section we showed that we can write a linear transformation as a matrix. In this section we prove that the matrix of the *composite* transformation is the *multiplication* of the

matrices that represent each transformation. Multiplying matrices is painless compared to combining linear transformations by using the given formula.

The next proposition is complicated because of the different bases used. If we used standard bases then the result would be a lot simpler – $(S \circ T)(\mathbf{u}) = \mathbf{BA}(\mathbf{u})$. However, we want to have the flexibility of using any bases.

Proposition (5.34). Let $T : U \rightarrow V$ and $S : V \rightarrow W$ be linear transforms and U , V and W be finite-dimensional vector spaces with bases B , C and D respectively. If

$$[T(\mathbf{u})]_C = \mathbf{A}[\mathbf{u}]_B \text{ and } [S(\mathbf{v})]_D = \mathbf{B}[\mathbf{v}]_C$$

where $\mathbf{u}$ is a vector in U and $\mathbf{v}$ is a vector in V then

$$[(S \circ T)(\mathbf{u})]_D = \mathbf{BA}[\mathbf{u}]_B$$

Here the bold $\mathbf{B}$ is a matrix and italic B is a basis. We could remove all the subscripts if we used the standard bases and the proposition would be $(S \circ T)(\mathbf{u}) = \mathbf{BA}(\mathbf{u})$.

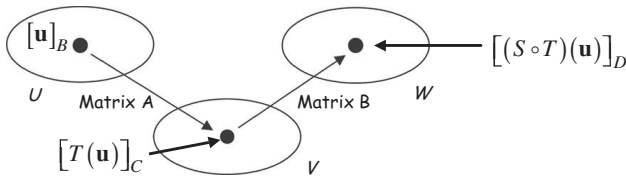


Figure 5.36

This means that the matrix multiplication $\mathbf{BA}$ represents the composite transform $S \circ T$.

Proof.

Let $\mathbf{u}$ be an arbitrary vector in the start vector space U . Then by assumption we have

$$[T(\mathbf{u})]_C = \mathbf{A}[\mathbf{u}]_B \quad (*)$$

Let the vector $\mathbf{v} = T(\mathbf{u})$ be in the vector space V . Then by the other assumption we have

$$[S(T(\mathbf{u}))]_D = \mathbf{B}[T(\mathbf{u})]_C \quad (**)$$

Note that $S(T(\mathbf{u})) = (S \circ T)(\mathbf{u})$ by the definition of composition. We have

$$\begin{aligned} [(S \circ T)(\mathbf{u})]_D &= [S(T(\mathbf{u}))]_D \\ &= \mathbf{B}[T(\mathbf{u})]_C && [\text{by } (**)] \\ &= \mathbf{B}(\mathbf{A}[\mathbf{u}]_B) = \mathbf{BA}[\mathbf{u}]_B && [\text{by } (*)] \end{aligned}$$

Hence this proves our required result that the composite transformation $S \circ T$ can be represented by the matrix multiplication $\mathbf{BA}$. ■

Example 5.38

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $S : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be linear operators given by:

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} x - y \\ x + y \end{pmatrix} \text{ and } S\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} 2x + 3y \\ x - 5y \end{pmatrix}$$

Determine the standard matrices for the compositions $(S \circ T)$ and $(T \circ S)$.

Also, find how the composite transforms act on vector $\mathbf{u} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, $(S \circ T)(\mathbf{u})$ and $(T \circ S)(\mathbf{u})$.

Solution

How do we find the standard matrix for the given transformation T ?

Read off the coefficients of x and y . Let $\mathbf{A}$ be this matrix then

$$\mathbf{A} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \quad \left[\text{because } T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} x - y \\ x + y \end{pmatrix} \right]$$

Let $\mathbf{B}$ be the standard matrix for the other given linear transformation S :

$$\mathbf{B} = \begin{pmatrix} 2 & 3 \\ 1 & -5 \end{pmatrix} \quad \left[\text{because } S\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} 2x + 3y \\ x - 5y \end{pmatrix} \right]$$

By the above Proposition (5.34) and using the standard basis we have

$$\begin{aligned} (S \circ T)(\mathbf{u}) &= \mathbf{BA}[\mathbf{u}] \\ &= \begin{pmatrix} 2 & 3 \\ 1 & -5 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} [\mathbf{u}] = \begin{pmatrix} 5 & 1 \\ -4 & -6 \end{pmatrix} [\mathbf{u}] \end{aligned}$$

The composite transform $S \circ T$ on vector $\mathbf{u} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ is given by $(S \circ T)\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right)$:

$$(S \circ T)\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{pmatrix} 5 & 1 \\ -4 & -6 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 13 \\ -26 \end{pmatrix}$$

Similarly, applying the composite transform $T \circ S$ on vector $\mathbf{u}$ gives

$$\begin{aligned} (T \circ S)(\mathbf{u}) &= \mathbf{AB}[\mathbf{u}] \\ &= \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 1 & -5 \end{pmatrix} [\mathbf{u}] = \begin{pmatrix} 1 & 8 \\ 3 & -2 \end{pmatrix} [\mathbf{u}] \end{aligned}$$

Also in the above manner we have

$$(T \circ S)\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{pmatrix} 1 & 8 \\ 3 & -2 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 26 \\ 0 \end{pmatrix}$$

Note that these answers agree with the previous Example 5.37 because we have the same transformations.

The calculations in this example are easier than Example 5.37 because we have reduced the combining of linear transformations to a matrix arithmetic problem.

If a linear transformation represented a rotation, then a number of rotations can be evaluated by multiplying matrices. This is a much smoother task than applying the formula to the composition of many linear transformations.

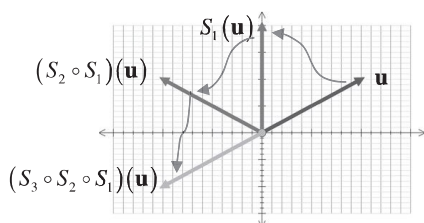


Figure 5.37

For example, consider the rotation transformations S_1 , S_2 and S_3 on vector $\mathbf{u}$ (Fig. 5.37). If the transformation S_j is represented by the matrix $\mathbf{A}_j$ then

$$(S_3 \circ S_2 \circ S_1)(\mathbf{u}) = \underbrace{\mathbf{A}_3 \mathbf{A}_2 \mathbf{A}_1}_{\text{multiplication of matrices}} (\mathbf{u})$$

Multiplying the matrices on the right hand side is a lot simpler than evaluation of $S_3 \circ S_2 \circ S_1$.

Since matrix multiplication is *not* commutative so the composition of transforms is *not* commutative:

$$\mathbf{BA} \neq \mathbf{AB} \text{ [not equal], therefore } (T \circ S) \neq (S \circ T)$$

Changing the order of the transformations changes the resulting vector as you should have noticed for the above examples.

Example 5.39

Let V be the space spanned by $B = \{e^x, xe^x, x^2e^x\}$. Let $T : V \rightarrow V$ and $S : V \rightarrow V$ be defined by

$$T(\mathbf{p}) = \mathbf{p}' \text{ (differential operator) and } S(\mathbf{p}) = \mathbf{p} \text{ (identity operator)}$$

where $\mathbf{p} = ae^x + bxe^x + cx^2e^x$. Determine the matrix for the composite transform $S \circ T$ and find $(S \circ T)(e^x + 2xe^x + 3x^2e^x)$.

Solution

We need to find the matrix which represents the given differential operator T .

How?

By finding the transformation of each vector in $\{e^x, xe^x, x^2e^x\}$:

$$T(e^x) = (e^x)' = e^x \quad \text{[differentiating]}$$

$$T(xe^x) = (xe^x)' = e^x + xe^x \quad \text{[differentiating by product rule]}$$

$$T(x^2e^x) = (x^2e^x)' = 2xe^x + x^2e^x \quad \text{[differentiating by product rule]}$$

We need to write these in terms of the vectors $\{e^x, xe^x, x^2e^x\}$:

$$T(e^x) = e^x = 1(e^x) + 0(xe^x) + 0(x^2e^x)$$

$$T(xe^x) = e^x + xe^x = 1(e^x) + 1(xe^x) + 0(x^2e^x)$$

$$T(x^2e^x) = 2xe^x + x^2e^x = 0(e^x) + 2(xe^x) + 1(x^2e^x)$$

What is the matrix that represents the transform T equal to?

We use (5.31) of the last section: $\mathbf{A} = ([T(\mathbf{v}_1)]_C \quad [T(\mathbf{v}_2)]_C \quad \cdots \quad [T(\mathbf{v}_n)]_C)$

$$\mathbf{A} = (T(e^x) \quad T(xe^x) \quad T(x^2e^x)) = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

What is the matrix for the identity transformation $S(\mathbf{p}) = \mathbf{p}$?

Let $\mathbf{B}$ be the matrix for this transformation S . By question 14 of Exercises 5.5:

If S is an identity linear operator then the matrix for S is the identity matrix $\mathbf{I}_n$.

$$\mathbf{B} = \mathbf{I} \quad [\text{Identity matrix}]$$

Thus the matrix for composite transform $S \circ T$ is $\mathbf{BA}$ which is

$$\mathbf{BA} = \mathbf{IA} = \mathbf{A}$$

We need to evaluate $(S \circ T)(e^x + 2xe^x + 3x^2e^x)$. Rewriting

$$e^x + 2xe^x + 3x^2e^x = 1(e^x) + 2(xe^x) + 3(x^2e^x) = \mathbf{p}$$

Hence $[\mathbf{p}]_B = (1 \ 2 \ 3)^T$:

$$(S \circ T)(1(e^x) + 2(xe^x) + 3(x^2e^x)) = \mathbf{BA}[\mathbf{p}]_B \quad \stackrel{\text{because } \mathbf{BA}=\mathbf{A}}{=} \quad \mathbf{A}[\mathbf{p}]_B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 8 \\ 3 \end{pmatrix}$$

Hence using these values gives

$$(S \circ T)(e^x + 2xe^x + 3x^2e^x) = 3e^x + 8xe^x + 3x^2e^x$$

Verify this result by applying the differential operator T to the vector $e^x + 2xe^x + 3x^2e^x$ and then the identity, and you should end up with the above result $3e^x + 8xe^x + 3x^2e^x$.

5.6.3 Invertible linear transformations



What is an invertible linear transformation?

From section 5.4.5 we have

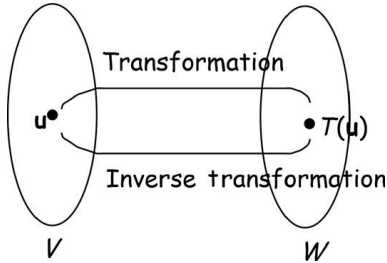


Figure 5.38

Fig. 5.38 shows a linear transformation $T : V \rightarrow W$ (from V to W) which takes u to $T(u)$. The inverse transformation denoted by $T^{-1} : W \rightarrow V$ (from W to V) takes $T(u)$ back to u .

Definition (5.35). Let $T : V \rightarrow W$ be a linear transform. Then T is invertible $\Leftrightarrow$ there exists a transformation T^{-1} such that

- (a) $(T^{-1} \circ T) = I_V$ where I_V is the identity transformation on the start vector space V
- (b) $(T \circ T^{-1}) = I_W$ where I_W is the identity transformation on the arrival vector space W

If a linear transform T is invertible we say it is an **invertible linear transform**.

Proposition (5.36). Let $T : V \rightarrow W$ be a linear transform of finite-dimensional spaces V and W . Then T is **invertible** $\Leftrightarrow$ the matrix A which represents the transformation T is invertible.

Proof – Exercises 5.6. ■

We found an inverse transformation in section 5.4.5. We use the same technique established in that section to find the matrix for T^{-1} in the following example.

Example 5.40

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear operator given by $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} x - y \\ x + y \end{pmatrix}$.

- (i) Determine the matrix for the linear operator $T^{-1} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$.
- (ii) Show that the identity matrix I represents the composite function $T \circ T^{-1}$.

T is the same transform given in Example 5.27 of section 5.4.5.

(continued...)

Solution

(i) By Example 5.27 we have the inverse transformation T^{-1} :

$$T^{-1} \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \frac{1}{2} \begin{pmatrix} x+y \\ -x+y \end{pmatrix}$$

What is the standard matrix, call it $\mathbf{A}$, which represents T^{-1} ?

It is the coefficients of x and y , which means the matrix is $\mathbf{A} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$.

What is the matrix, call it $\mathbf{B}$, which represents the given transformation $T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{pmatrix} x-y \\ x+y \end{pmatrix}$?

Again, this is the coefficients of x and y as column vectors. We have $\mathbf{B} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$.

(ii) How do we show that the identity matrix $\mathbf{I}$ represents the composite transform $T \circ T^{-1}$?

By showing $\mathbf{BA} = \mathbf{I}$ because $\mathbf{BA}$ represents the composite transform $T \circ T^{-1}$. We have

$$\mathbf{BA} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}$$

In the next proposition, we state and prove that we can find the *inverse* transformation of a given transform by determining the *inverse* matrix $\mathbf{A}^{-1}$.

Proposition (5.37). Let $T : V \rightarrow V$ be an invertible linear transform (actually a linear operator) of finite-dimensional space V . Let $\mathbf{A}$ be a matrix representing the transform T with respect to a basis for V . Then $\mathbf{A}^{-1}$ is the matrix representing the inverse transform T^{-1} with respect to the same basis.



How do we prove this result?

We use the above Proposition (5.36) which says:

T is invertible $\Leftrightarrow$ the matrix $\mathbf{A}$ which represents the transformation T is invertible.

Proof.

By (5.36) we know that the matrix $\mathbf{A}$ is invertible. Since T is invertible we have

$$T^{-1} \circ T = I_V$$

This means that $T^{-1} \circ T$ is the identity transformation. Let $\mathbf{B}$ be the matrix representing T^{-1} then we need to show $\mathbf{B} = \mathbf{A}^{-1}$.

By Proposition (5.35) we have that the matrix multiplication $\mathbf{BA}$ represents the composite transform $T^{-1} \circ T$ which is the identity. Thus by the result of question 14 of the last Exercises 5.5 which says:

If T is an identity linear operator then the matrix for T is the identity matrix $\mathbf{I}_n$. We have $\mathbf{BA} = \mathbf{I}$, because $T^{-1} \circ T$ is the identity transformation.



What is the matrix $\mathbf{B}$ equal to?

Because $\mathbf{BA} = \mathbf{I}$, therefore $\mathbf{B} = \mathbf{A}^{-1}$ which is our required result.

Example 5.41

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear operator given by $T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} x + 5y + 3z \\ 2x + 3y + z \\ 3x + 4y + z \end{pmatrix}$.

Determine the inverse transformation $T^{-1} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$.

Solution

How do we find the inverse transformation T^{-1} ?

First we write out the standard matrix for T and then we find the inverse of this matrix. The standard matrix for T is given by reading the coefficients of x , y and z :

$$\mathbf{A} = \begin{pmatrix} 1 & 5 & 3 \\ 2 & 3 & 1 \\ 3 & 4 & 1 \end{pmatrix} \quad \left[\text{Because } T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} x + 5y + 3z \\ 2x + 3y + z \\ 3x + 4y + z \end{pmatrix} \right]$$

For the inverse transformation T^{-1} we need to find the inverse matrix $\mathbf{A}^{-1}$.

How?

We can use row operations or apply MATLAB. You can verify (in your own time) that the following is the inverse matrix $\mathbf{A}^{-1}$:

$$\mathbf{A}^{-1} = \begin{pmatrix} -1 & 7 & -4 \\ 1 & -8 & 5 \\ -1 & 11 & -7 \end{pmatrix}$$

What is the inverse transformation T^{-1} equal to?

The entries of the first column of this matrix $\mathbf{A}^{-1}$ are the x coefficients, the entries in the second column are the y coefficients and the entries in the last column are the z coefficients:

$$T^{-1} \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} -x + 7y - 4z \\ x - 8y + 5z \\ -x + 11y - 7z \end{pmatrix} \quad \left[\text{because } \mathbf{A}^{-1} = \begin{pmatrix} -1 & 7 & -4 \\ 1 & -8 & 5 \\ -1 & 11 & -7 \end{pmatrix} \right]$$

Finding the inverse matrix is *no* easy task but it is still simpler than trying to find the inverse transformation by transposing formulae.

Example 5.42

Let $T : P_2 \rightarrow P_2$ be the linear operator given by

$$T(\mathbf{p}) = (\mathbf{p}x)' \text{ where } \mathbf{p} = ax^2 + bx + c$$

and $B = [x^2, x, 1]$ be an ordered basis (axes) for P_2 . Determine an expression for the inverse transformation $T^{-1}(\mathbf{p})$ with respect to the same basis B .

Solution

What do we need to find first?

The matrix which represents the given transformation T .

How can we locate this matrix?

By finding the transformation $T(\mathbf{p}) = (\mathbf{p}x)'$ of each basis (axes) vector in $B = [x^2, x, 1]$:

$$T(x^2) = (x^2x)' = (x^3)' = 3x^2$$

$$T(x) = (xx)' = (x^2)' = 2x$$

$$T(1) = (1x)' = (x)' = 1$$

We need to write these in terms of the basis vectors $B = [x^2, x, 1]$:

$$T(x^2) = 3x^2 = 3(x^2) + 0(x) + 0(1)$$

$$T(x) = 2x = 0(x^2) + 2(x) + 0(1)$$

$$T(1) = 1 = 0(x^2) + 0(x) + 1(1)$$

What is the matrix $\mathbf{A}$ which represents the given transformation T equal to?

$$\mathbf{A} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

To find the inverse transformation T^{-1} we need to find $\mathbf{A}^{-1}$. By using row operations we have:

$$\mathbf{A}^{-1} = \begin{pmatrix} 1/3 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

With $\mathbf{p} = ax^2 + bx + c$ we can evaluate the coefficients of x^2 , x and constants of the inverse transformation T^{-1} as follows:

$$\begin{pmatrix} 1/3 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a/3 \\ b/2 \\ c \end{pmatrix}$$

The entries in the right hand vector are the coefficients of x^2 , x and constants respectively. This means that the inverse transformation is given by

$$T^{-1}(\mathbf{p}) = T^{-1}(ax^2 + bx + c) = \frac{a}{3}x^2 + \frac{b}{2}x + c$$

Note that in the above Example 5.42 we were given the transformation

$$\begin{aligned} T(\mathbf{p}) &= (\mathbf{p}x)' = ([ax^2 + bx + c]x)' \\ &= (ax^3 + bx^2 + cx)' = 3ax^2 + 2bx + c \end{aligned}$$

In this transformation we have multiplied a general quadratic $ax^2 + bx + c$ by x and then found the derivative of this result. The inverse transformation is

$$T^{-1}(\mathbf{p}) = T^{-1}(ax^2 + bx + c) = \frac{a}{3}x^2 + \frac{b}{2}x + c$$

This is the same as integrating a general quadratic $ax^2 + bx + c$ which gives $\frac{a}{3}x^3 + \frac{b}{2}x^2 + cx$ and then dividing this result by x which yields $\frac{a}{3}x^2 + \frac{b}{2}x + c = T^{-1}(\mathbf{p})$. Note that this is the reverse process of the given transformation T . We can illustrate this as shown in Fig. 5.39.

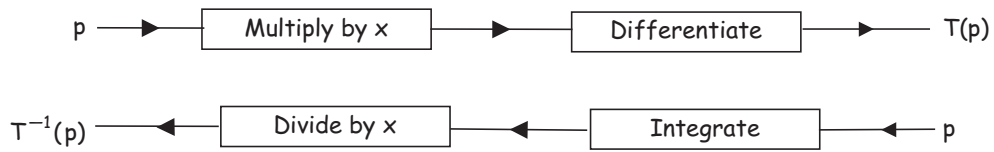


Figure 5.39

5.6.4 Operations of invertible linear operators

Proposition (5.38). Let $T : V \rightarrow V$ and $S : V \rightarrow V$ be invertible linear operators of finite-dimensional space V . Then $S \circ T$ is invertible and

$$(S \circ T)^{-1} = T^{-1} \circ S^{-1}$$

Remember, inverses undo the last operation first as illustrated in Fig. 5.39 above.



How do we prove this result?

We use the above propositions and the matrix result of chapter 1:

$$(1.27) \quad (\mathbf{BA})^{-1} = \mathbf{A}^{-1}\mathbf{B}^{-1}$$

Proof.

Let the matrices $\mathbf{A}$ and $\mathbf{B}$ represent the given linear transformations T and S respectively. By Proposition (5.34)

$$(S \circ T)(\mathbf{u}) = \mathbf{BA}(\mathbf{u})$$

We know that the matrix multiplication $\mathbf{BA}$ represents the composite transform $S \circ T$. By applying the above Proposition (5.37):

If matrix $\mathbf{A}$ represents the transform T then $\mathbf{A}^{-1}$ represents the inverse T^{-1} . We have that the matrix representation of $(S \circ T)^{-1}$ is $(\mathbf{BA})^{-1}$. By the above (1.27) we have

$$(\mathbf{BA})^{-1} = \mathbf{A}^{-1}\mathbf{B}^{-1}$$

The matrix $\mathbf{A}^{-1}$ represents the transform T^{-1} and $\mathbf{B}^{-1}$ represents the transform S^{-1} . Thus $\mathbf{A}^{-1}\mathbf{B}^{-1}$ represents the composite transform $T^{-1} \circ S^{-1}$. This means that we have our required result

$$(S \circ T)^{-1} = T^{-1} \circ S^{-1}$$

Summary

The composite transform denoted $S \circ T$ is defined as

$$(5.32) \quad (S \circ T)(\mathbf{u}) = S(T(\mathbf{u}))$$

(5.37) Let $\mathbf{A}$ be a matrix representing the transform T then $\mathbf{A}^{-1}$ is the matrix representing the inverse transform T^{-1} .



EXERCISES 5.6

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

1. Consider the linear operators $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $S : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} y \\ x \end{pmatrix} \text{ and } S\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} y \\ 0 \end{pmatrix}$$

- (i) Determine the vector $(S \circ T)\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right)$ and $(T \circ S)\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right)$.
 (ii) Determine the vector $(S \circ T)\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$ and $(T \circ S)\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$.

2. Let $T : P_2 \rightarrow P_1$ and $S : P_2 \rightarrow P_2$ be defined by

$$T(\mathbf{p}) = \mathbf{p}' \text{ and } S(\mathbf{p}) = ax^2 + bx \text{ where } \mathbf{p} = ax^2 + bx + c$$

Determine (a) $(T \circ S)(\mathbf{p})$ (b) $(S \circ T)(\mathbf{p})$ (c) $(T \circ T)(\mathbf{p})$ (d) $(S \circ S)(\mathbf{p})$

3. Let the linear operators $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $S : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} x+2y \\ x-3y \end{pmatrix} \text{ and } S\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} 2x+y \\ -x-y \end{pmatrix}$$

Determine the standard matrices for the compositions $S \circ T$ and $T \circ S$. Also find

$$(i) (S \circ T) \left(\begin{bmatrix} 1 \\ 5 \end{bmatrix} \right) \quad (ii) (T \circ S) \left(\begin{bmatrix} 1 \\ 5 \end{bmatrix} \right) \quad (iii) (T \circ T) \left(\begin{bmatrix} 1 \\ 5 \end{bmatrix} \right) \quad (iv) (S \circ S) \left(\begin{bmatrix} 1 \\ 5 \end{bmatrix} \right)$$

4. Let the linear transformations $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ and $S : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be given by

$$T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} 2x - y + 10z \\ 3x - 5y + 6z \\ x + 3y - 9z \end{pmatrix} \quad \text{and} \quad S \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} -x - y - z \\ x + y + 2z \\ x + 2y + 3z \end{pmatrix}$$

Determine the standard matrices for the compositions $S \circ T$ and $T \circ S$. Also find

$$(i) (S \circ T) \left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right) \quad (ii) (T \circ S) \left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right) \quad (iii) (T \circ T) \left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right) \\ (iv) (S \circ S) \left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right)$$

5. (a) Let V be the space spanned by $\{e^x, xe^x, x^2e^x\}$. Let $T : V \rightarrow V$ and $S : V \rightarrow V$ be linear operators defined by

$$T(\mathbf{p}) = \mathbf{p}' \quad \text{and} \quad S(\mathbf{p}) = ce^x + bxe^x + ax^2e^x \quad \text{where} \quad \mathbf{p} = ae^x + bxe^x + cx^2e^x$$

Determine the matrix for

$$(i) S \circ T \quad (ii) T \circ S \quad (iii) T \circ T \quad (iv) S \circ S$$

What do you notice about the transformation $S \circ S$? Find $S^{-1}(\mathbf{p})$.

- (b) Show that the linear operator T is invertible and find T^{-1} . Use T^{-1} to determine

$$\int x^2 e^x dx$$

6. Let $T : P_2 \rightarrow P_2$ and $S : P_2 \rightarrow P_2$ be linear operators defined by

$$T(ax^2 + bx + c) = -ax^2 - bx - c \quad \text{and} \quad S(ax^2 + bx + c) = -bx - c$$

and $B = [1, x, x^2]$ be an ordered basis for P_2 .

Determine the matrix which represents the following transformations.

$$(i) S \circ T \quad (ii) T \circ S \quad (iii) T \circ T \quad (iv) S \circ S$$

Also find $(S \circ T)(1 + 2x + 3x^2)$, $(T \circ S)(1 + 2x + 3x^2)$, $(T \circ T)(1 + 2x + 3x^2)$ and $(S \circ S)(1 + 2x + 3x^2)$.

7. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear operator given by $T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{pmatrix} x - y \\ x + y \end{pmatrix}$.

Determine the standard matrices for transformations T and $T^{-1} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$.

8. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear operator given by $T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} x + y + z \\ x + y - z \\ x - y - z \end{pmatrix}$. Find the inverse transformation $T^{-1} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. [Use MATLAB to find the inverse.]

9. Let $T : P_3 \rightarrow P_3$ be the linear operator defined by

$$T(\mathbf{p}) = (\mathbf{p}x)' \text{ where } \mathbf{p} = ax^3 + bx^2 + cx + d$$

and $B = [x^3, x^2, x, 1]$ be an ordered basis for P_3 . Determine an expression for $T^{-1}(\mathbf{p})$ with respect to the same basis B .

10. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear operator given by

$$T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} x + y + z \\ 2x + 2y + z \\ y + z \end{pmatrix}$$

Find $T^{-1} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$.

11. Determine which of the following transformations are invertible. If they are invertible then find the inverse transformation.

(a) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ given by $T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} x - z \\ x + y + z \end{pmatrix}$

(b) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by $T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{pmatrix} 0 \\ x + y + z \\ x - y - z \end{pmatrix}$

(c) $T : P_3 \rightarrow P_3$ given by $T(\mathbf{p}) = \mathbf{p}$ where $\mathbf{p} = ax^3 + bx^2 + cx + d$.

In computer graphics, each pixel on a screen can be viewed as a point or a vector. The following matrices carry out the corresponding operations on the vector or point:

$\mathbf{R}_\theta = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$ rotates a vector by an angle of θ anti-clockwise about the origin.

$\mathbf{RF}_x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ reflects a vector in the x axis.

12. Let $T, Q : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear operators acting on $\mathbf{x} = (x \ y)^T$ be given by

$$T(\mathbf{x}) = \mathbf{R}_\theta \mathbf{x} \quad \text{and} \quad Q(\mathbf{x}) = \mathbf{RF}_x \mathbf{x}$$

- (i) What is the net effect of $(Q \circ Q)(\mathbf{x})$?
- (ii) Show that if we apply the linear operator Q an *odd* number of times on the vector $\mathbf{x}$ then the net effect is reflection in the x axis.
- (ii) Show that the composite transform $T \circ T$ rotates the vector $\mathbf{x}$ by an angle of 2θ .
13. The probabilities of injury (I) and death (D) on urban roads (U) and motorways (M) are given by the matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{pmatrix} \text{I} & \text{D} \\ 1/3 & 2/3 \\ 2/3 & 1/3 \end{pmatrix} \begin{matrix} \text{M} \\ \text{U} \end{matrix}$$

A linear operator $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is given by $T(\mathbf{x}) = \mathbf{A}\mathbf{x}$. The n th vector $\mathbf{x}_n$ is defined as

$$\mathbf{x}_n = \underbrace{T \circ T \circ T \circ \cdots \circ T}_{n \text{ copies}}(\mathbf{x}_0) \text{ where } \mathbf{x}_0 = (1 \ 0)^T$$

- (i) Use MATLAB or some other software to find x_1, x_2, x_3, x_4 and x_5 correct to 4dp.
 - (ii) By using MATLAB, or otherwise, predict a value for x_n for large n .
- 14.** Let $T : V \rightarrow W$ be a linear transformation. Let I_V and I_W be the identity linear transformations on V and W respectively. Prove that
- (i) $T \circ I_V = T$ (ii) $I_W \circ T = T$
- 15.** Let $T : U \rightarrow V$ and $S : V \rightarrow W$ be linear transformations and U, V and W be finite-dimensional vector spaces. Prove that $(S \circ T) \neq (T \circ S)$.
- 16.** Let $T : U \rightarrow V$ and $S : V \rightarrow W$ be linear transformations, and U, V and W be finite-dimensional vector spaces. Prove that

$$k(S \circ T) = (kS) \circ T = S \circ (kT) \text{ where } k \text{ is any scalar.}$$

- 17.** Let $T : V \rightarrow V$ be an invertible linear operator. Prove that T^{-1} is linear.
- 18.** Let $T : V \rightarrow V$ be an invertible linear operator. Prove that $(T^{-1})^{-1} = T$.
- 19.** Let $T : V \rightarrow W$ be an invertible linear transformation. Prove that the inverse transformation $T^{-1} : W \rightarrow V$ is unique.
- 20.** Prove Proposition (5.36).



MISCELLANEOUS EXERCISES 5

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

[In chapter 5 we used the term transformation to mean map.]

- 5.1.** Show that the function $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is a linear transformation.

$$T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + 5x_2 \\ 0 \\ 2x_1 - 3x_2 \end{bmatrix}$$

University of Puget Sound, USA

- 5.2.** Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the map defined by

$$T[x_1, x_2, x_3]^T = [2x_1 - 3x_2 + 4x_3, -x_1 + x_2]^T$$

Show that T is a linear map and find its standard matrix.

University of Ottawa, Canada

5.3. Give an example or state no such example exists of:

A function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that is not a linear transformation.

Illinois State University, USA
(part question)

5.4. Let $T(x_1, x_2, x_3) = (3x_2 + 2x_3, 3x_1 - 4x_2)$.

- (a) Find the standard matrix for T .
- (b) Is T one-to-one?
- (c) Is T onto?

University of South Carolina, USA

5.5. (a) Let S denote the subset of **all** vectors $\vec{x}$ in $\mathbb{R}^2$ for which $\vec{x} \cdot \vec{u} = 0$ for some fixed $\vec{u} \in \mathbb{R}^2$. Is S a subspace of $\mathbb{R}^2$ or not? Give reasons for your answer.

(b) Let $\vec{u} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$ be two vectors of $\mathbb{R}^2$, and let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation for which $T(\vec{u}) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $T(\vec{v}) = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$.

- (i) Find $T(\vec{e}_1)$ and $T(\vec{e}_2)$.
- (ii) Find the matrix representation of T with respect to the standard basis $\{\vec{e}_1, \vec{e}_2\}$.
- (iii) For what vector $\vec{w}$ of $\mathbb{R}^2$ is $T(\vec{w}) = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$?

University of New Brunswick, Canada

5.6. Let the operator $S : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ be defined by

$$S(x_1, x_2, x_3) = (x_1 - 4x_2 + 2x_3, 2x_1 + 7x_2 - x_3, -x_1 - 8x_2 + 2x_3, 2x_1 + x_2 + x_3)$$

- (a) Find the standard matrix of S .
- (b) Find a basis for the range of S .

University of Western Ontario, Canada

5.7. (a) Given the vector spaces, V, W , over the field K , state what is meant by a linear transformation $T : V \rightarrow W$.

- (b) What is meant by the *kernel* of T , $\ker T$?
- (c) Show that T is injective if and only if $\ker T = \{\mathbf{0}\}$.
- (d) If $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is given by the formula,

$$T(x, y, z) = (x + y, x + y + z),$$

find $\ker T$.

University of Sussex, UK

5.8. Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be a linear transformation defined by

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right) = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & -1 & 1 \\ 0 & 0 & 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

(a) Find a basis for the set of vectors in $\mathbb{R}^3$ that are in the image of T .

Do the columns of the matrix associated with T span $\mathbb{R}^3$?

(b) Is T onto? Explain clearly why or why not.

(c) Find a basis for the null space of T .

(d) Is T one-to-one? Explain why or why not.

Mount Holyoke College, Massachusetts, USA

5.9. Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is a linear transformation such that $T \begin{bmatrix} 3 \\ -5 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$ and

$$T \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ -2 \end{bmatrix}. \text{ Determine } T \begin{bmatrix} 7 \\ -11 \end{bmatrix}.$$

University of Puget Sound, USA

5.10. Let $L : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation such that

$$L \left(\begin{bmatrix} -2 \\ 1 \\ -2 \end{bmatrix} \right) = \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix}, \quad L \left(\begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}, \quad L \left(\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}$$

$$\text{Find } L \left(\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right).$$

Purdue University, USA

5.11. Let $\mathbf{A}$ be a 6×5 matrix such that the nullity of $\mathbf{A}$ is 0. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be the linear transformation defined by $T(\mathbf{x}) = \mathbf{Ax}$. Answer each of the following questions.

Be sure to justify your answers.

(a) What is the value of m ? What is the value of n ?

(b) What is the maximum number of linearly independent vectors in the range of T ?

(c) Is T onto?

(d) Is T one-to-one?

Illinois State University, USA

5.12. The images of the unit vectors in $\mathbb{R}^2$ under the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$

are given as $T(\mathbf{e}_1) = \begin{bmatrix} 2 \\ 1 \\ h \end{bmatrix}$, and $T(\mathbf{e}_2) = \begin{bmatrix} 3 \\ k \\ 0 \end{bmatrix}$. Determine all the values of the parameters h and k for which T is one-to-one.

Washington State University, USA

- 5.13.** Let $C^1[0, 1]$ represent the vector space of continuously differentiable functions defined on the interval $[0, 1]$ and let $C[0, 1]$ be the vector space of continuous functions on $[0, 1]$. Define the transformation $T : C^1[0, 1] \rightarrow C[0, 1]$ by

$$f \rightarrow Tf \text{ where } (Tf)(x) = \frac{df}{dx}(x) - 2f(x) \text{ for } 0 \leq x \leq 1$$

$$\text{Thus, } T(e^x) = \frac{d(e^x)}{dx} - 2e^x = -e^x, \text{ i.e., } e^x \xrightarrow{T} -e^x.$$

- (a) What is the zero vector in the space $C[0, 1]$?
- (b) Show that T is a linear transformation.
- (c) Determine the kernel of T . What is $\dim \ker T$?

Harvey Mudd College, California, USA

- 5.14.** If $T : V \rightarrow W$ is a linear mapping, show that

- (a) $\ker(T)$ is a subspace of V .
- (b) $\text{im}(T)$ is a subspace of W .
- (c) For the linear transformation T given below, find the dimensions of the kernel and the image respectively.

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3 \text{ given by } T(x, y, z) = (x + 2y - z, y + z, x + y - 2z);$$

University of Sussex, UK

- 5.15.** Let $T : V \rightarrow W$ be a linear transformation from the vector spaces V to the vector space W . Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a set of vectors in V . Suppose that the set of vectors $\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\}$ is a *linearly independent* set of vectors in W . Prove that S must be a linearly independent set in V . Produce a counter-example to show that the converse is generally false.

Harvey Mudd College, California, USA

- 5.16.** Let $T : P_2 \rightarrow P_3$ be defined by

$$T(a_0 + a_1t + a_2t^2) := 3a_1 + 2a_2t + a_0t^2 + (a_1 + a_2)t^3$$

- (a) This is a linear mapping. What are the two conditions that one has to check in order to prove this? Check one of them.
- (b) Equip these two spaces P_2 and P_3 with the bases $B = \{1, t, t^2\}$ and $C = \{1, t, t^2, t^3\}$, respectively. Compute the matrix A that represents T with respect to these two bases.
- (c) Demonstrate how to work with this matrix by computing $T(2 + 5t - t^2)$, using A .
- (d) Is it possible to do the same with $T(p) := t^3 + p(t)$? (If yes, don't do it!)

University of New York, USA

5.17. (a) Give the definition of a linear transformation T from $\mathbb{R}^n \rightarrow \mathbb{R}^k$.

(b) (i) Show that $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ given by

$$T([x, y, z]) = [x + 2y, 2y - 3z, z + x, x]$$

is a linear transformation.

(ii) Find the standard matrix representation of T .

(c) Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^k$ be a linear transformation and let $\beta = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ be a basis of $\mathbb{R}^n$. Show that, for any $\mathbf{v} \in \mathbb{R}^n$, the vector $T(\mathbf{v})$ is uniquely determined by the vectors $T(\mathbf{b}_1), T(\mathbf{b}_2), \dots, T(\mathbf{b}_n)$.

University of Toronto, Canada

5.18. Consider the subspace V of $C(\mathbb{R})$ spanned by the set

$$\beta = \{1 + x, 1 - x, e^x, (1 + x)e^x\}$$

(a) Show that β is a basis for V (by showing that it is linearly independent).

(b) Find the coordinates $[\mathbf{u}]^\beta$ in the ordered basis β , where

$$\mathbf{u} = 1 - xe^x$$

(c) Let $D : V \rightarrow V$ be the differentiation operator $D : f \mapsto \frac{df}{dx}$.

Find the matrix representation $[D]_\beta^\beta$.

(d) Verify that

$$[D\mathbf{u}]^\beta = [D]_\beta^\beta [\mathbf{u}]^\beta$$

Loughborough University, UK

5.19. (a) Determine whether the following maps are linear or not. Justify your answers.

(i) $f : \mathbb{R}^2 \rightarrow P_5 : (a, b) \rightarrow (a + b)x^5$.

(ii) $f : M(2, 2) \rightarrow \mathbb{R}^2 : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \rightarrow (ad, bc)$.

(b) Let V, W be two real finite-dimensional vector spaces and let $f : V \rightarrow W$ be a linear map. Define what is meant by the image, the kernel, the rank and the nullity of f and state the rank-nullity theorem.

(c) Let $f : P_2 \rightarrow \mathbb{R}^3$ be a linear map with nullity $f = 1$. Determine whether the map f is injective, surjective, both or neither. Justify your answer.

(d) Consider the map $f : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ given by

$$f(x, y, z, t) = (x + y, 0, z + t)$$

for all $(x, y, z, t) \in \mathbb{R}^4$. Determine whether f is injective, surjective, both or neither and find a basis for the kernel of f and a basis for the image of f .

City University, London, UK

5.20. Let $T_1 \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} 2x + y \\ -x + y \end{bmatrix}$ and $T_2 \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} x \\ x + 2y \end{bmatrix}$.

Which of the following matrices represents the composition $T_1 \circ T_2$ with respect to the standard basis for $\mathbb{R}^2$?

A. $\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$

B. $\begin{bmatrix} 3 & 4 \\ 0 & 1 \end{bmatrix}$

C. $\begin{bmatrix} 3 & 2 \\ 0 & 2 \end{bmatrix}$

D. $\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$

E. $\begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$

University of Toronto, Canada

5.21. Suppose S and T are the linear transformations given by

$$S((x_1, x_2, x_3)) = (3x_1 + 5x_2 - x_3, 4x_2 + 3x_3, x_1 - x_2 + 4x_3)$$

and $T(\mathbf{x}) = \mathbf{Ax}$, where

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 5 \\ 2 & -1 & 0 \\ 4 & 1 & 0 \end{pmatrix}$$

Find the matrix $\mathbf{C}$ so that $(S \circ T)(\mathbf{x}) = S(T(\mathbf{x})) = \mathbf{Cx}$.

University of Maryland, Baltimore County, USA

5.22. Let $\phi : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ be the mapping given by $\phi(\mathbf{x}) = \mathbf{Ax}$ with

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & -1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ -1 & 5 & 2 & 2 \end{pmatrix}$$

- Is ϕ a linear mapping? Justify your answer. Let $\mathbf{e}_1 = (1, 0, 0, 0)^T$. What is $\phi(\mathbf{e}_1)$?
- Compute the determinant of $\mathbf{A}$. [Determinants are covered in the next chapter.]
- Find a basis for the subspaces $\ker(\phi)$ and $\text{im}(\phi)$. Is the mapping ϕ invertible?

Jacobs University, Germany

5.23. (a) Let V be a vector space of dimension n with a basis

$$B = \{\vec{u}_1, \dots, \vec{u}_n\}$$

- Give definition of a linear transformation on V .
 - If T is a linear transformation on V , define what is meant by the matrix of T with respect to the basis B .
- (b) Let V be the vector space of all functions of the form $f(x) = a \sin x + b \cos x$ for arbitrary real constants a and b . Select in V the basis $B = \{\sin x, \cos x\}$. Find the matrix $[T]_B$ of the linear transformation

$$T(f(x)) = f'(x) + f''(x)$$

with respect to the basis B . Here, $f'(x)$ and $f''(x)$ denote the first and the second derivative of the function $f(x)$,

$$f'(x) = \frac{d}{dx}f(x), \quad f''(x) = \frac{d}{dx}f'(x) = \frac{d^2}{dx^2}f(x)$$

- (c) Is the operator T from part (b) invertible? If so, find the matrix of the inverse operator T^{-1} in the basis B .
- (d) Using the operator T^{-1} , find a function $f(x)$ in V such that

$$f''(x) + f'(x) = 2 \sin x + 3 \cos x$$

University of Manchester, UK

- 5.24.** Let V be a vector space over $\mathbb{R}$, and let $T : V \rightarrow \mathbb{R}^n$ be a one-to-one linear transformation. Show that for $\mathbf{u}, \mathbf{v} \in V$ the formula

$$\langle \mathbf{u}, \mathbf{v} \rangle_V := \langle T(\mathbf{u}), T(\mathbf{v}) \rangle$$

defines an inner product on V . (Here $\langle -, - \rangle$ is the standard inner product on $\mathbb{R}^n$.)

University of California, Berkeley, USA

- 5.25.** Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be independent vectors in V and let $T : V \rightarrow W$ be a one-to-one linear transformation of V into W . Prove that $T(\mathbf{v}_1)$ and $T(\mathbf{v}_2)$ are independent vectors in W .

University of Toronto, Canada (part question)

- 5.26.** True or false:

If V is a vector space and $T : V \rightarrow V$ is an injective linear transformation, then T is surjective. (Be careful.)

University of California, Berkeley, USA (part question)

- 5.27.** Suppose A is a linear operator on a 10-dimensional space V , such that $A^2 = 0$.

- (a) Show that $\text{Im}(A) \subset \text{Ker}(A)$.
- (b) Show that the rank of A is at most 5. (Hint: (a) and the rank-nullity theorem might help.)

Stanford University, USA (part question)

Sample questions

- 5.28.** Let M_{nn} be the vector space of size n by n matrices. Let $\mathbf{B}$ be a matrix of M_{nn} and $T : M_{nn} \rightarrow M_{nn}$ be a transform such that $T(\mathbf{A}) = \mathbf{A}\mathbf{B} + \mathbf{B}\mathbf{A}$ where $\mathbf{A} \in M_{nn}$. Show that T is a linear transform.

- 5.29.** Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} e^x \\ e^y \end{pmatrix}$. Show that T is **not** linear.

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Tell us about yourself and your work.

Linear algebra is a fundamental tool in the design and analysis of algorithms for data mining and statistical data analysis applications: numerous modern, massive datasets are modelled as matrices. Naturally, a solid background in linear algebra is a must if your work will eventually involve processing large-scale data, and many highly rewarding jobs in the future will involve the analysis of Big Data. Think, for example, of the following data matrix: the rows of the matrix correspond to customers of your favourite online store, and the columns correspond to the products available at the store. Let the (i,j) -th entry of the matrix be the ‘utility’ of product j for customer i . Then, a recommender system like the ones used by many such online stores, seeks to recommend high utility products to customers, with the obvious hope that the customer will purchase the product. Since it is not possible to ask every customer to reveal his or her preference for every product, a recommender system tries to *infer* products of high utility for a customer from a small sample of known entries of the matrix. A lot of my work focuses on problems of this type: how can we reconstruct missing entries of a matrix from a small sample of rows and/or columns and/or elements of the matrix? Solving this problem better than the current state of the art would have made you a millionaire in the Netflix challenge a couple of years ago!

A related question focuses on identifying influential rows and/or columns and/or elements of a matrix. Imagine, for example, that you are interested in approximating the product of two matrices by selecting a few rows of the first matrix and a few columns of the second matrix. Which rows and or columns will return the best approximation? Such problems are of particular interest when dealing with massive matrices that cannot be easily stored in random access memory (RAM). Similar sampling-based algorithms are very useful in approximating singular values and singular vectors of matrices. These operations are of particular importance in data mining, since they lie at the heart of principal components analysis (PCA), a fundamental dimensionality reduction technique that helps visualize, denoise and interpret high-dimensional datasets.

The challenges in my line of work lie at the intersection of linear algebra and probability theory. This ‘marriage’ of the two domains as well as the formal study of the effects of sampling rows/columns/elements of matrices is relatively recent, and you can trace it to a few influential papers in the late 1990s. On the one hand, probability theory

provides the fundamental tools that help quantify the ‘distance’ between the original matrix and the sampled one. On the other hand, linear algebraic tools (for example, matrix perturbation theory), allow us to quantify how characteristic properties (e.g. singular values, singular vectors, etc.) of the original matrix can be approximated using the sampled matrix.

Have you any particular messages that you would like to give to students starting off studying linear algebra?

You need to understand singular value decomposition (SVD)! This will probably be towards the end of an introductory linear algebra course, but it really should be the take-home message of any such course. It lies at the heart of so many things, from computing the rank of a matrix, to solving least-square problems and computing the inverse or pseudoinverse of a matrix. I will quote Prof. Dianne O’Leary of the University of Maryland: SVD is the Swiss Army knife and the Rolls Royce of numerical linear algebra. It has numerous uses (like a Swiss Army knife), but it is computationally expensive (like a Rolls Royce). I hope that some of my own work on randomized algorithms for the SVD has helped make it a bit more affordable!